

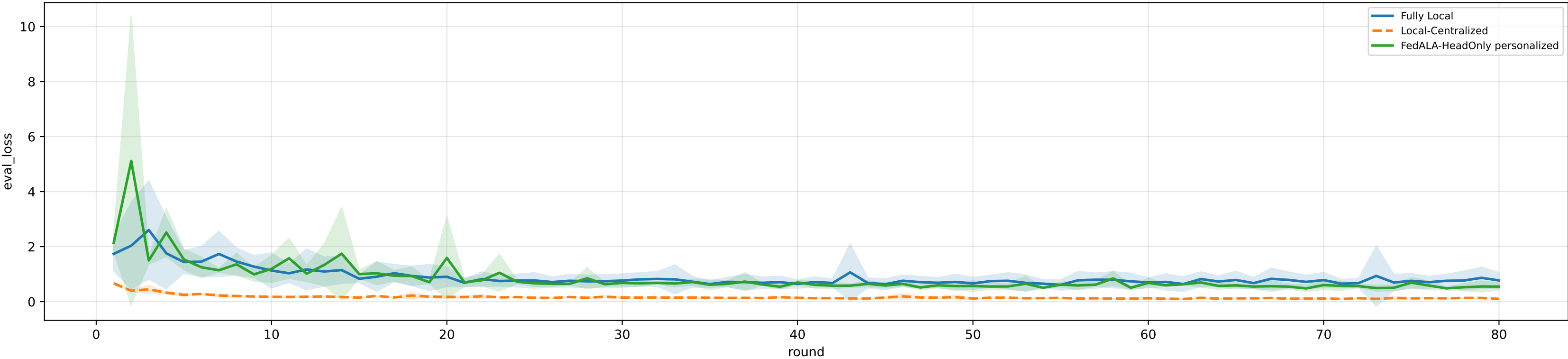
Controlled q-label heterogeneity: iid | FedALA-HeadOnly diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

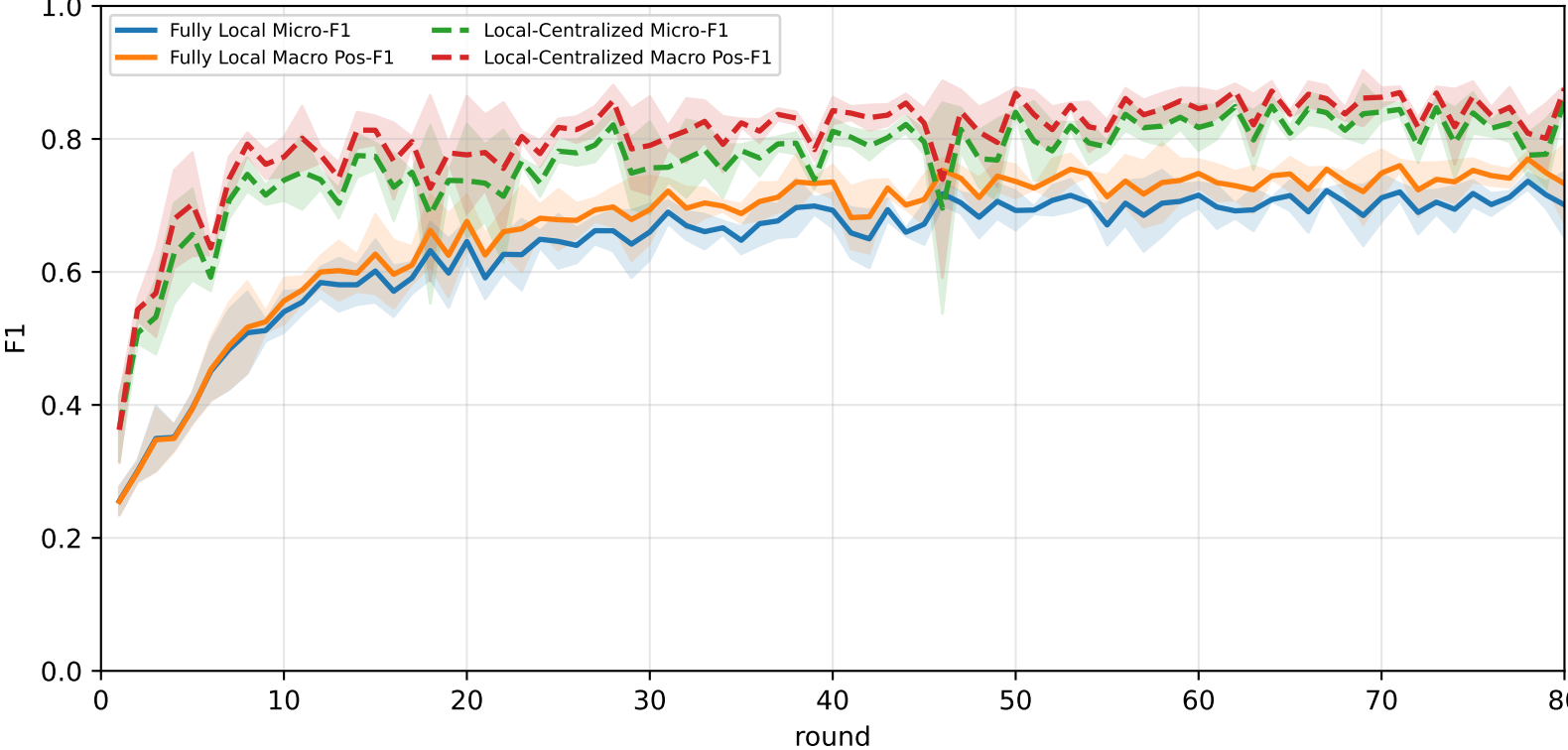
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	67.9% ± 1.6%	72.3% ± 1.5%	<u>90.0% ± 1.5%</u>	<u>85.0% ± 1.7%</u>	67.5% ± 3.5%	66.8% ± 2.2%	50.9% ± 1.2%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	<u>70.9% ± 2.4%</u>	<u>73.4% ± 2.0%</u>	84.7% ± 1.2%	83.8% ± 4.6%	<u>71.5% ± 2.8%</u>	<u>67.8% ± 3.5%</u>	<u>58.3% ± 3.9%</u>

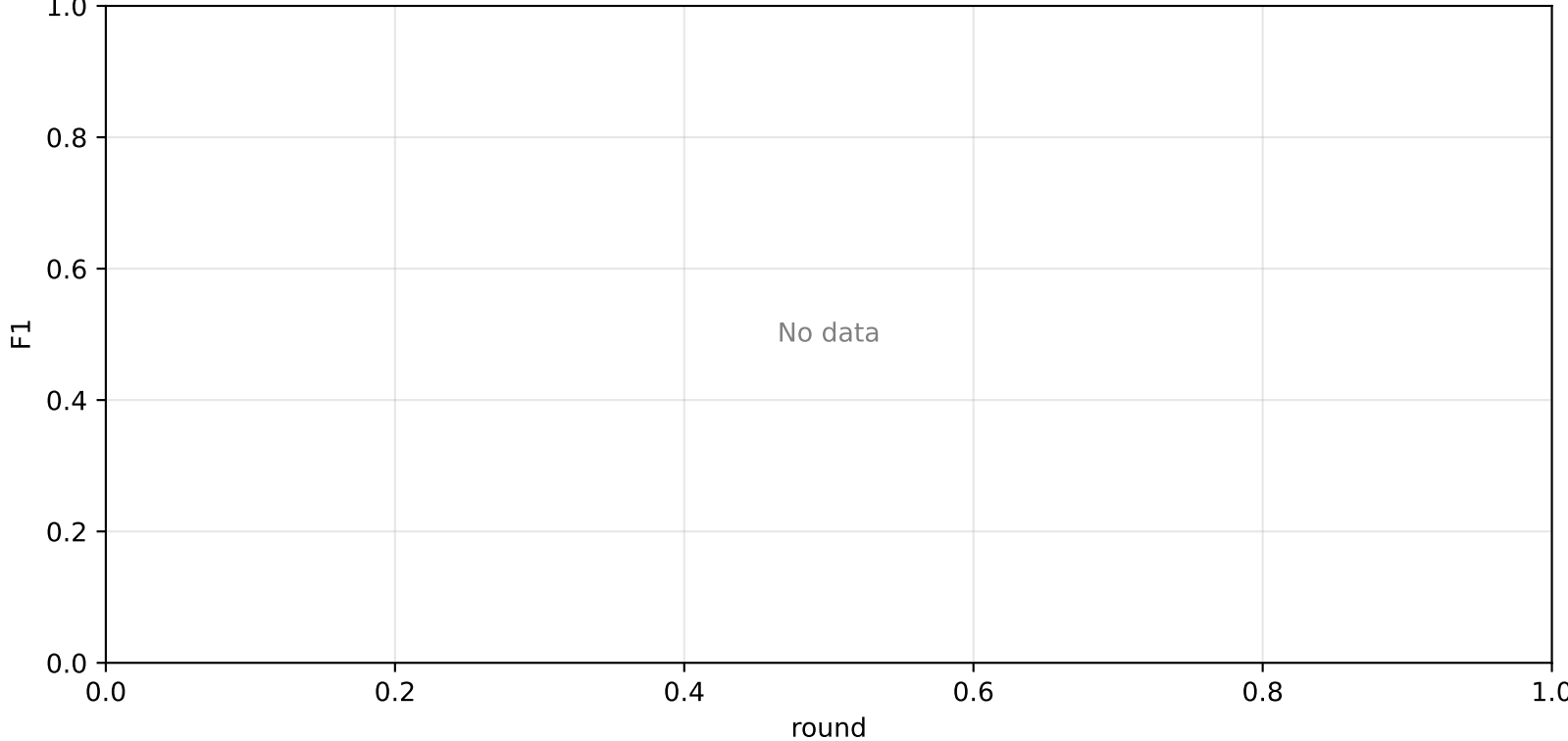
Validation loss comparison



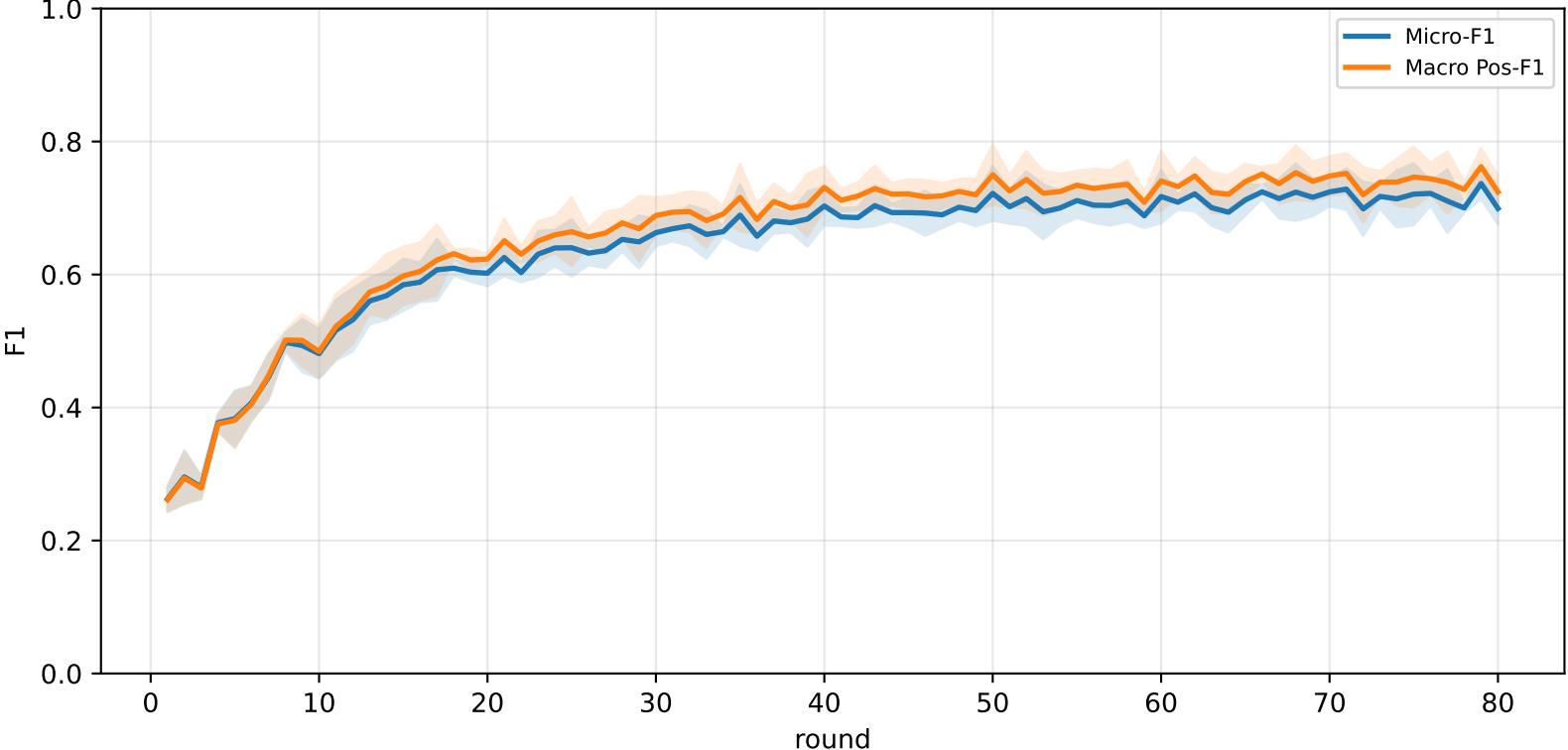
Pooled baseline micro / macro F1



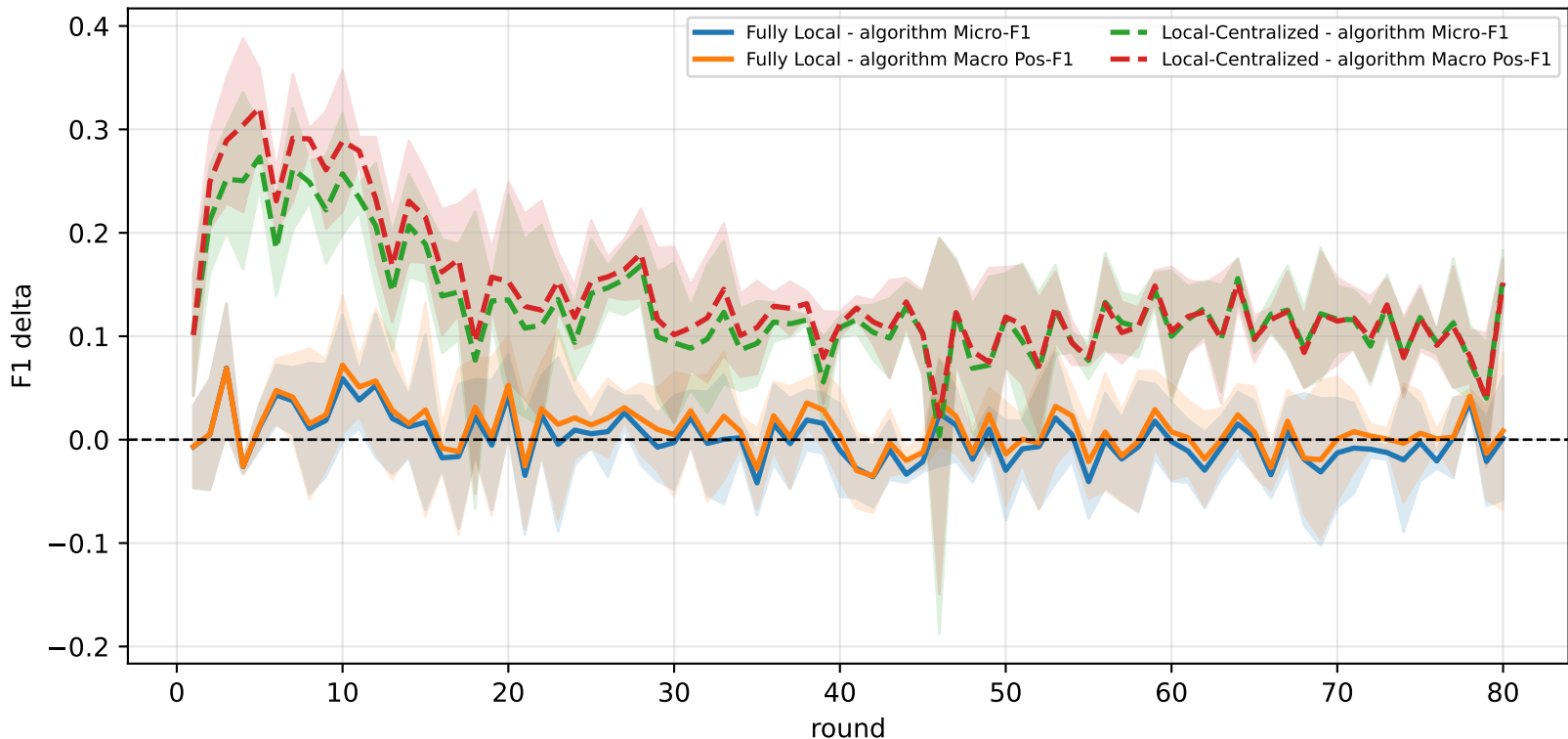
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly



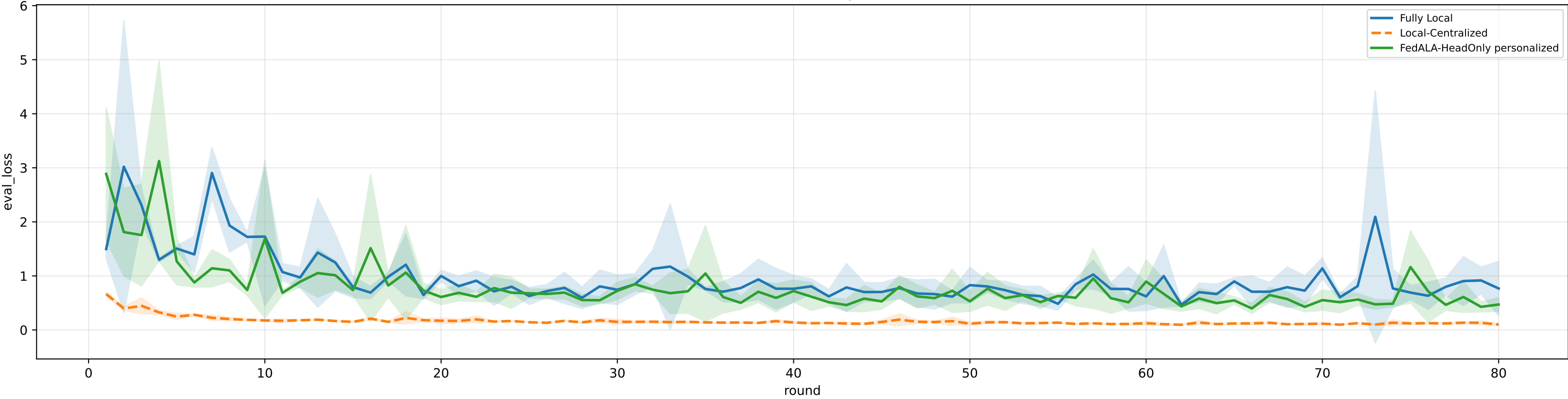
Controlled q-label heterogeneity: iid | FedALA-HeadOnly diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

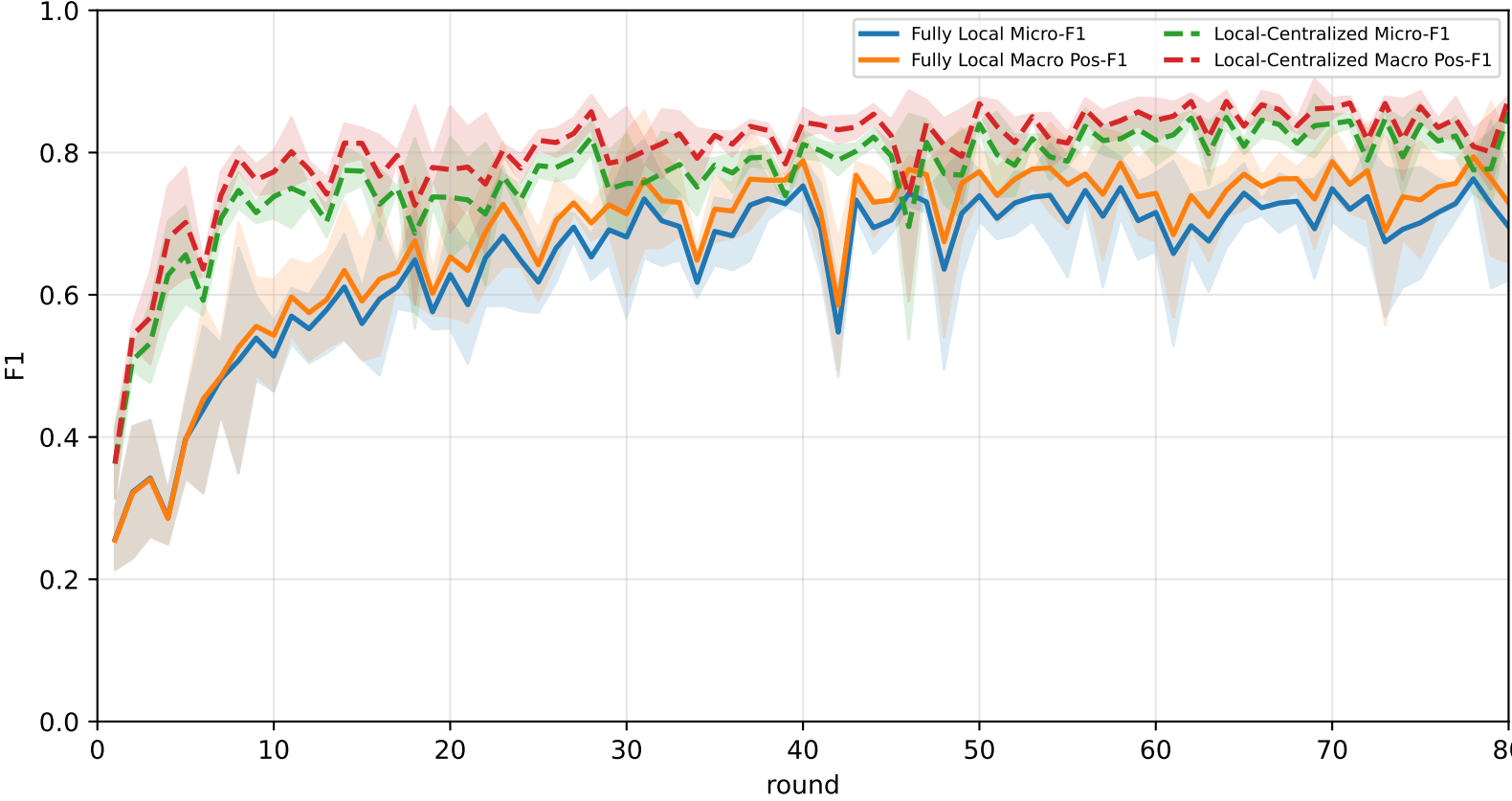
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.1% ± 3.2%	<u>74.3% ± 2.5%</u>	<u>93.1% ± 1.1%</u>	<u>86.7% ± 1.6%</u>	<u>70.3% ± 4.6%</u>	67.5% ± 2.2%	53.8% ± 5.0%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	<u>71.5% ± 2.5%</u>	<u>74.3% ± 2.3%</u>	88.5% ± 2.9%	86.6% ± 2.1%	69.7% ± 3.6%	<u>67.7% ± 3.0%</u>	<u>59.1% ± 3.5%</u>

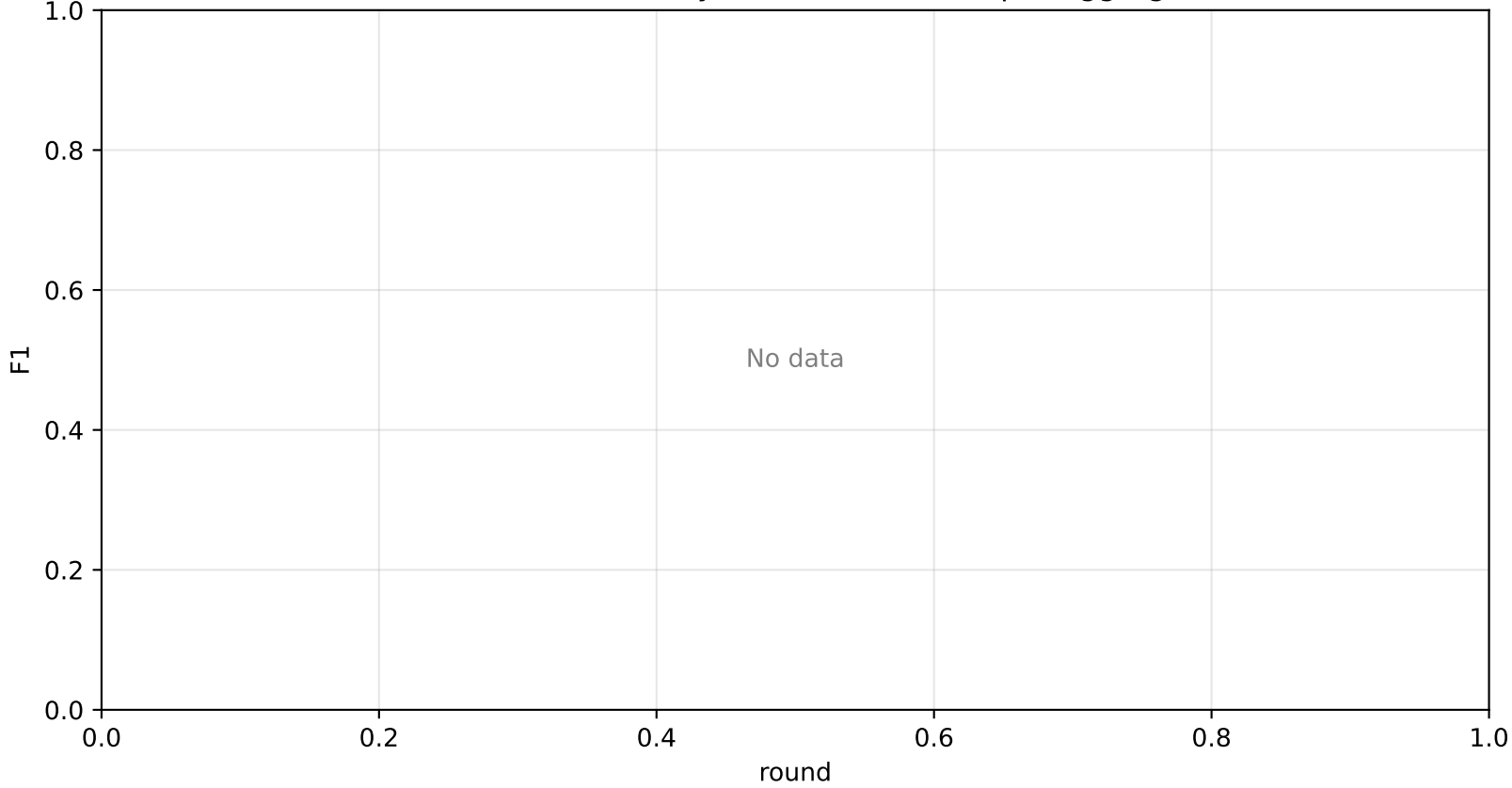
Validation loss comparison



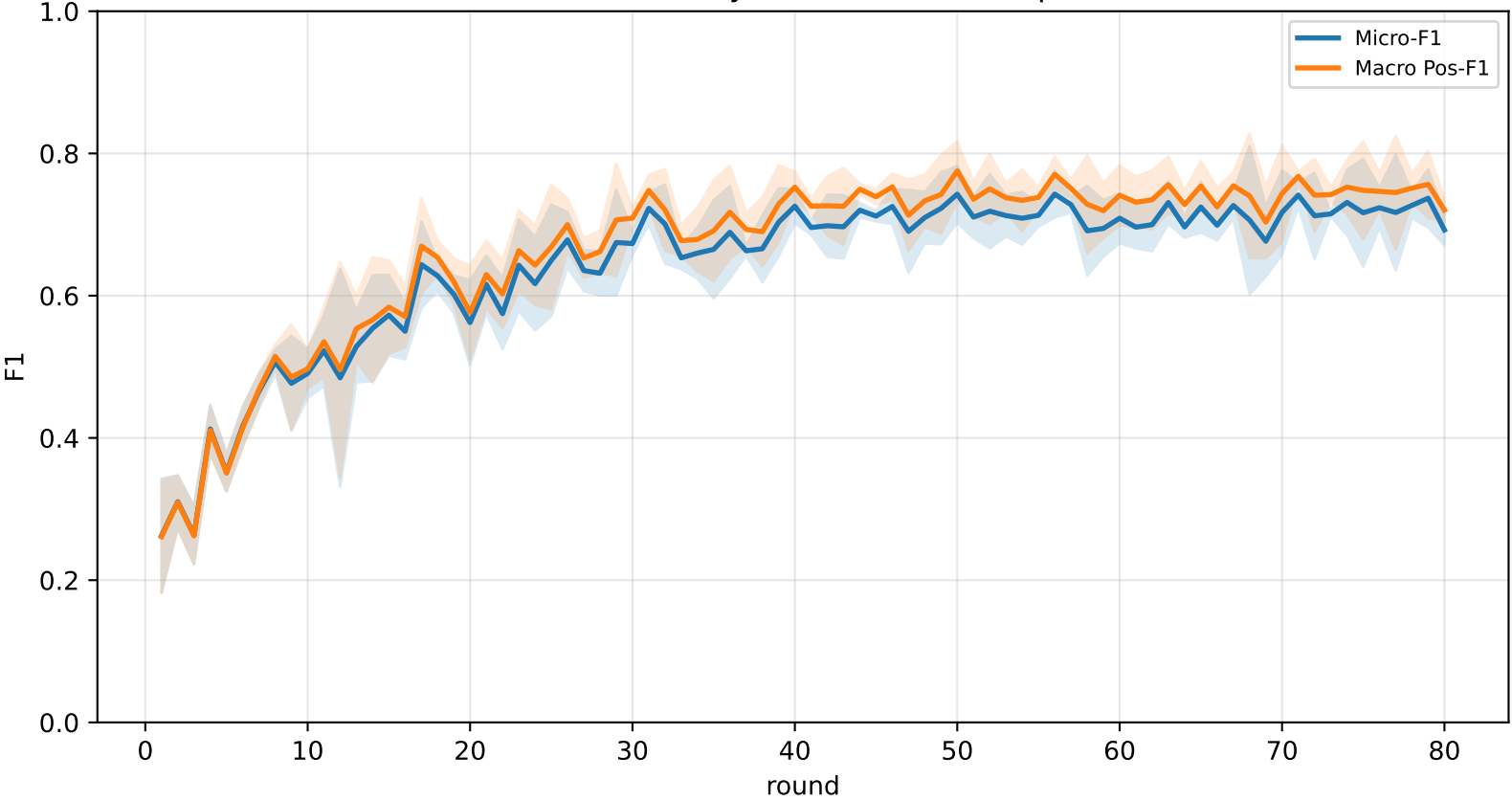
Pooled baseline micro / macro F1



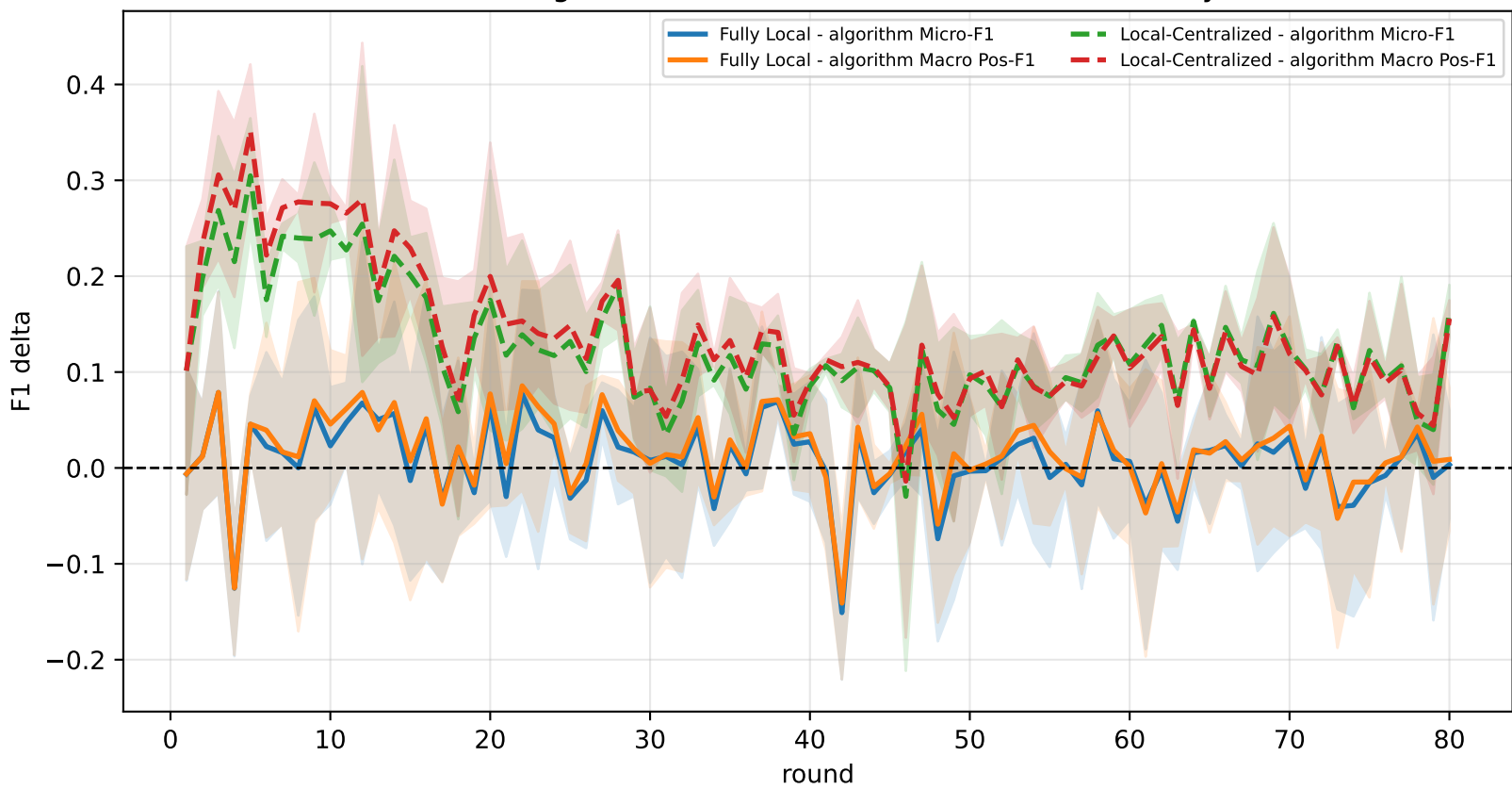
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

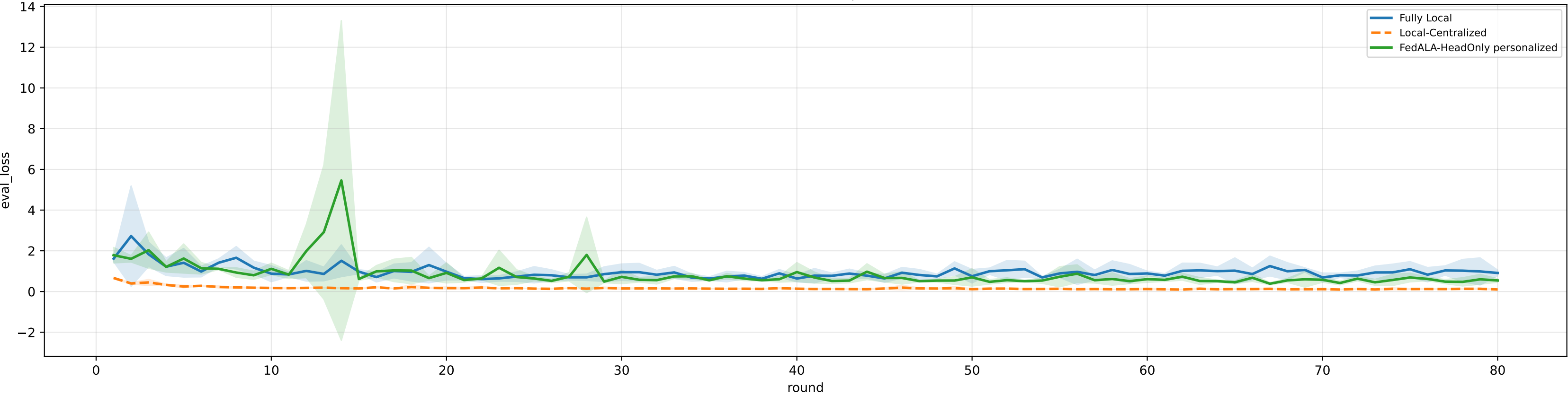


family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

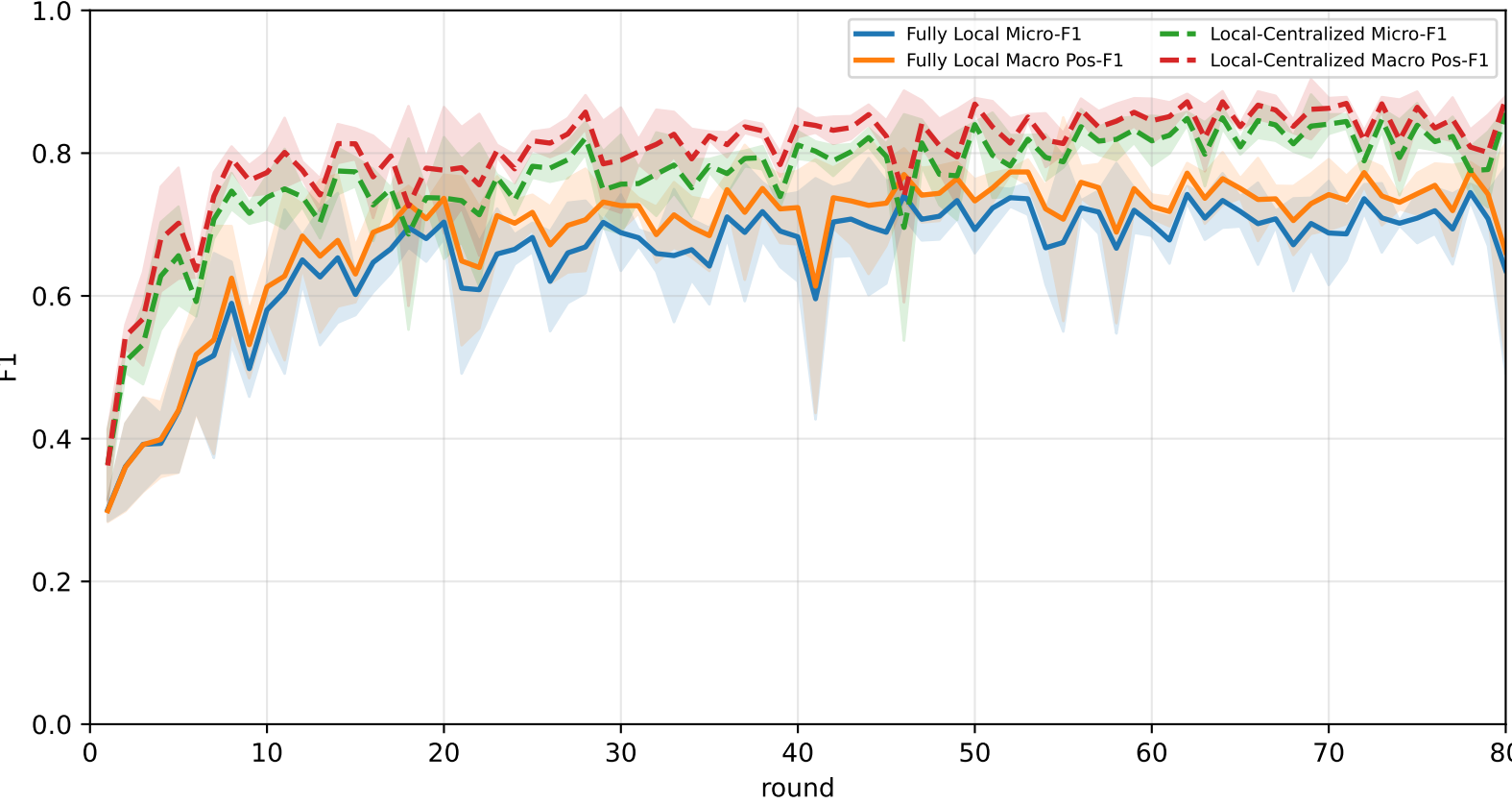
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	69.5% ± 0.8%	73.5% ± 0.6%	<u>90.4% ± 0.9%</u>	<u>85.2% ± 1.6%</u>	67.7% ± 0.6%	71.2% ± 1.1%	52.9% ± 2.1%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	<u>72.4% ± 3.4%</u>	<u>74.9% ± 3.3%</u>	87.3% ± 6.2%	84.3% ± 5.0%	<u>72.0% ± 4.1%</u>	<u>71.3% ± 2.7%</u>	<u>59.7% ± 6.5%</u>

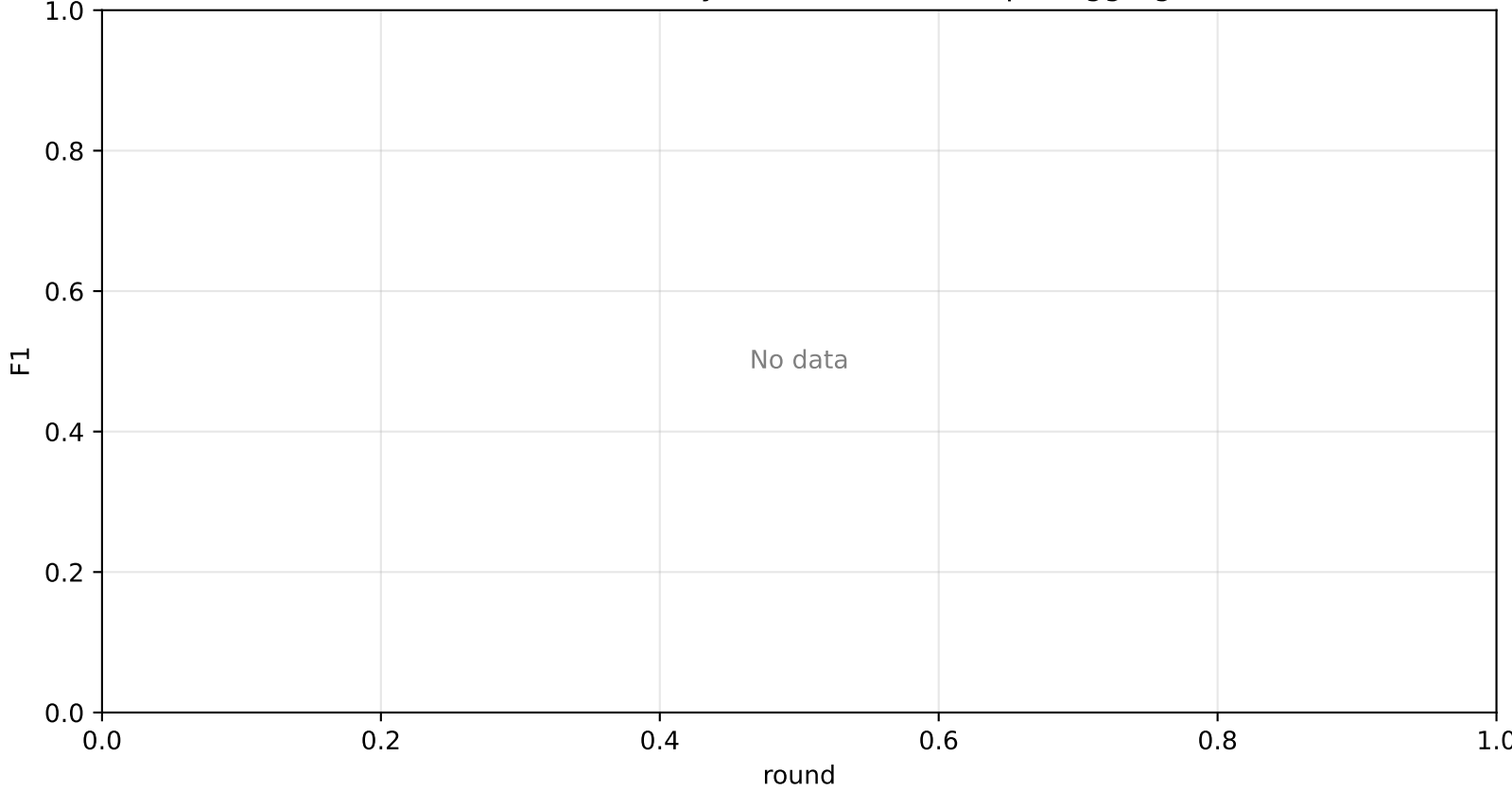
Validation loss comparison



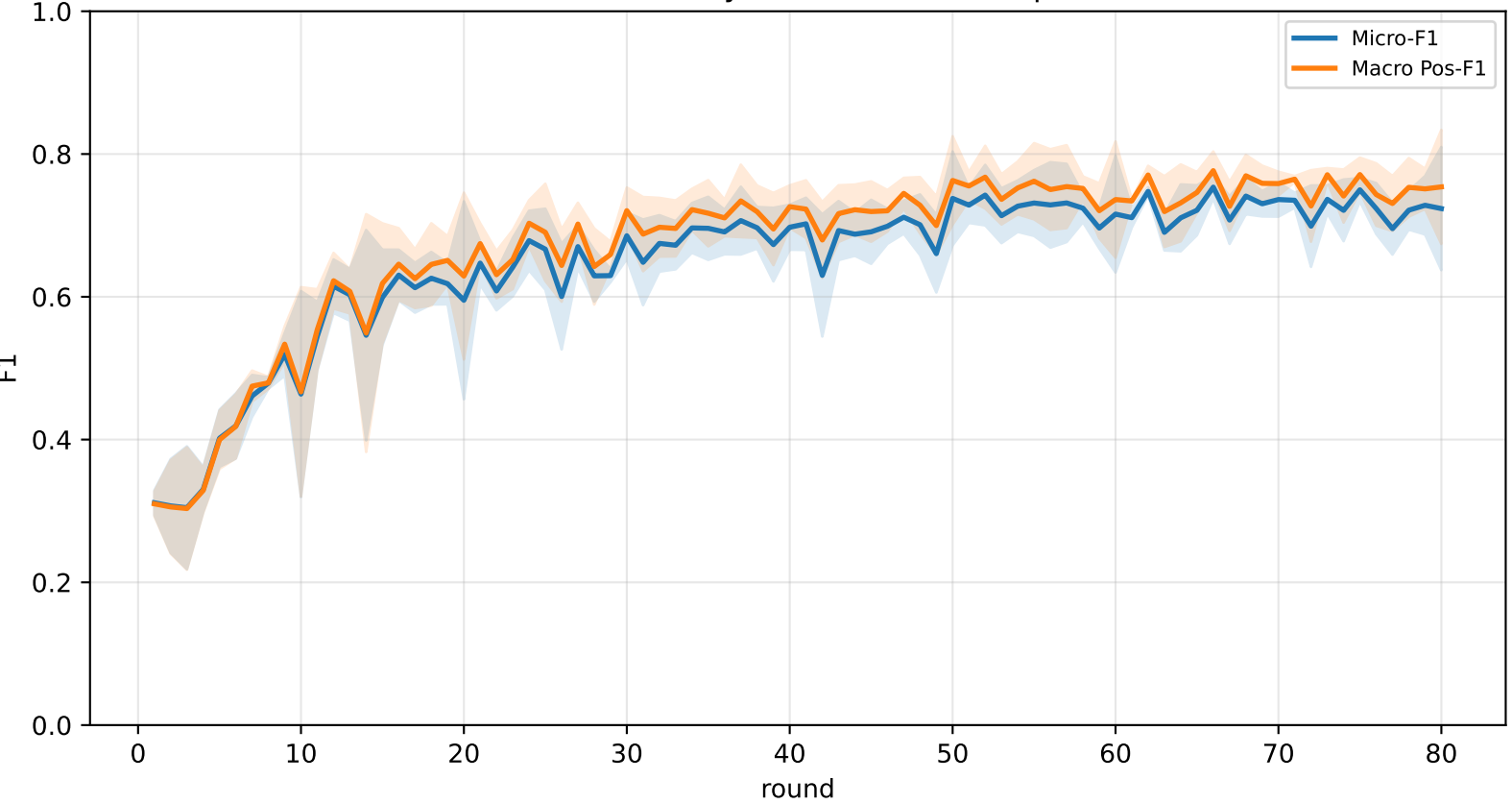
Pooled baseline micro / macro F1



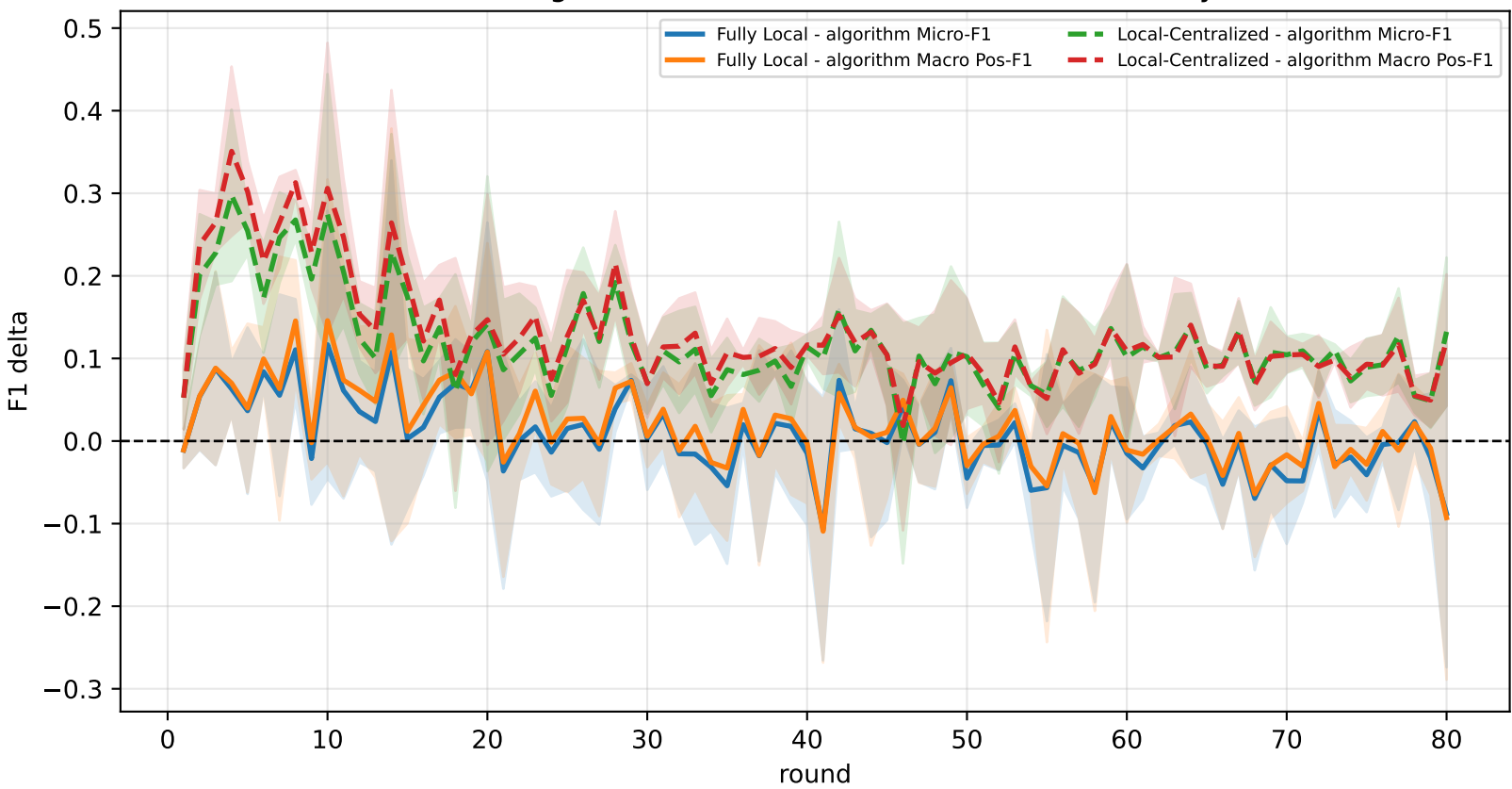
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

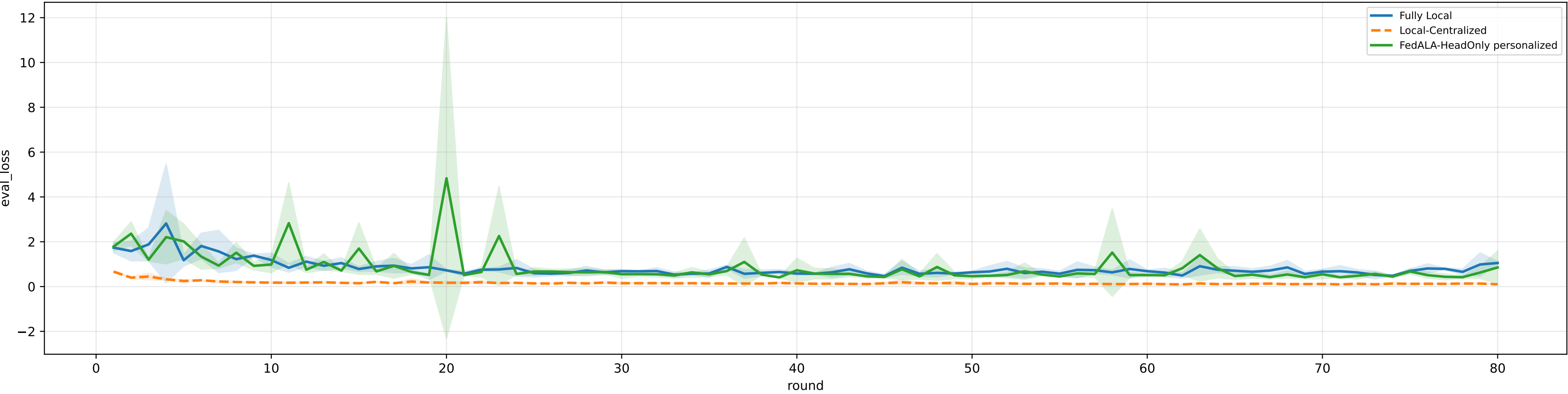


family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

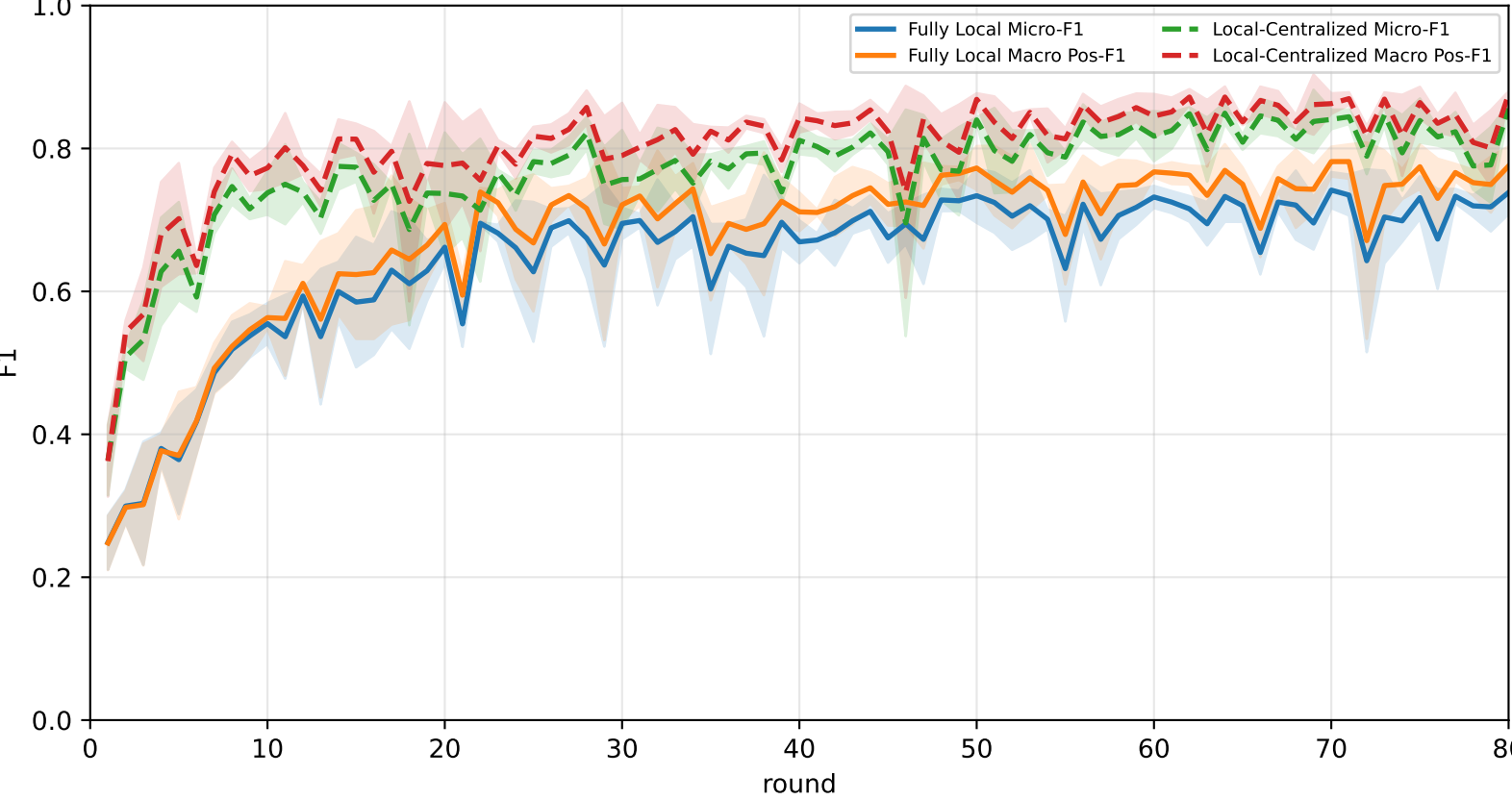
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.8% ± 3.3%	70.1% ± 2.7%	<u>92.2% ± 3.4%</u>	<u>82.7% ± 3.5%</u>	65.7% ± 8.3%	60.9% ± 7.0%	48.9% ± 2.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	<u>68.9% ± 2.3%</u>	<u>71.6% ± 2.9%</u>	83.9% ± 3.4%	82.2% ± 8.8%	<u>69.5% ± 3.7%</u>	<u>66.2% ± 4.2%</u>	<u>56.4% ± 1.4%</u>

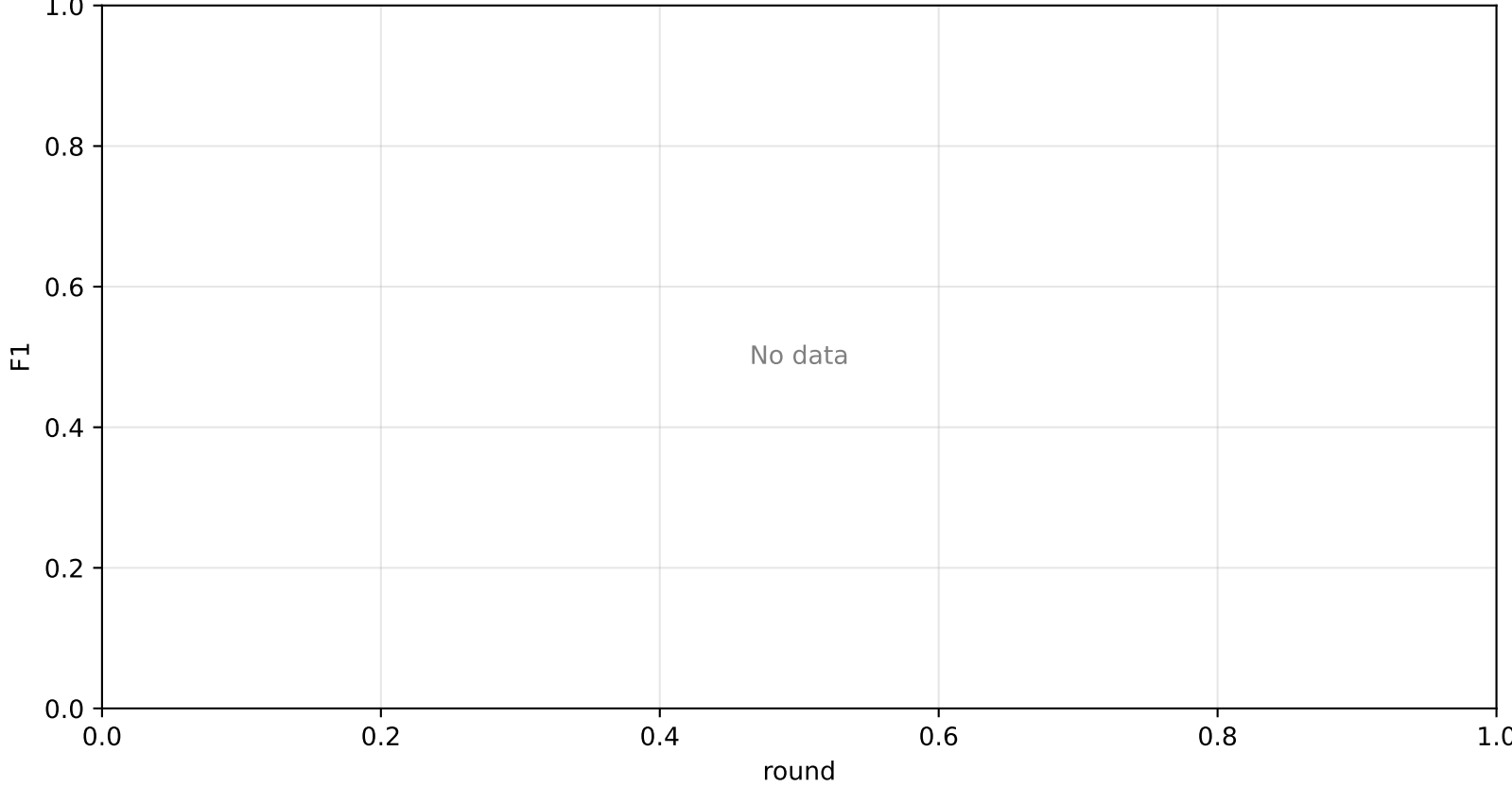
Validation loss comparison



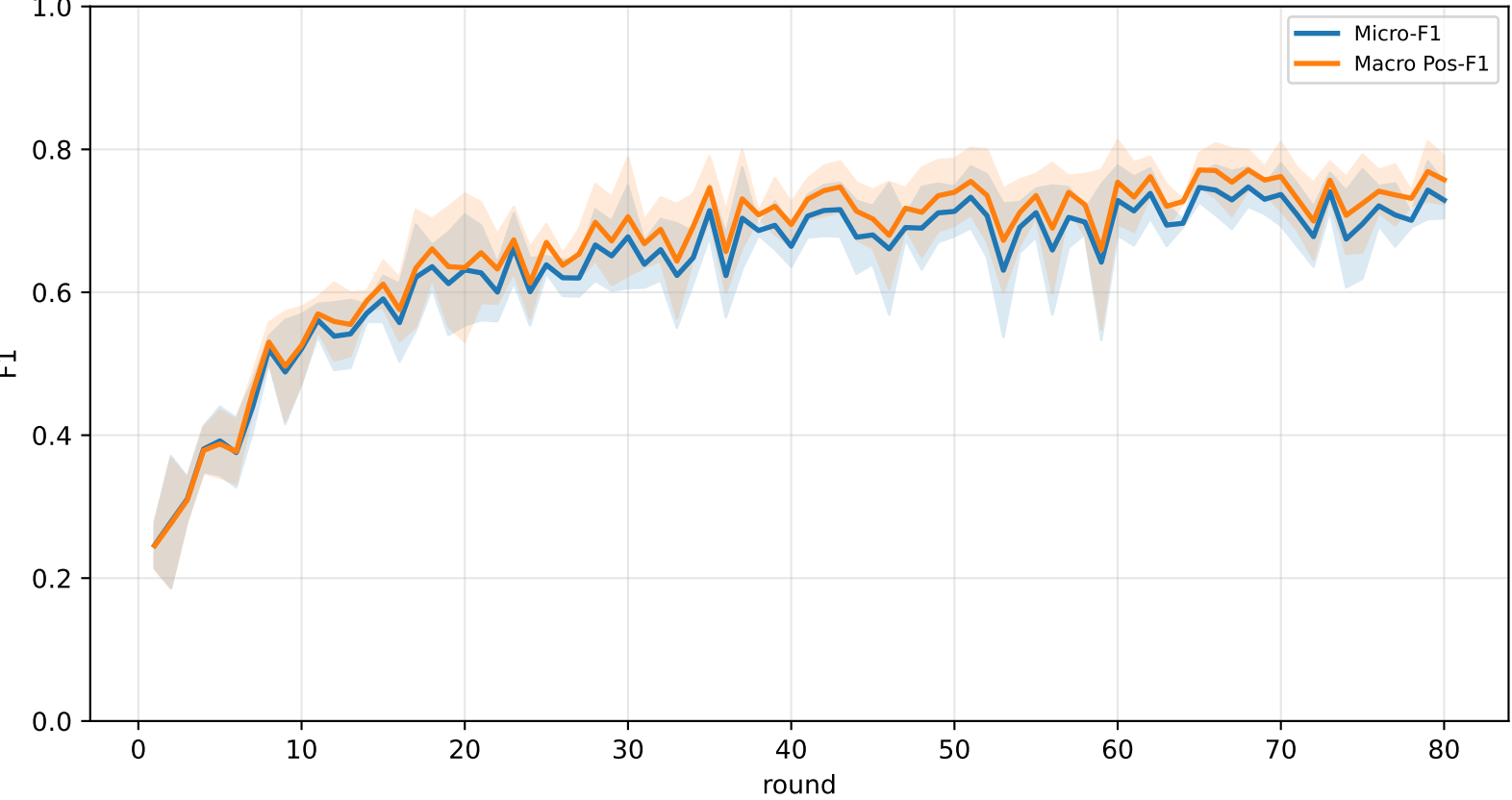
Pooled baseline micro / macro F1



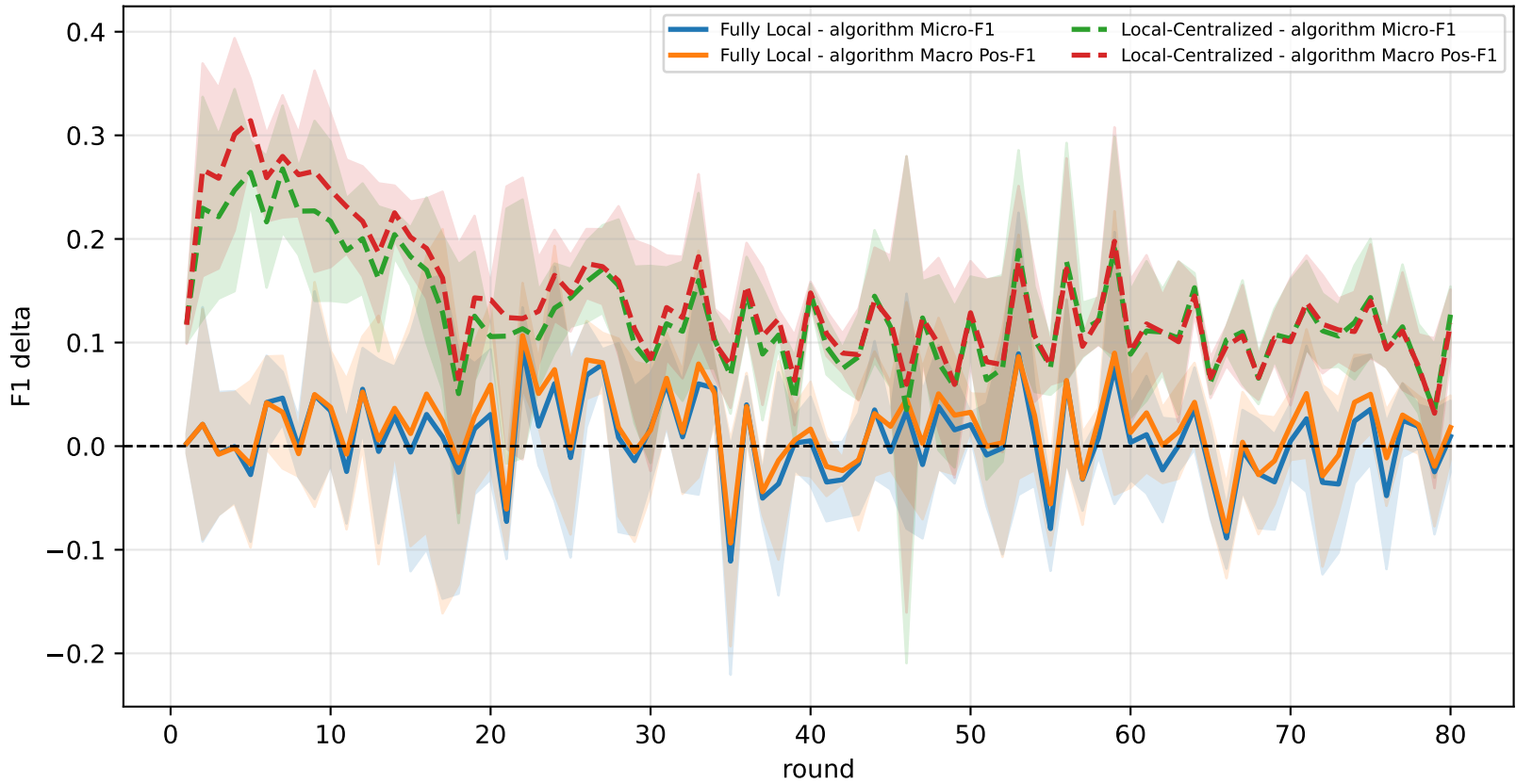
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

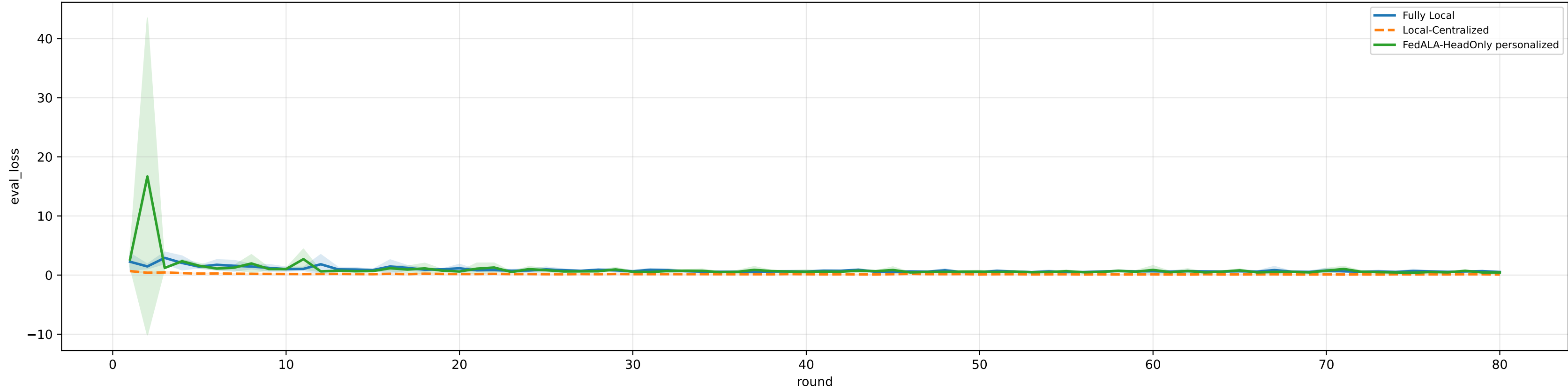


family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

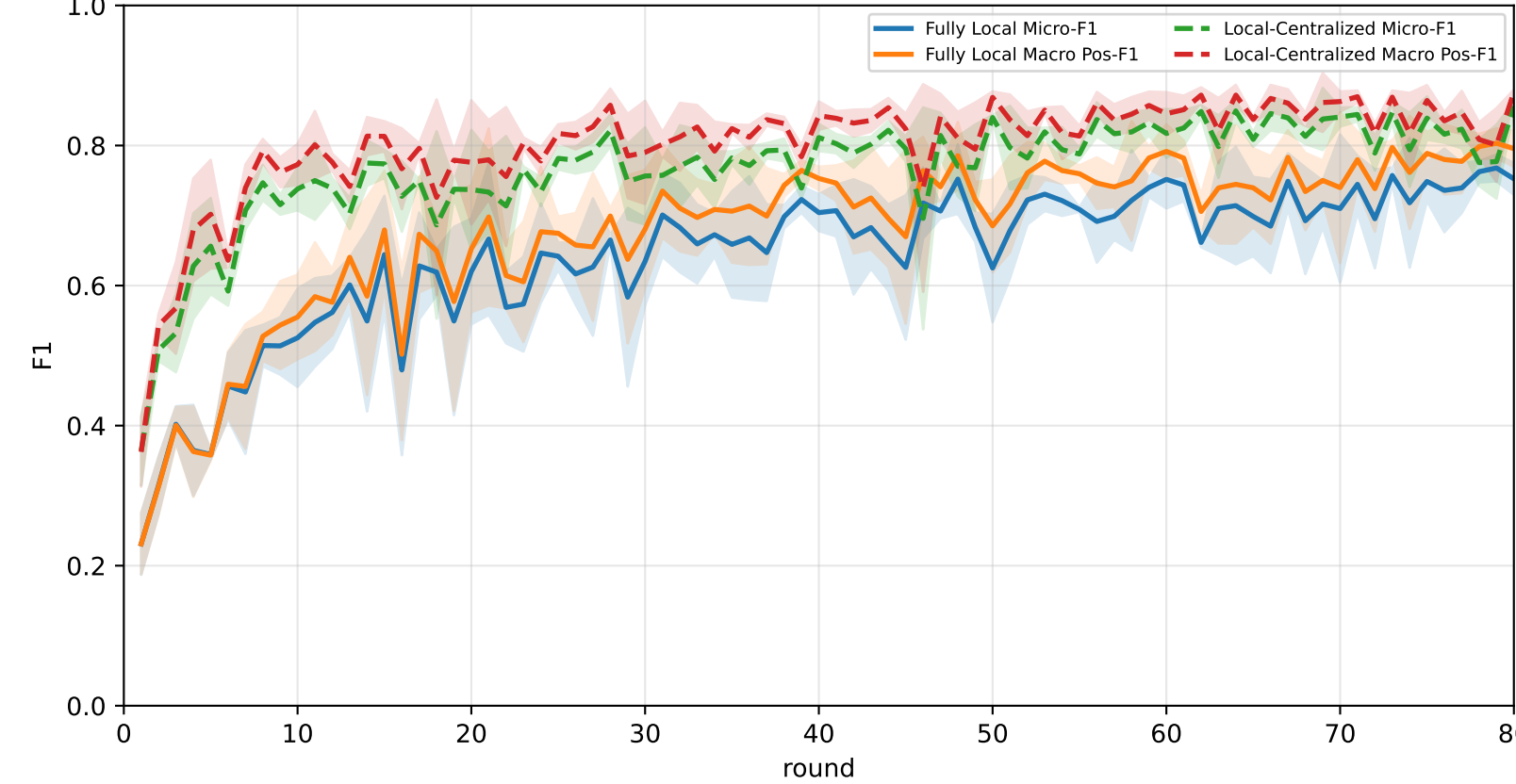
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	<u>70.5% ± 1.9%</u>	<u>74.2% ± 2.2%</u>	<u>87.8% ± 6.8%</u>	<u>86.4% ± 2.4%</u>	<u>74.0% ± 2.0%</u>	<u>69.7% ± 1.8%</u>	53.3% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	70.2% ± 5.3%	72.3% ± 4.9%	82.3% ± 6.7%	82.1% ± 5.5%	73.1% ± 3.0%	65.6% ± 7.1%	<u>58.4% ± 7.1%</u>

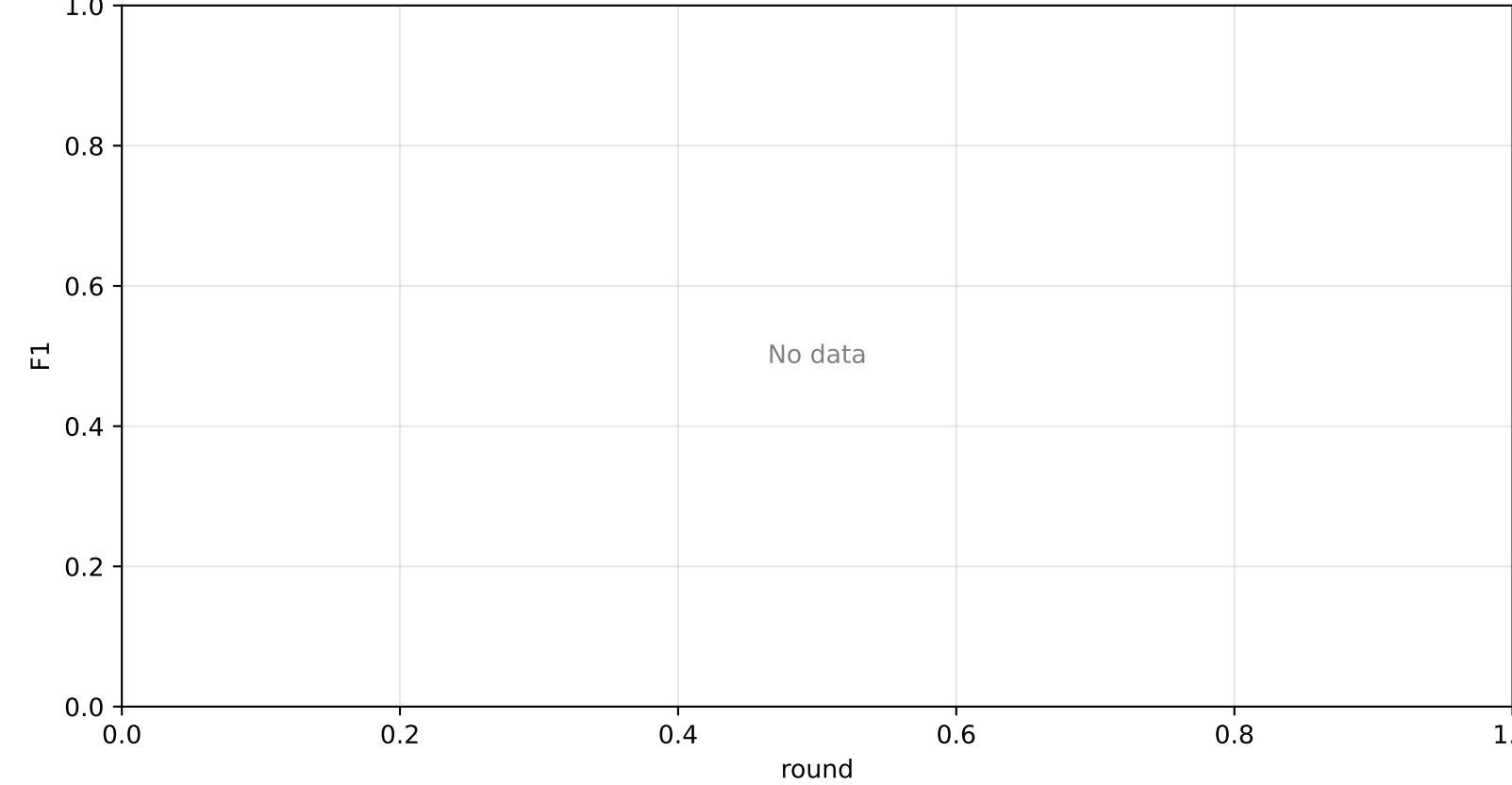
Validation loss comparison



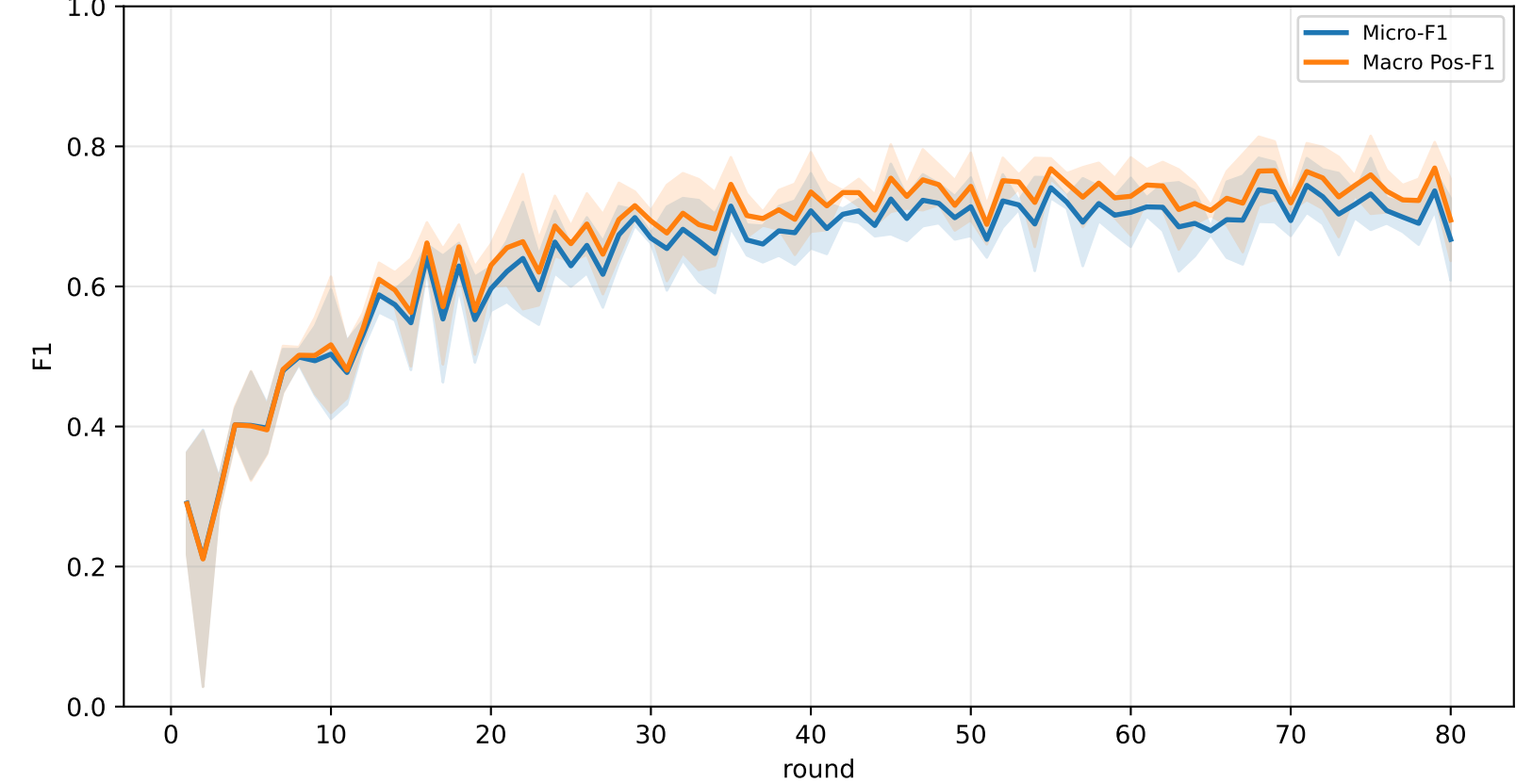
Pooled baseline micro / macro F1



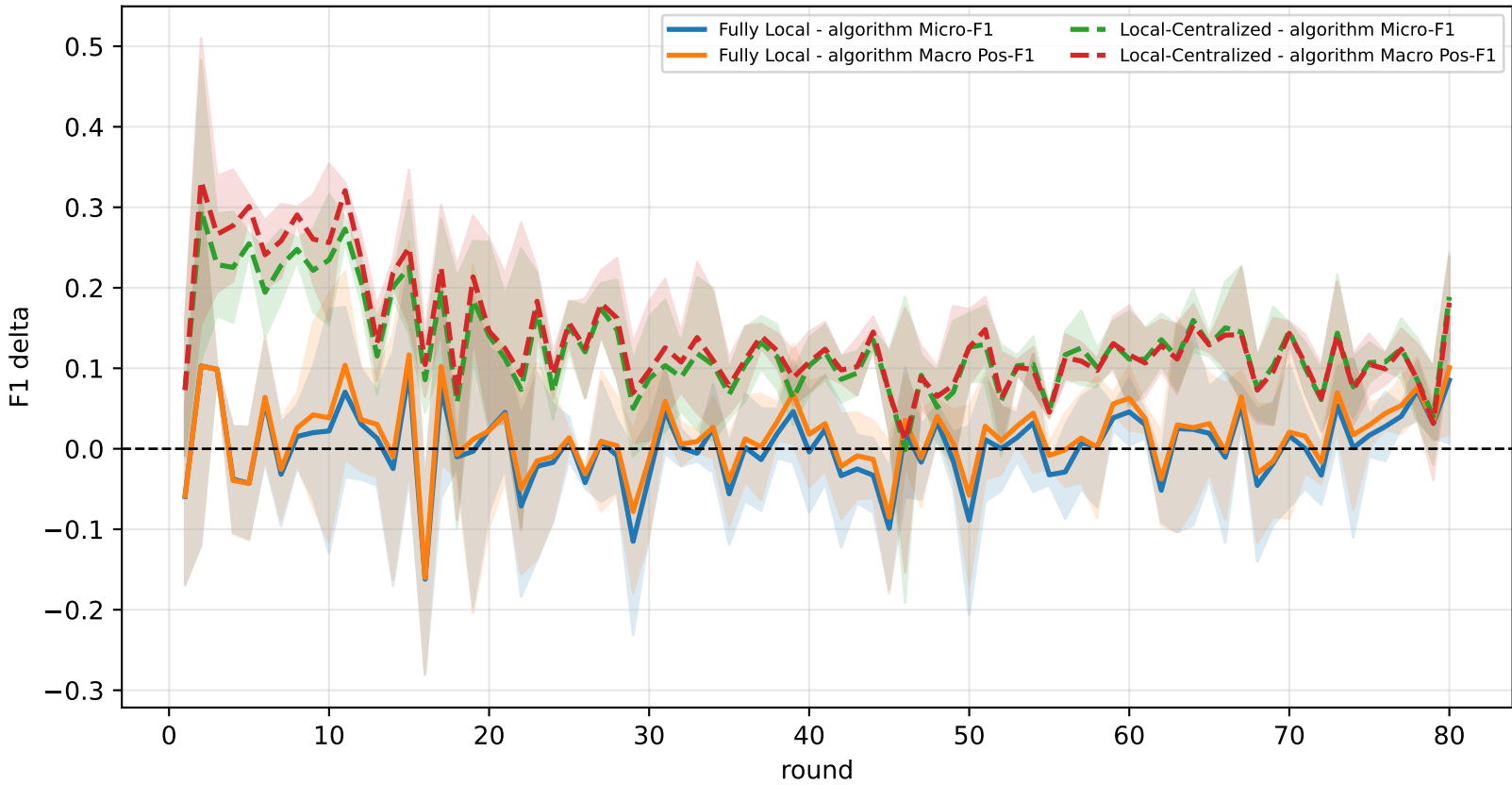
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

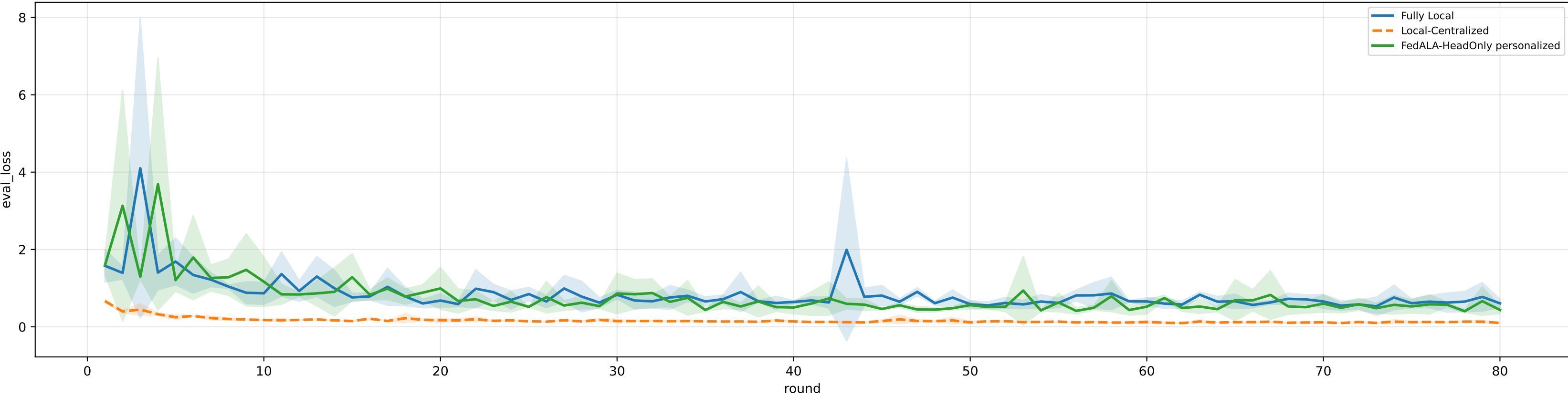


family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

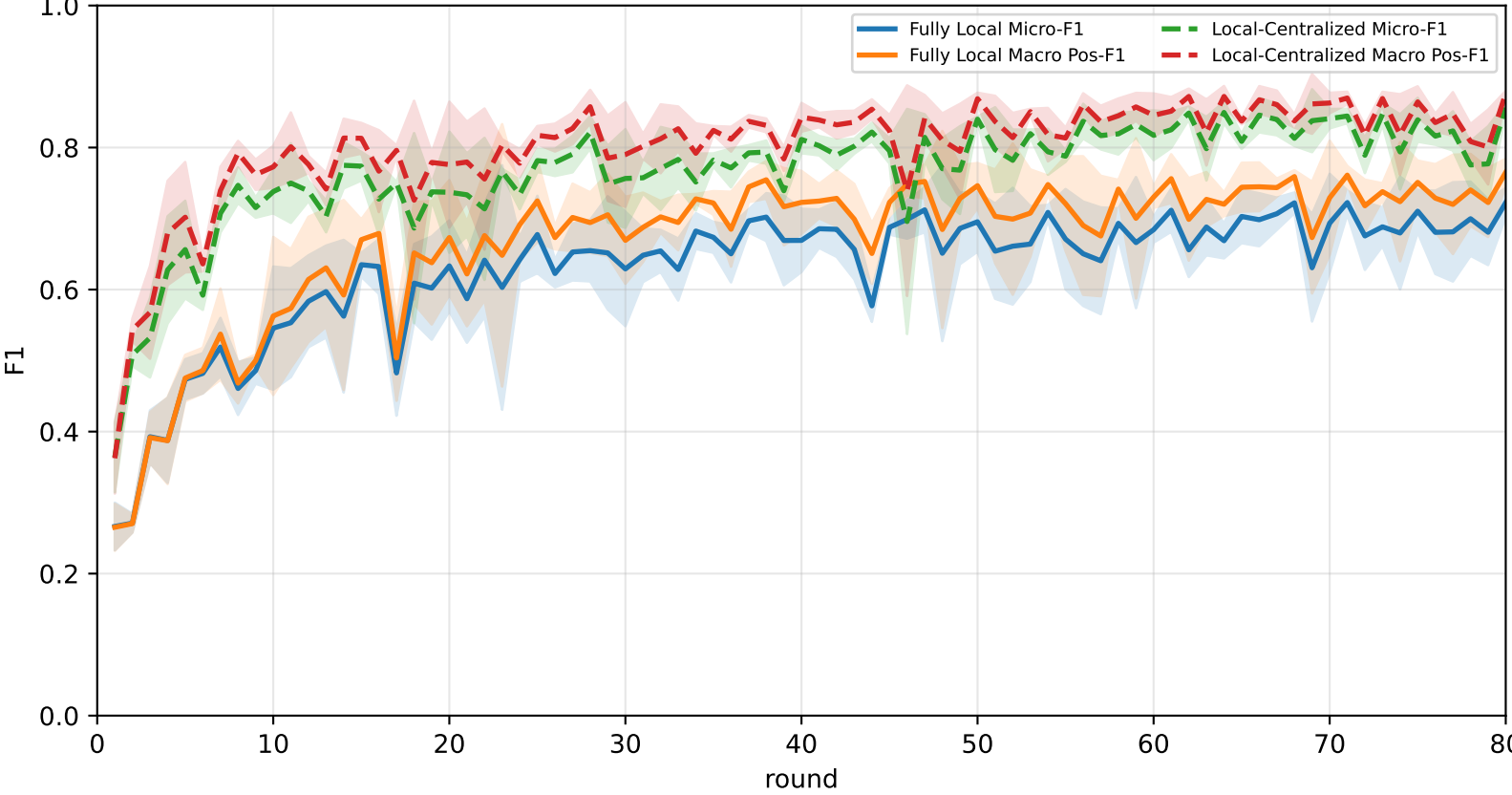
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.5% ± 3.5%	69.6% ± 3.3%	<u>87.3% ± 2.7%</u>	84.4% ± 3.4%	62.5% ± 7.2%	67.0% ± 5.0%	46.8% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedALA-HeadOnly	<u>71.5% ± 1.2%</u>	<u>73.9% ± 1.0%</u>	83.5% ± 6.0%	<u>84.8% ± 2.8%</u>	<u>74.0% ± 2.7%</u>	<u>68.8% ± 2.1%</u>	<u>58.5% ± 3.4%</u>

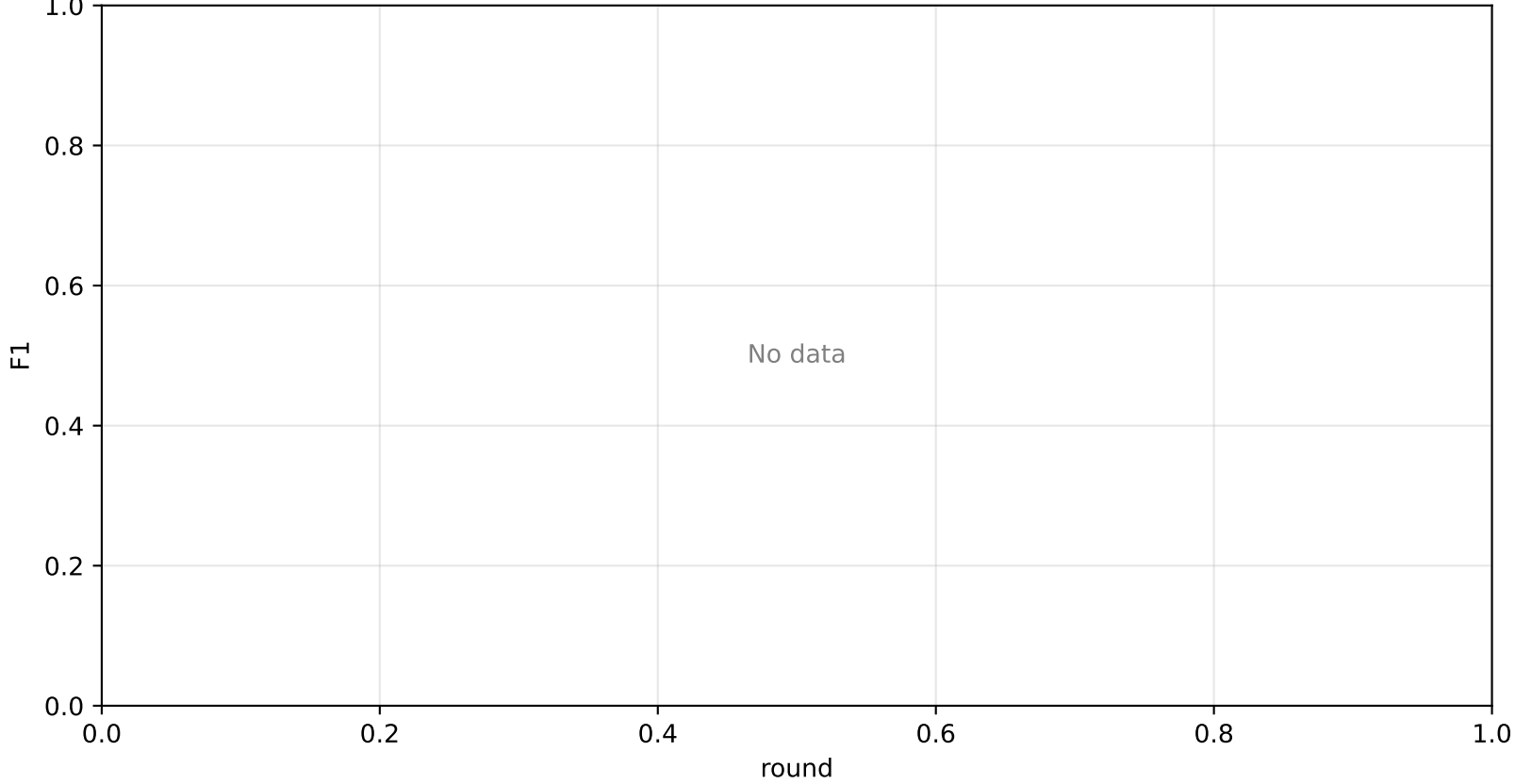
Validation loss comparison



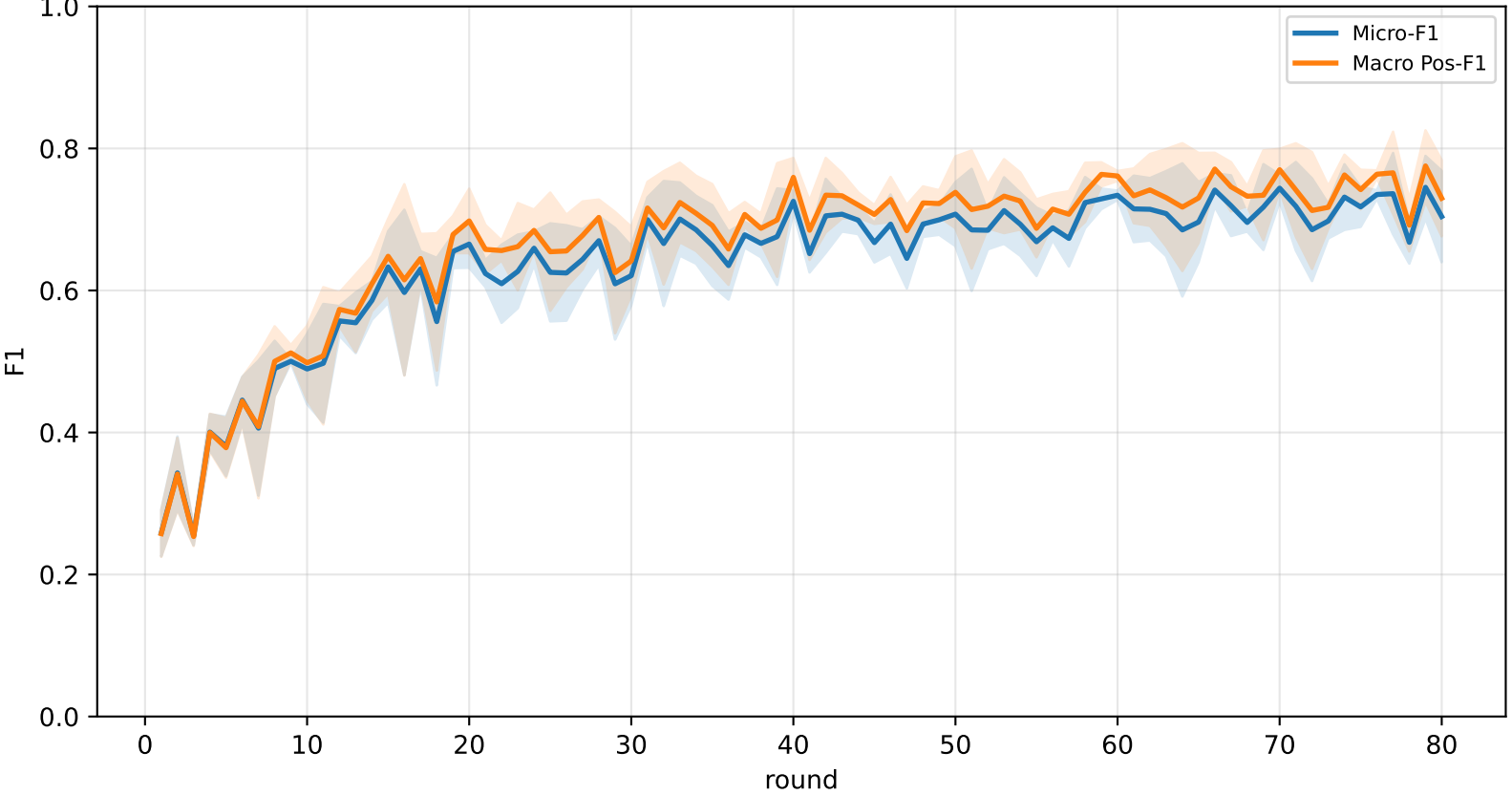
Pooled baseline micro / macro F1



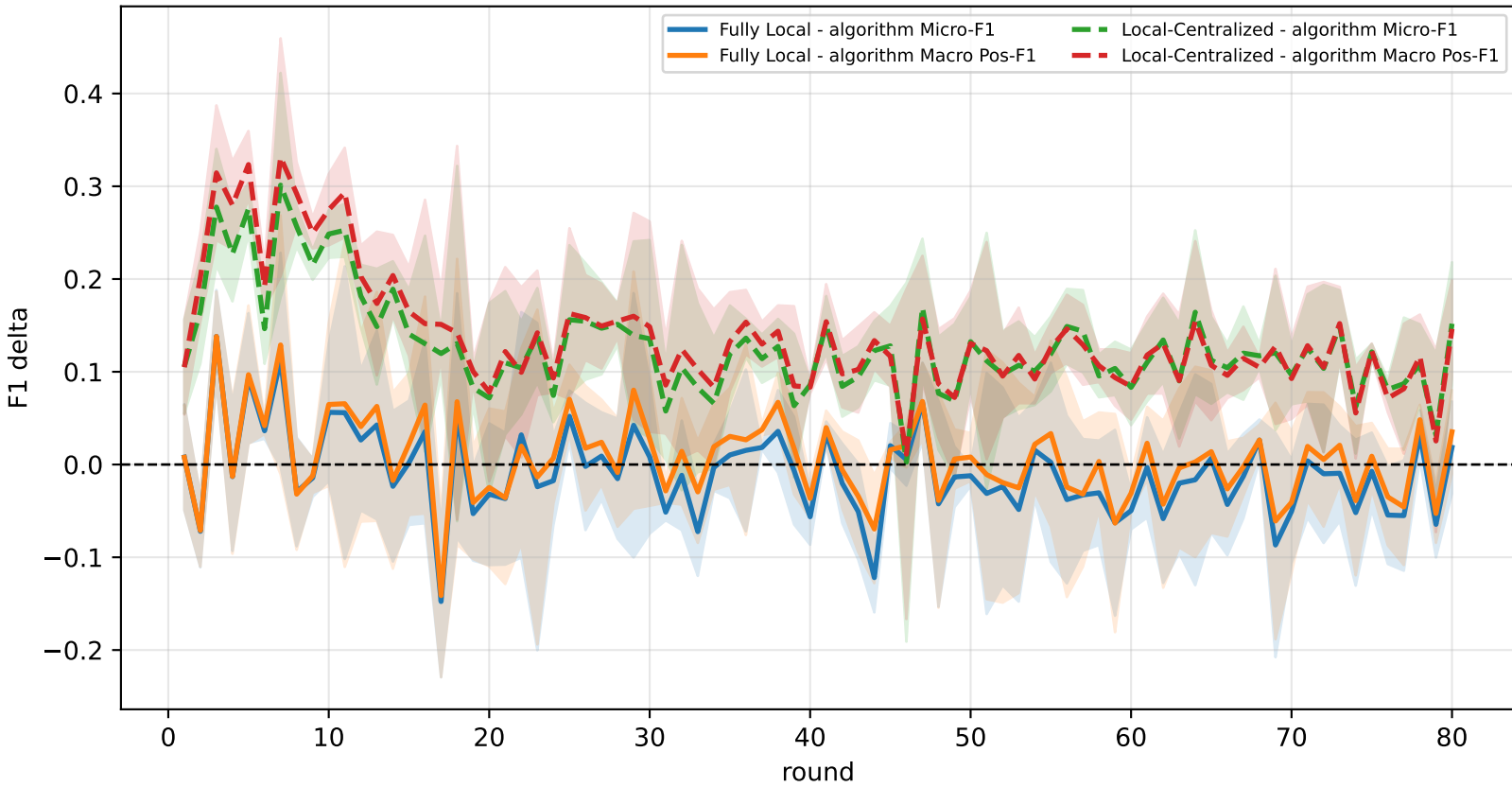
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



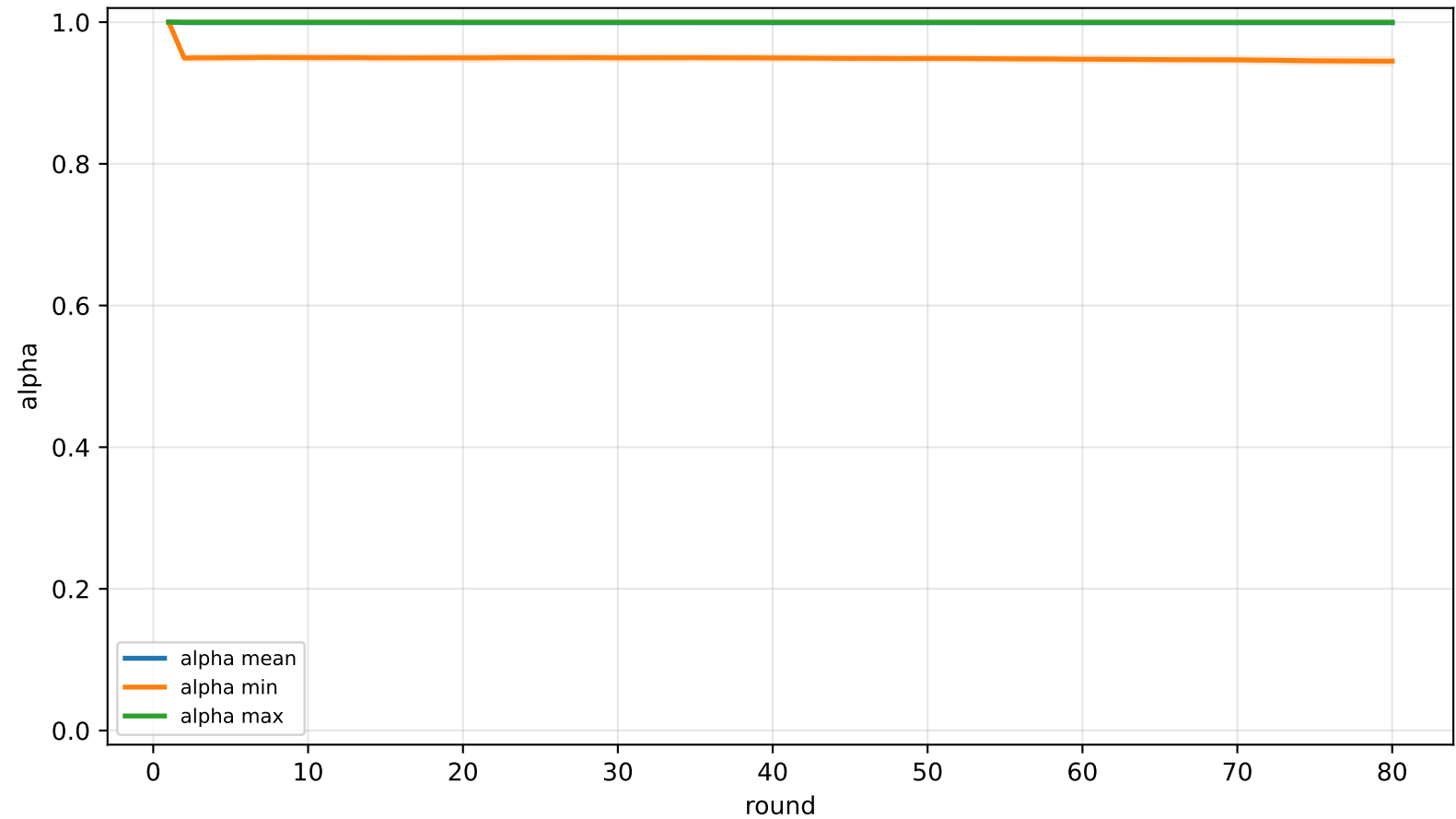
Delta diagnostics: baselines minus FedALA-HeadOnly



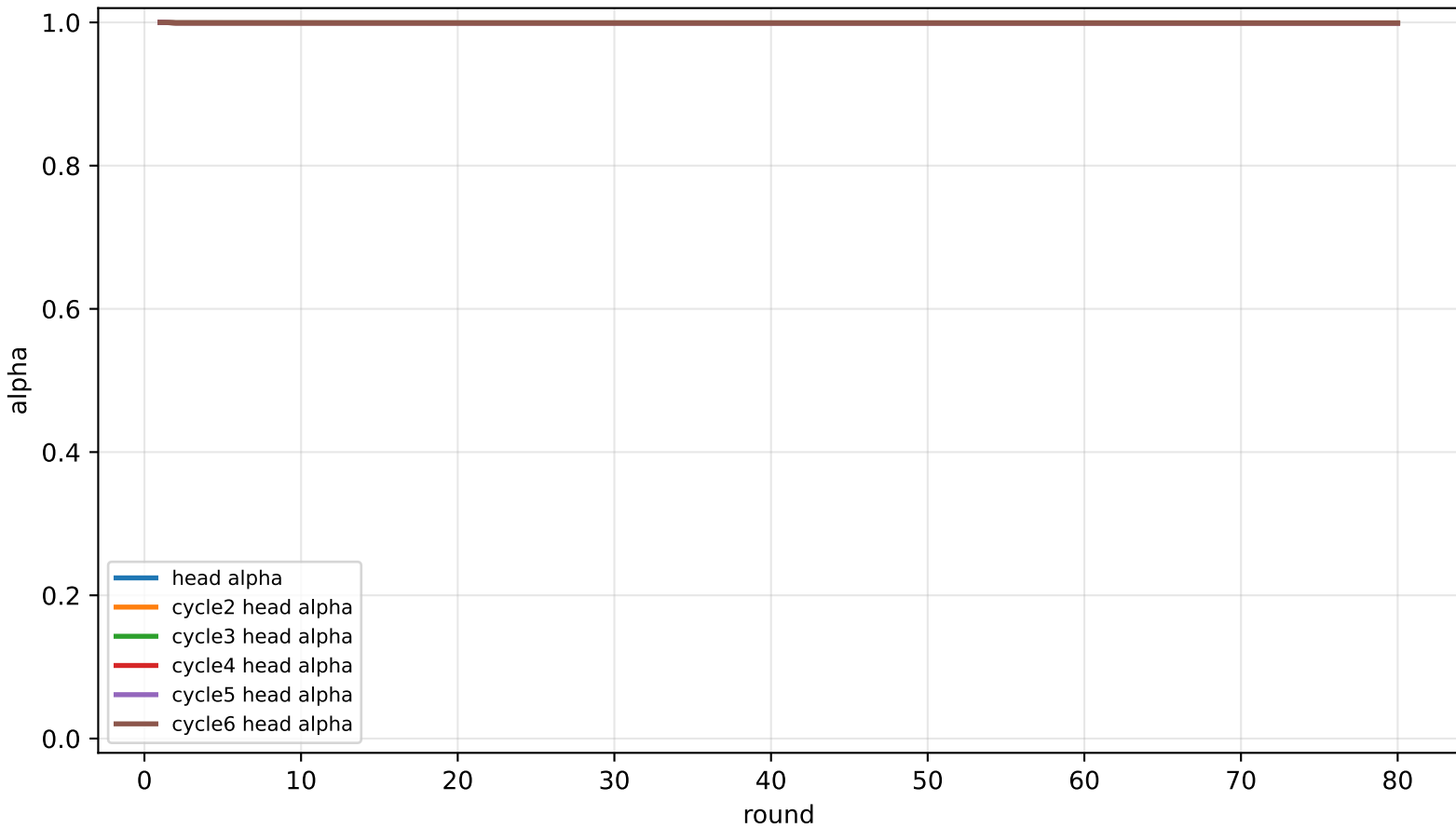
Controlled q-label heterogeneity: iid | FedALA-HeadOnly ALA diagnostics | aggregated / pooled | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

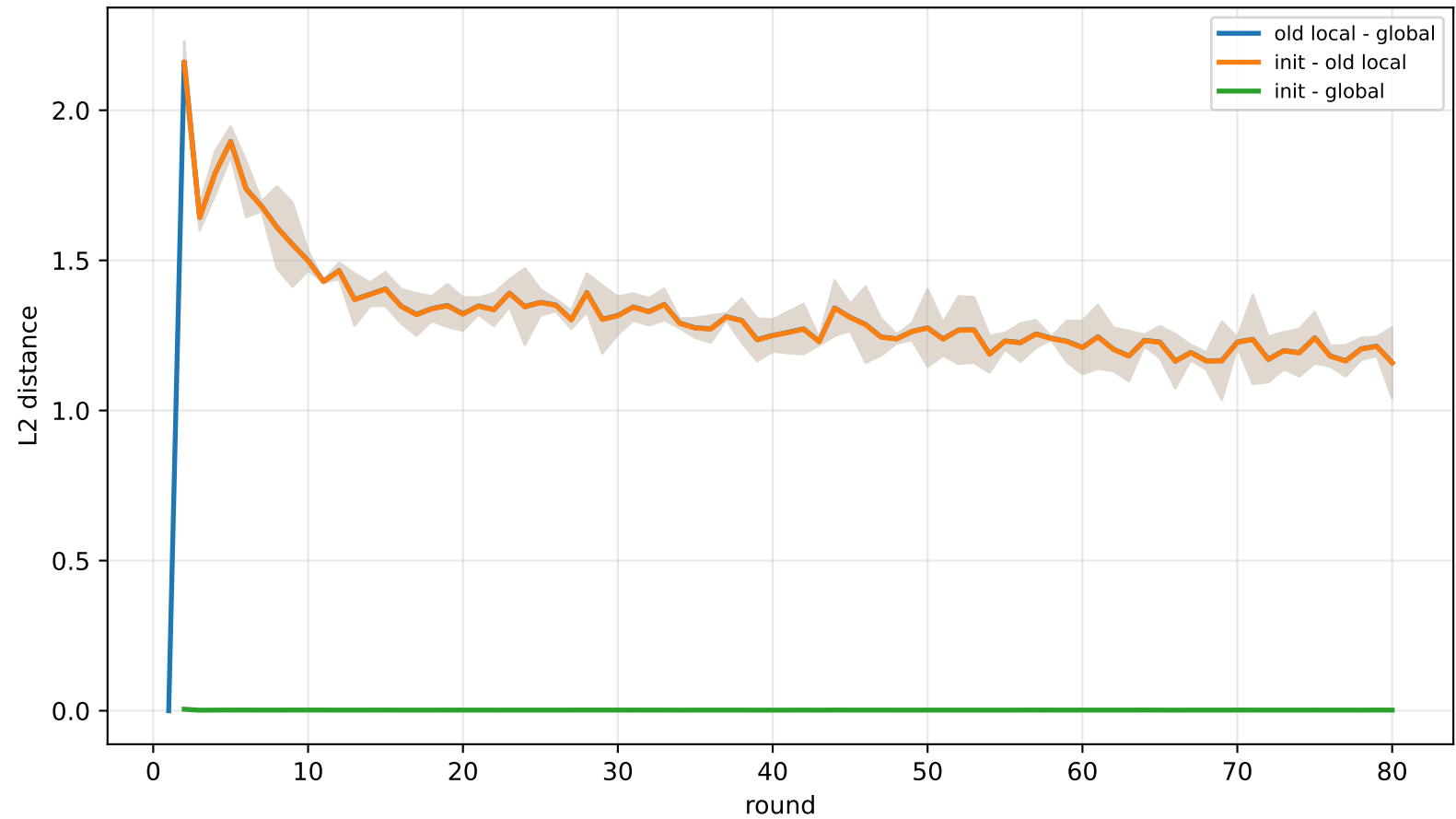
ALA interpolation weights



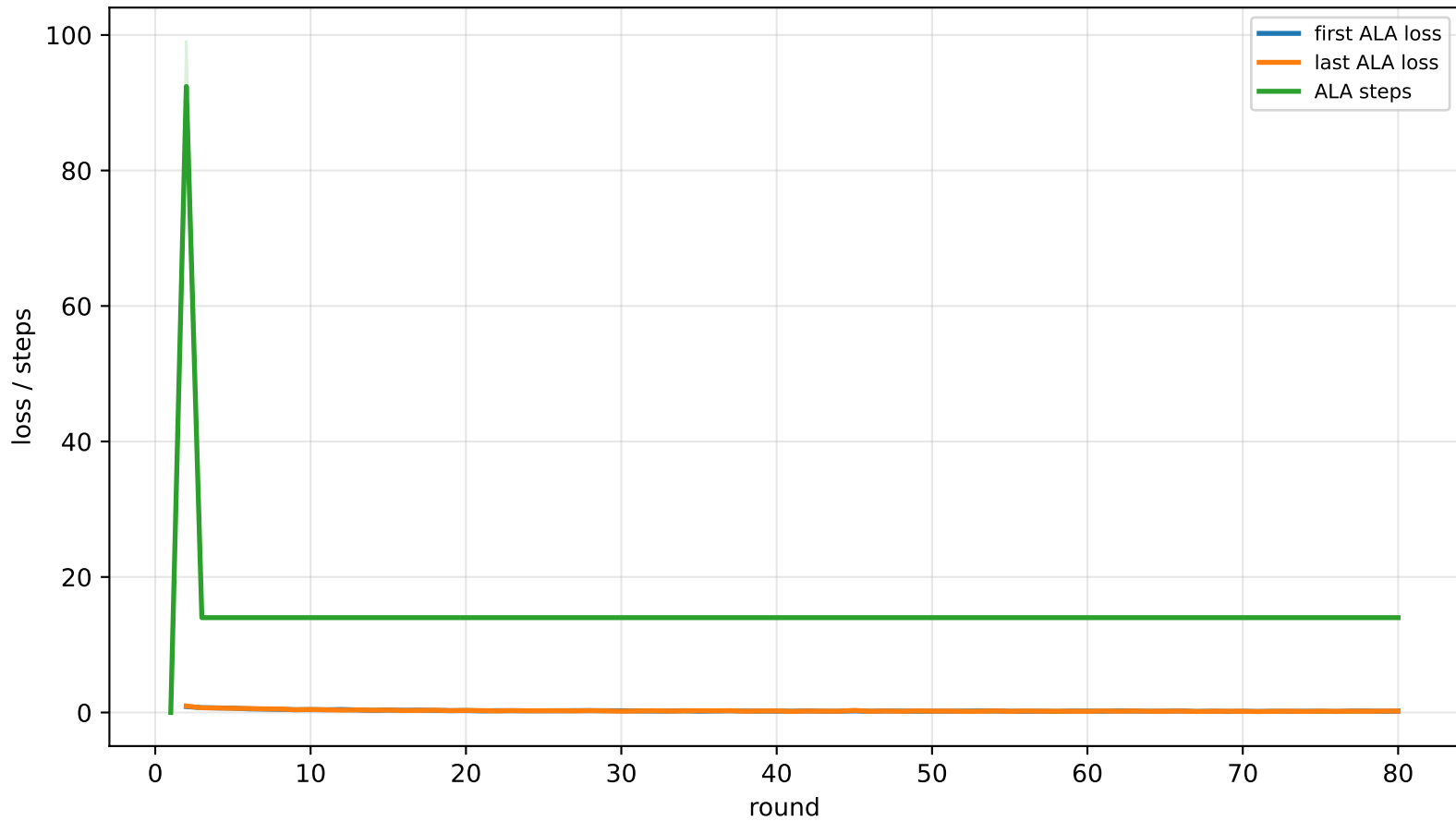
Backbone/head/task alpha diagnostics



Model distance diagnostics



ALA optimization diagnostics

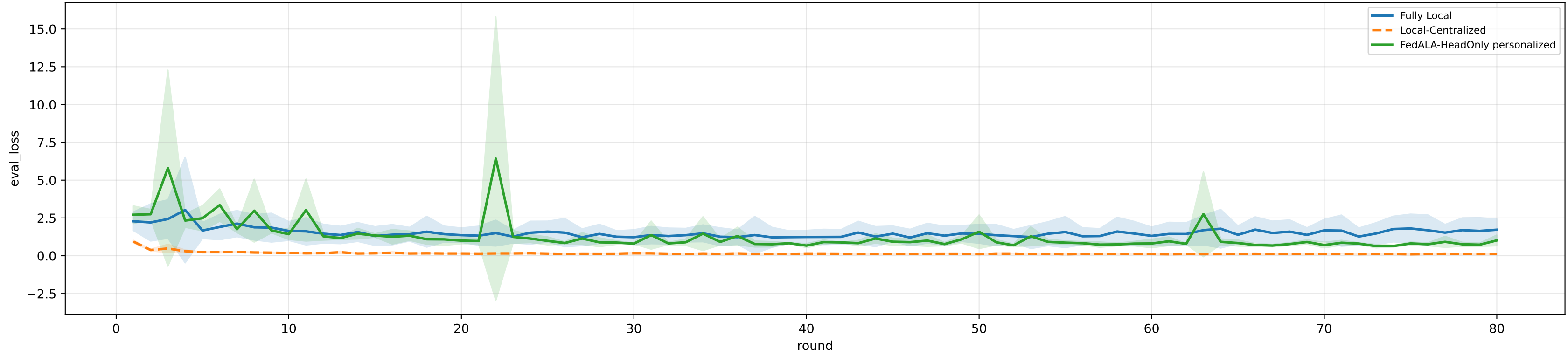


family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

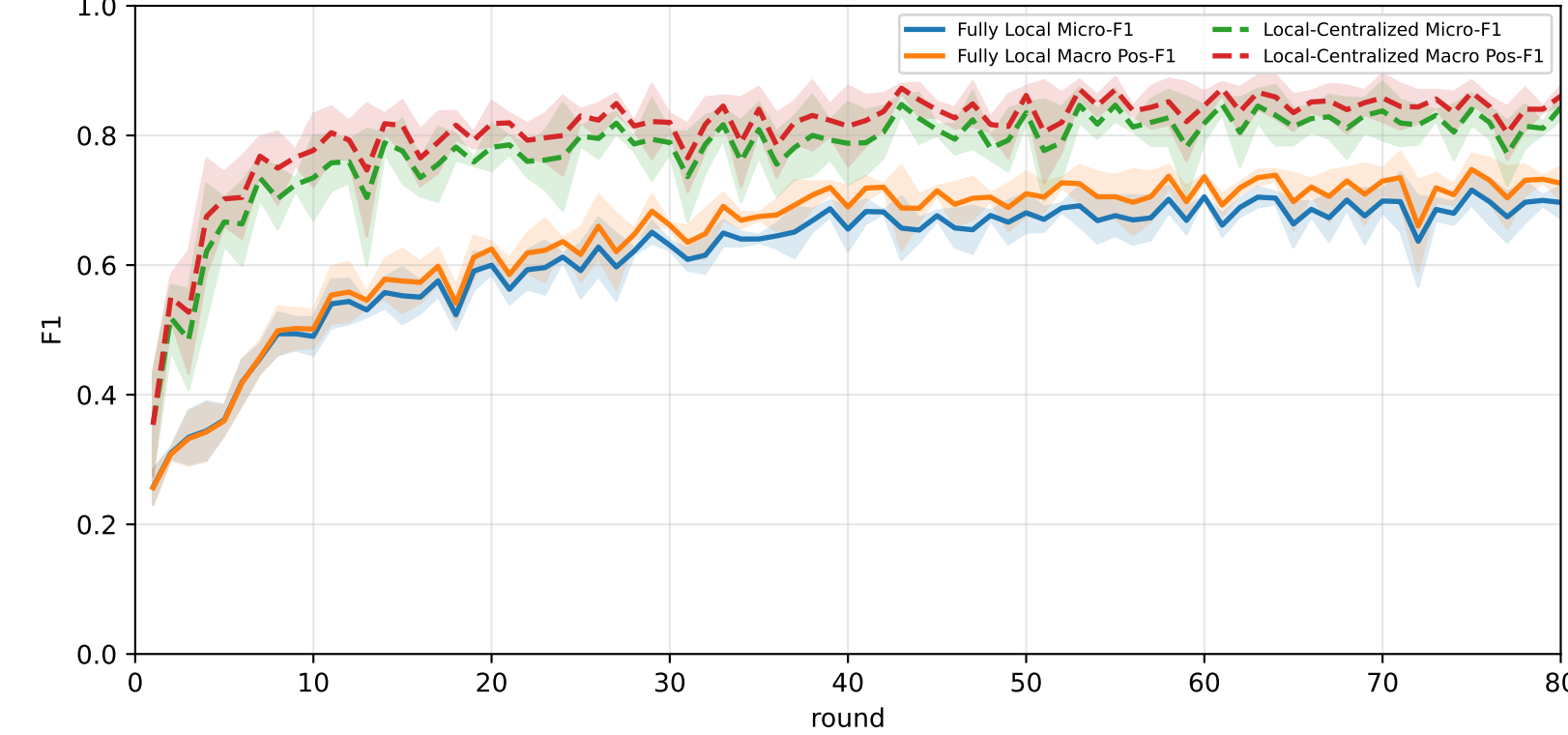
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.6% ± 4.0%	66.0% ± 4.3%	<u>81.6% ± 7.0%</u>	<u>78.2% ± 5.1%</u>	64.0% ± 3.6%	58.6% ± 5.2%	44.4% ± 3.0%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>67.0% ± 2.9%</u>	<u>68.6% ± 2.9%</u>	75.8% ± 4.6%	75.8% ± 3.3%	<u>66.3% ± 3.5%</u>	<u>65.9% ± 4.3%</u>	<u>57.0% ± 2.2%</u>

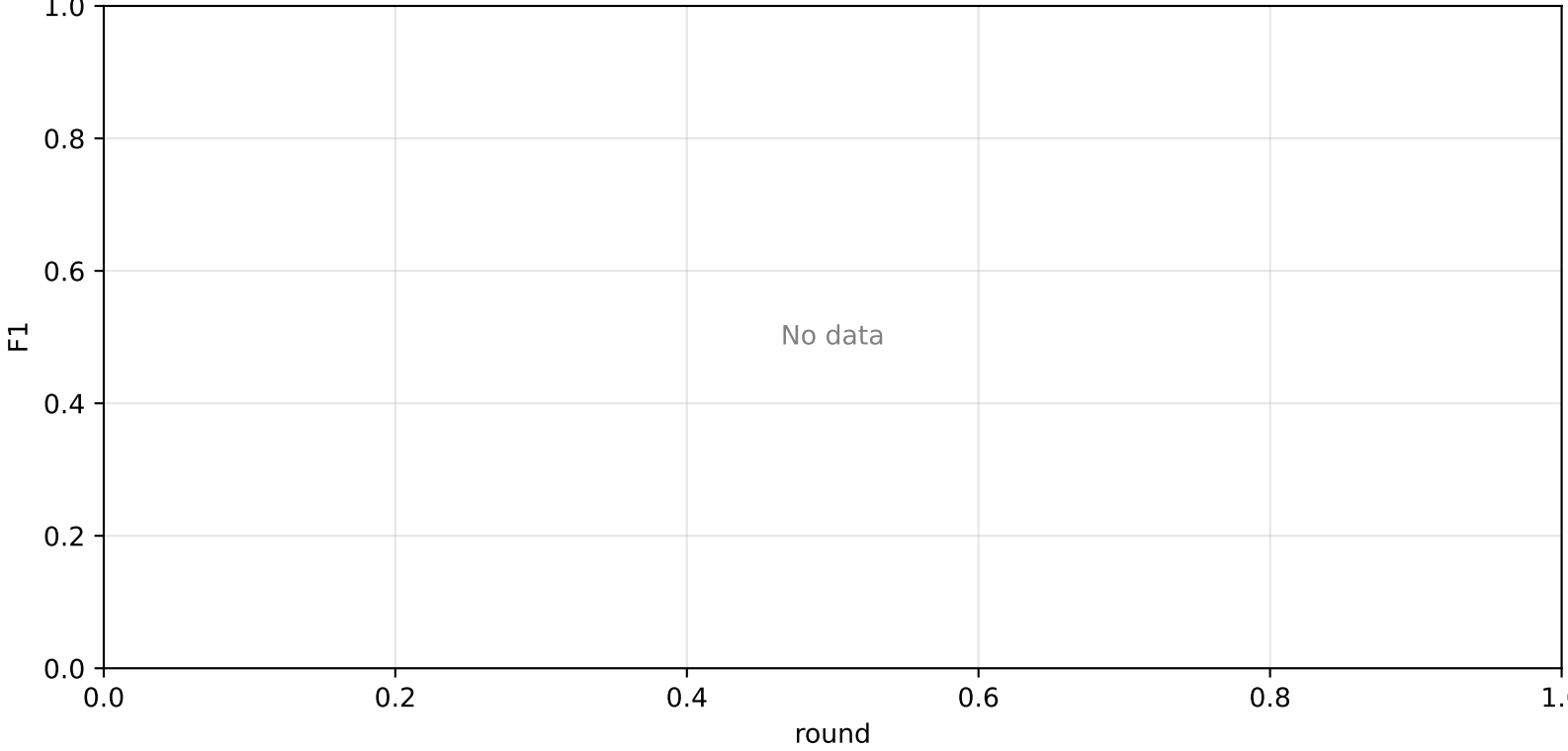
Validation loss comparison



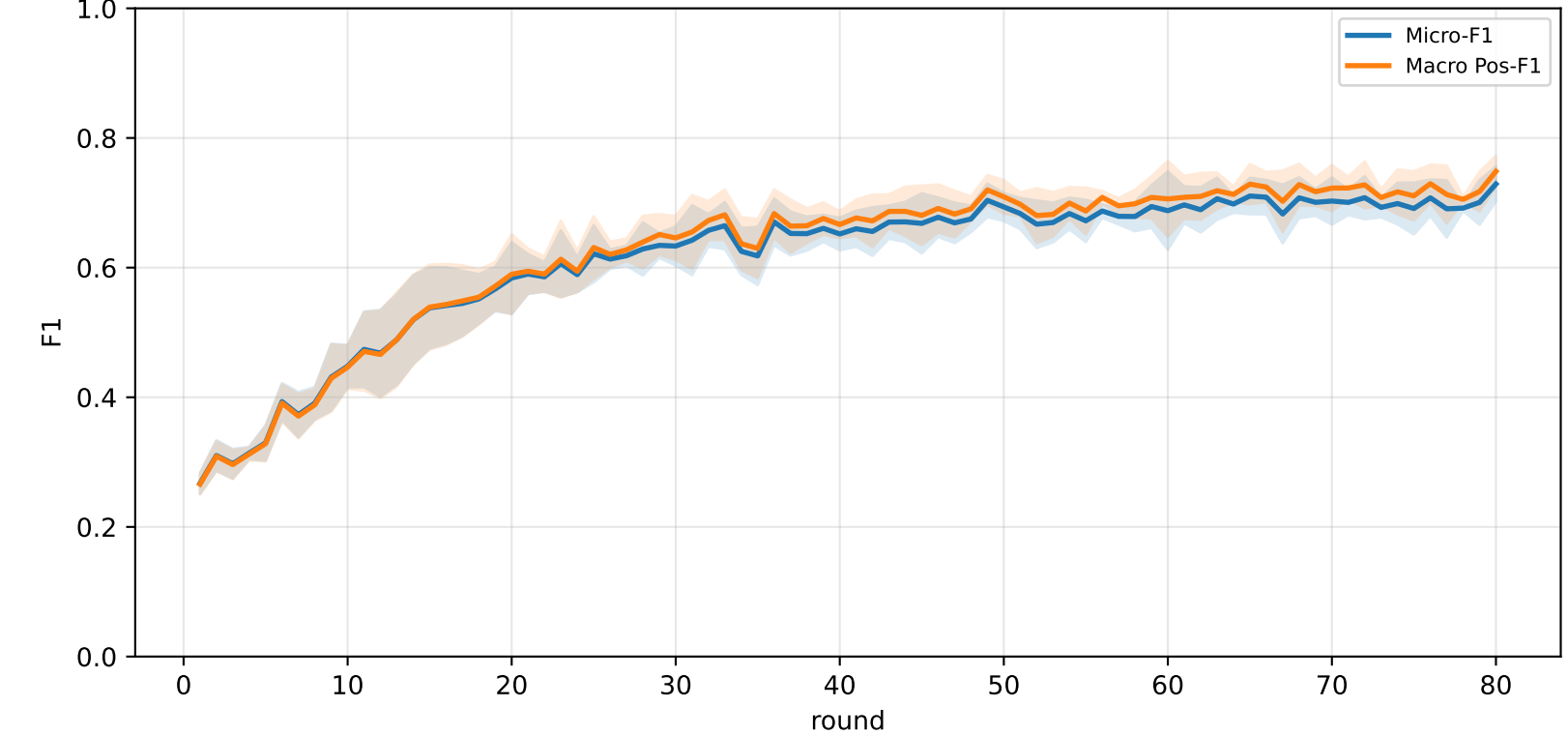
Pooled baseline micro / macro F1



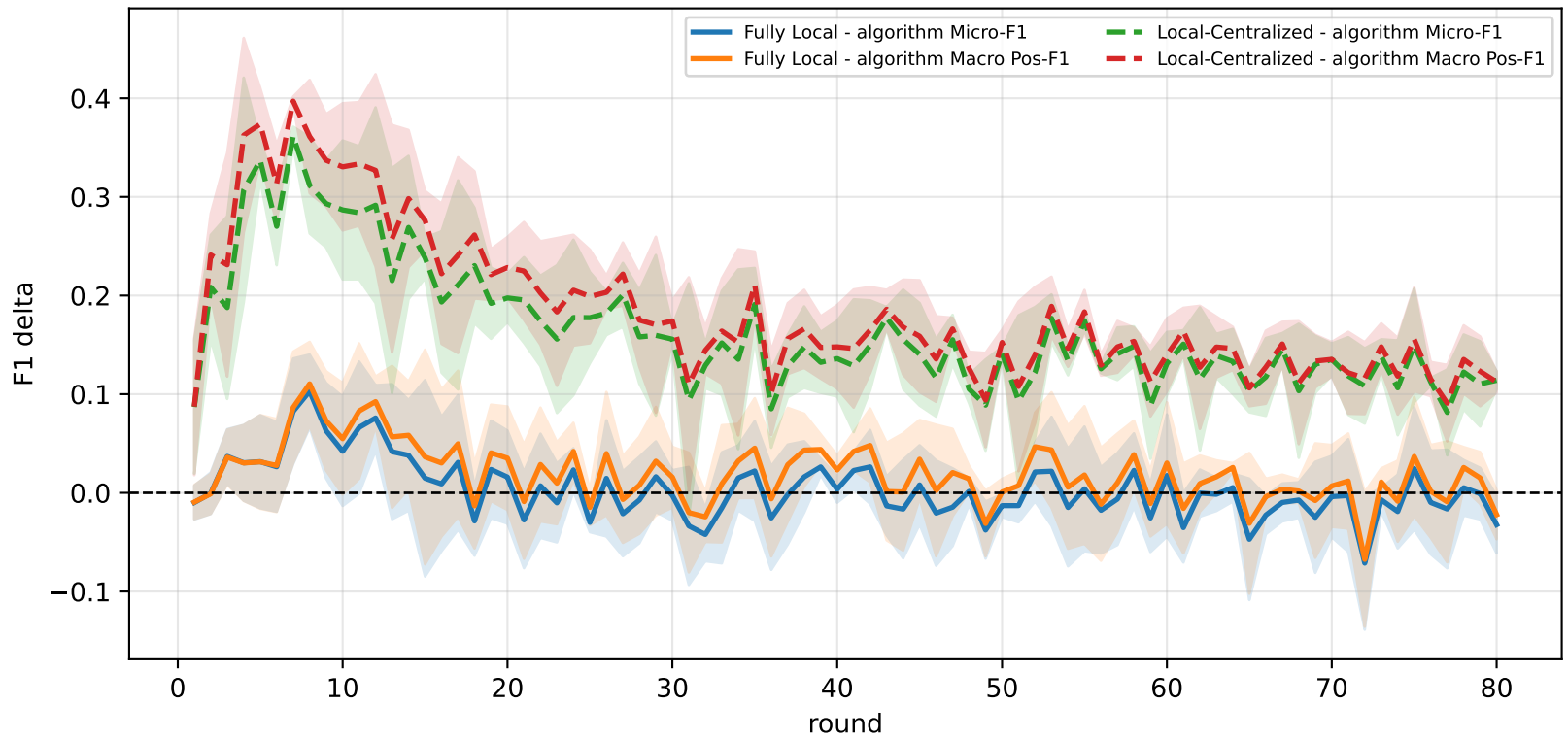
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

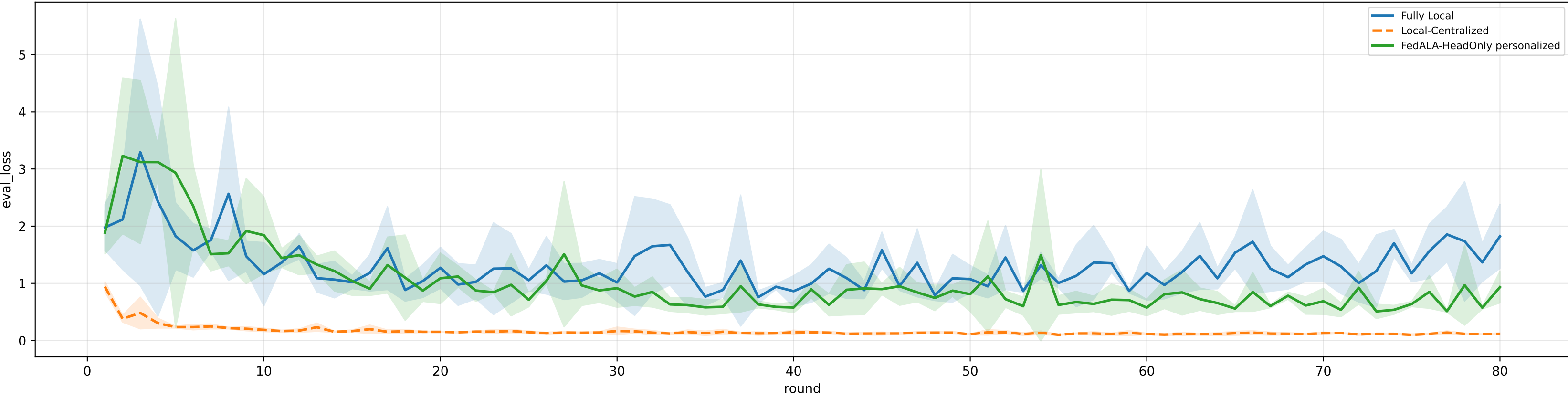


family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

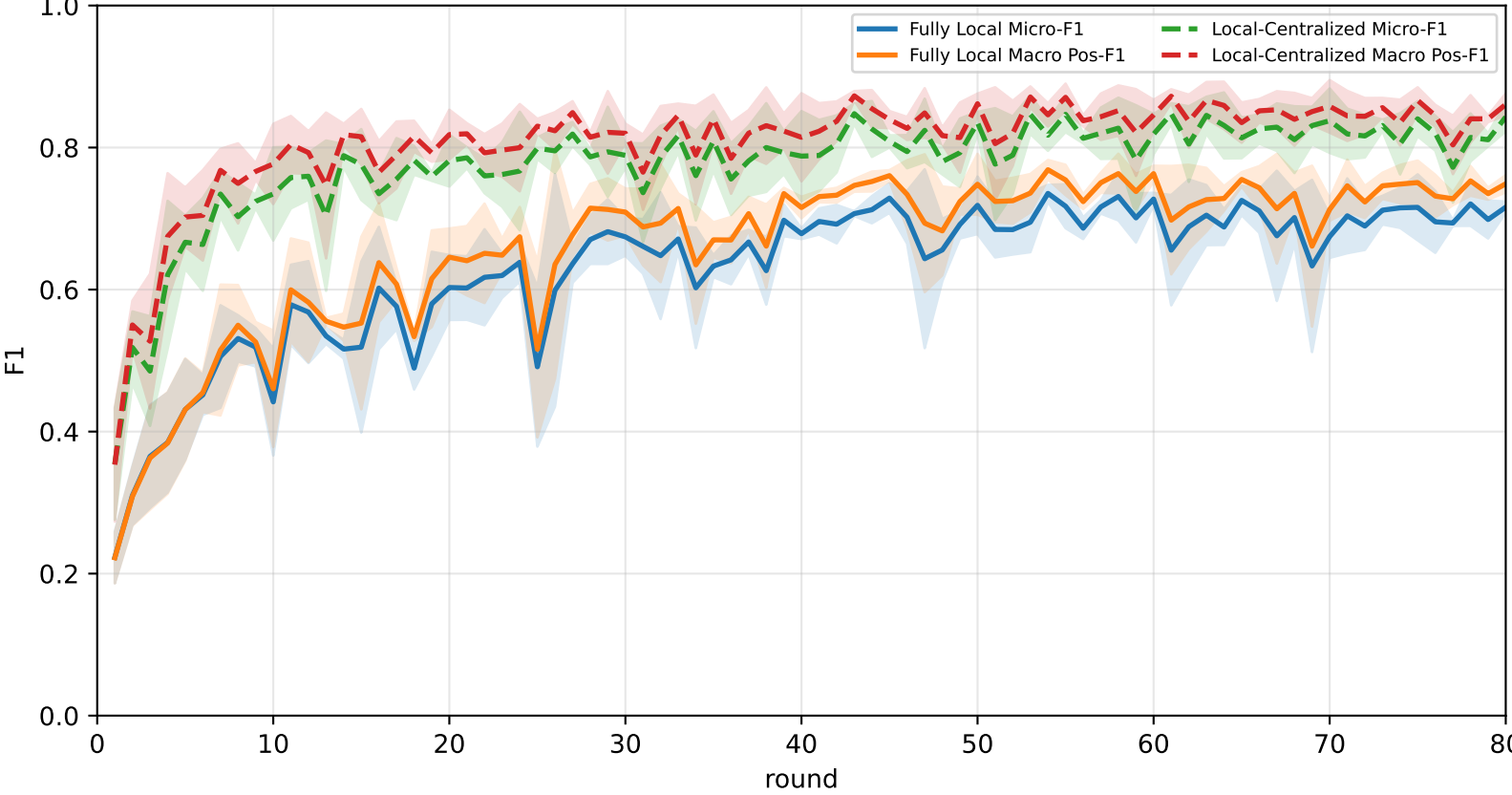
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	63.8% ± 5.4%	68.0% ± 5.8%	<u>85.6% ± 8.0%</u>	<u>80.6% ± 4.3%</u>	<u>65.4% ± 6.2%</u>	59.3% ± 9.8%	49.2% ± 2.7%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>66.2% ± 2.6%</u>	<u>68.2% ± 2.1%</u>	79.1% ± 4.0%	74.7% ± 1.5%	64.9% ± 1.9%	<u>66.1% ± 6.6%</u>	<u>56.0% ± 2.6%</u>

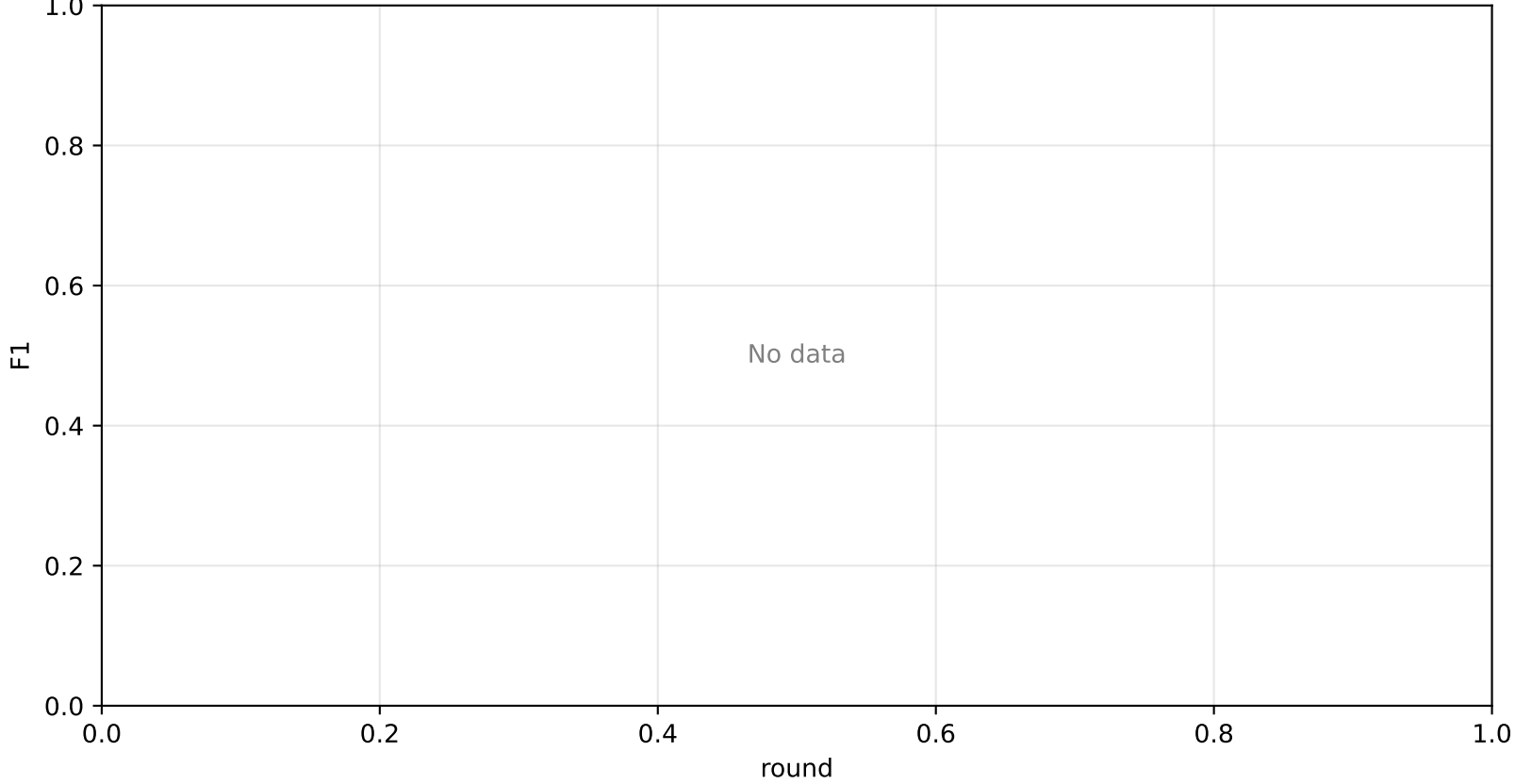
Validation loss comparison



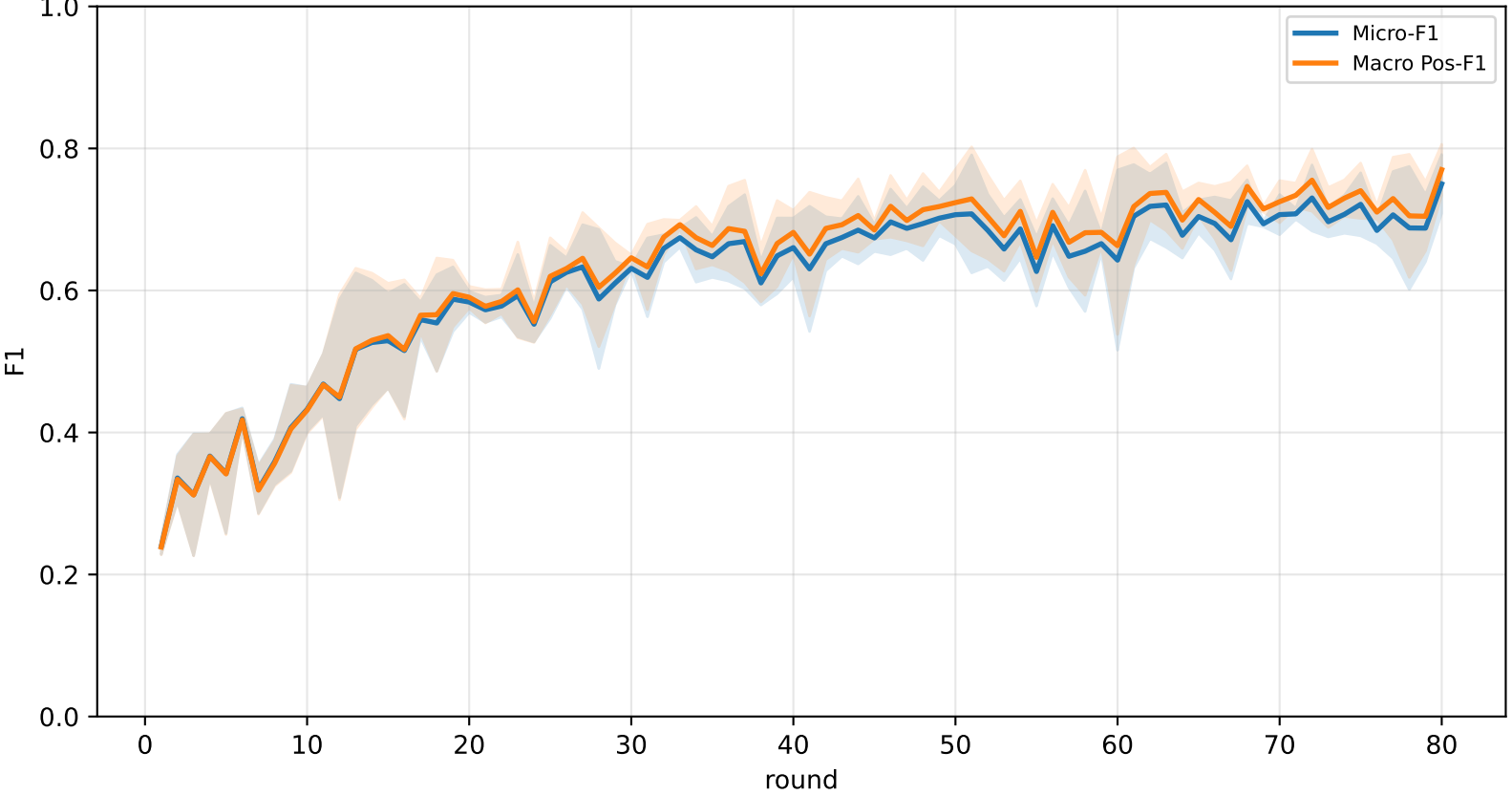
Pooled baseline micro / macro F1



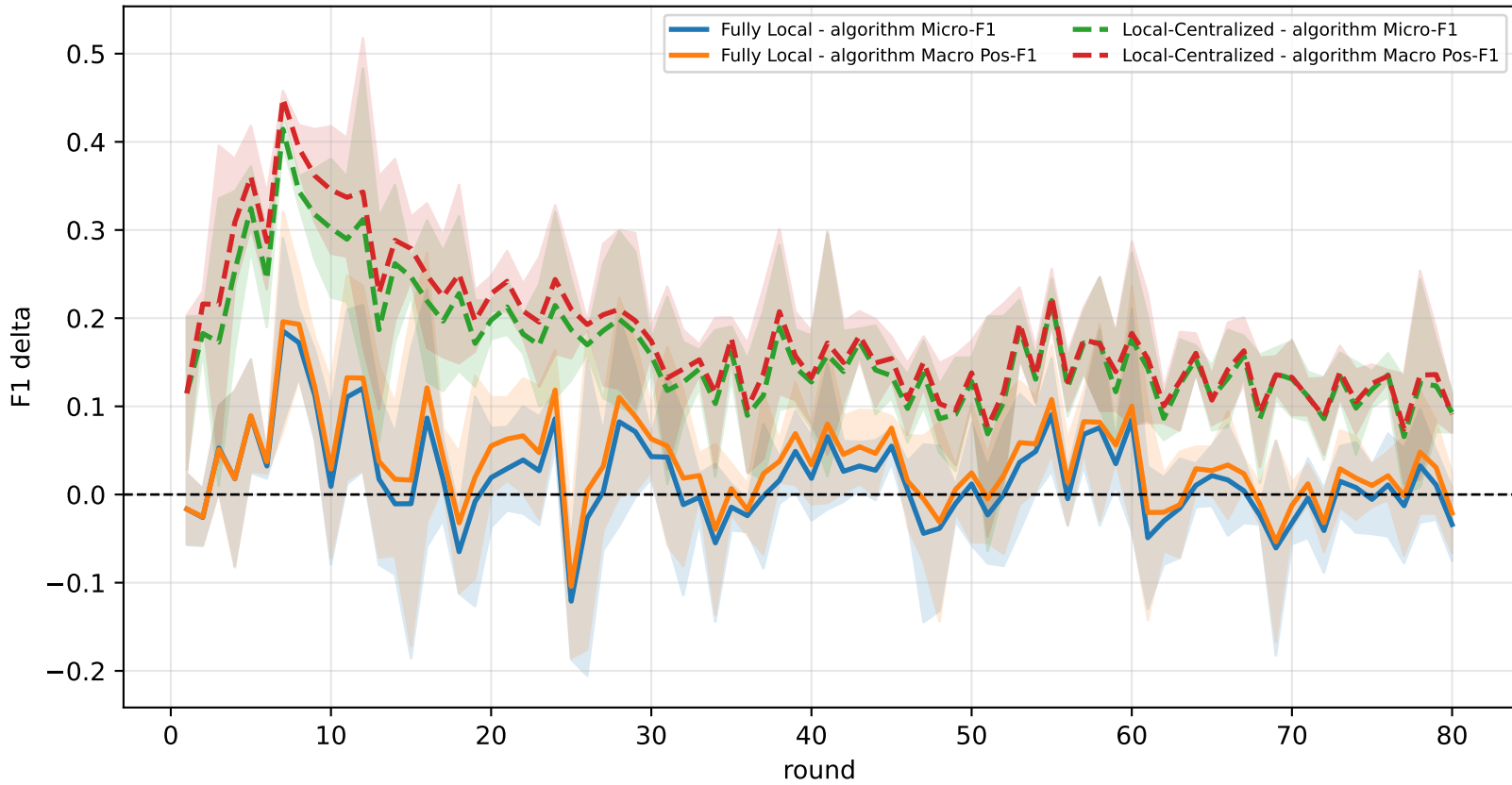
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

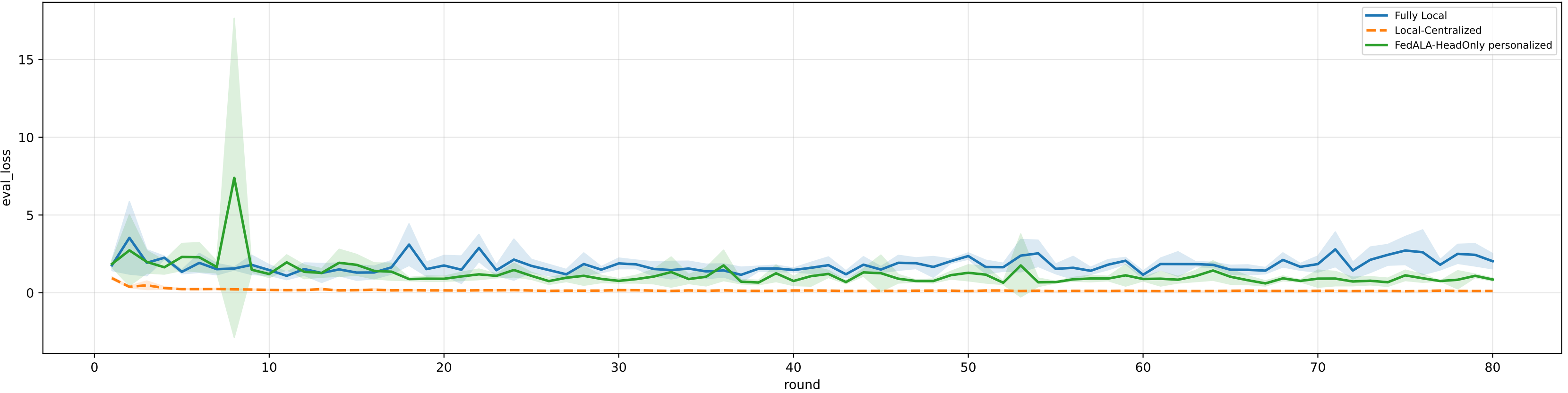


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

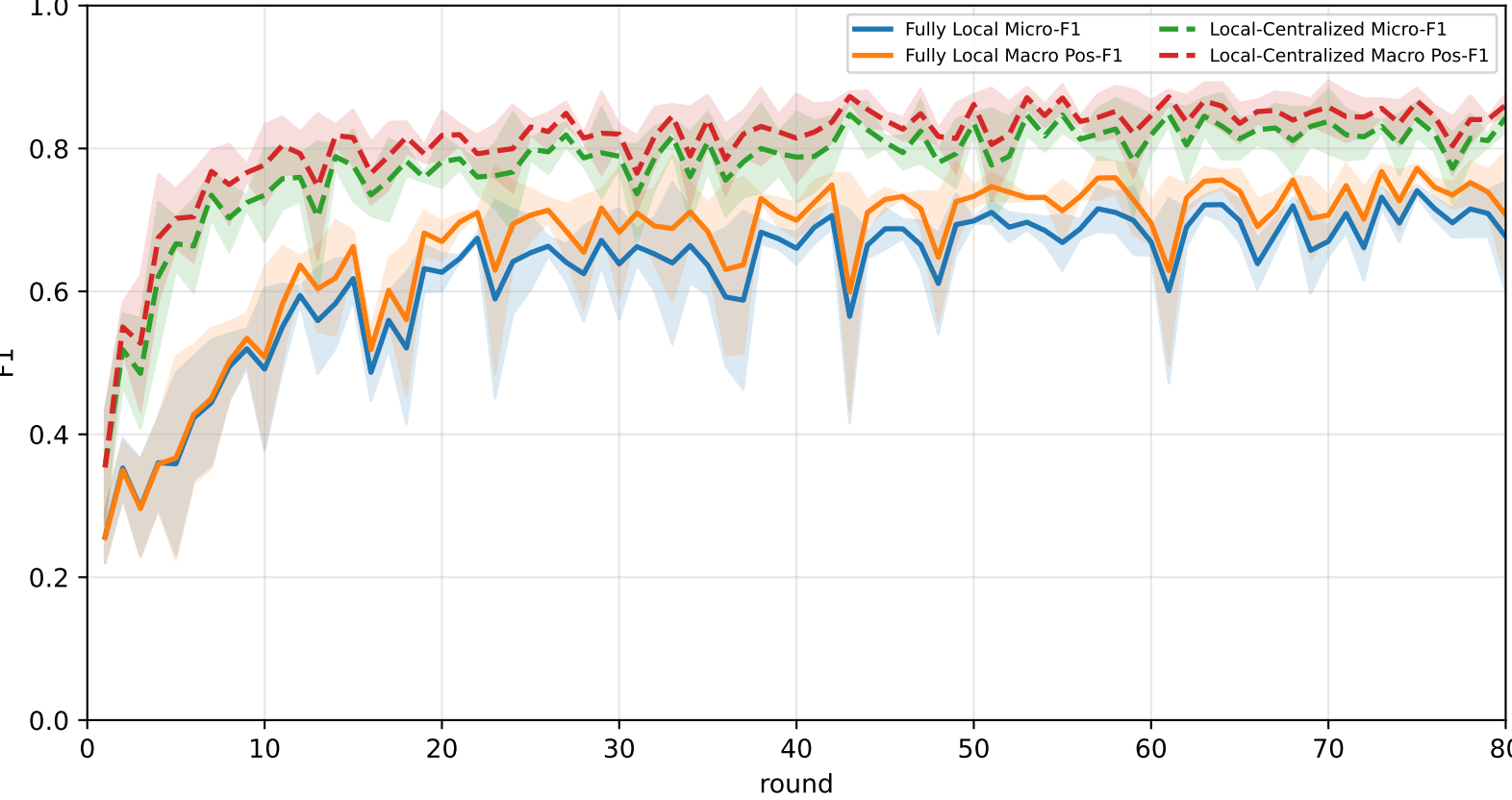
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	60.1% ± 11.6%	<u>64.2% ± 10.9%</u>	<u>75.3% ± 14.5%</u>	<u>77.9% ± 12.0%</u>	<u>63.8% ± 6.1%</u>	<u>60.7% ± 11.5%</u>	43.1% ± 11.1%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>62.3% ± 8.1%</u>	63.7% ± 8.8%	73.9% ± 11.7%	67.1% ± 11.8%	63.0% ± 7.9%	60.0% ± 8.7%	<u>54.6% ± 4.5%</u>

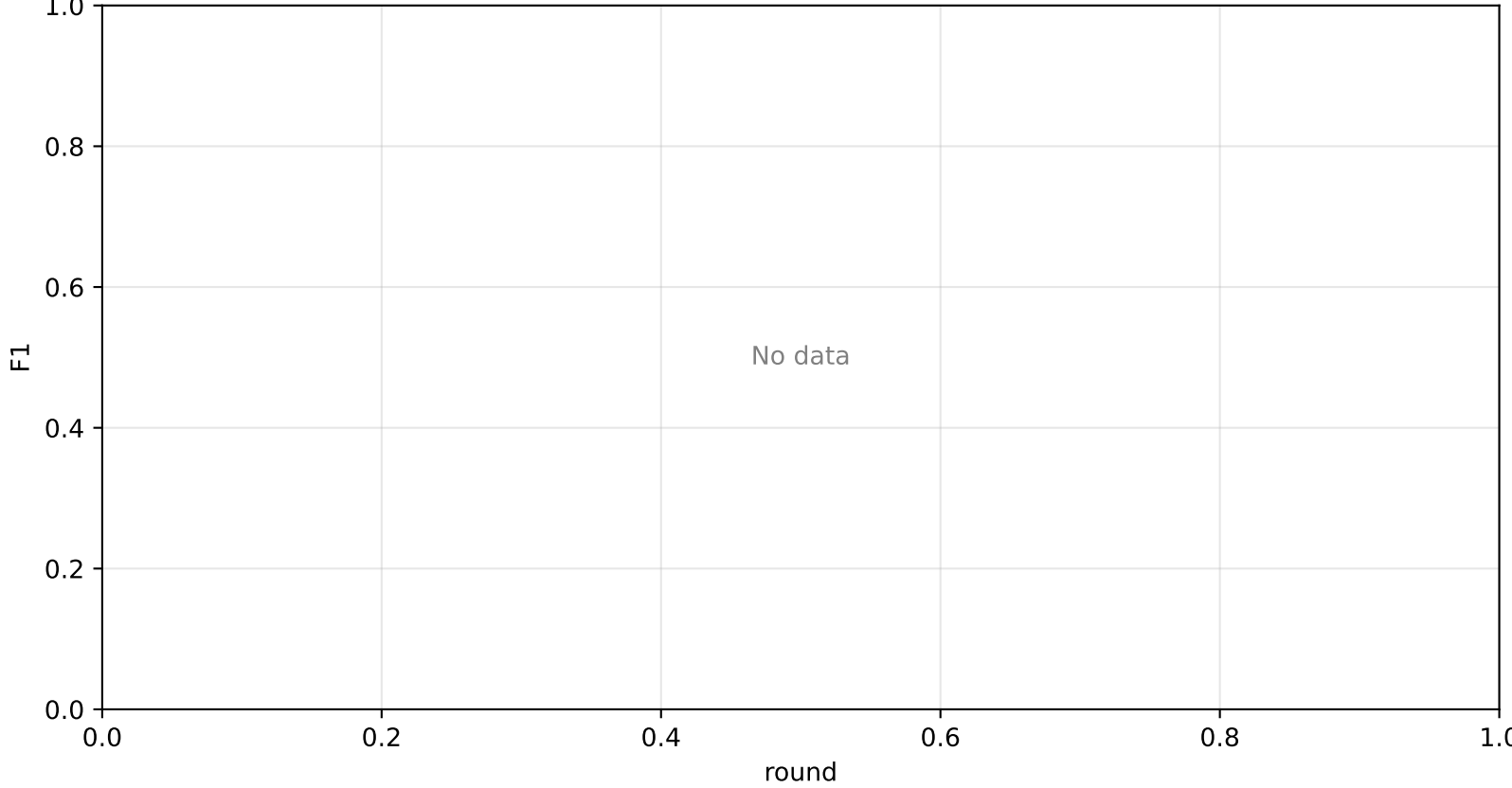
Validation loss comparison



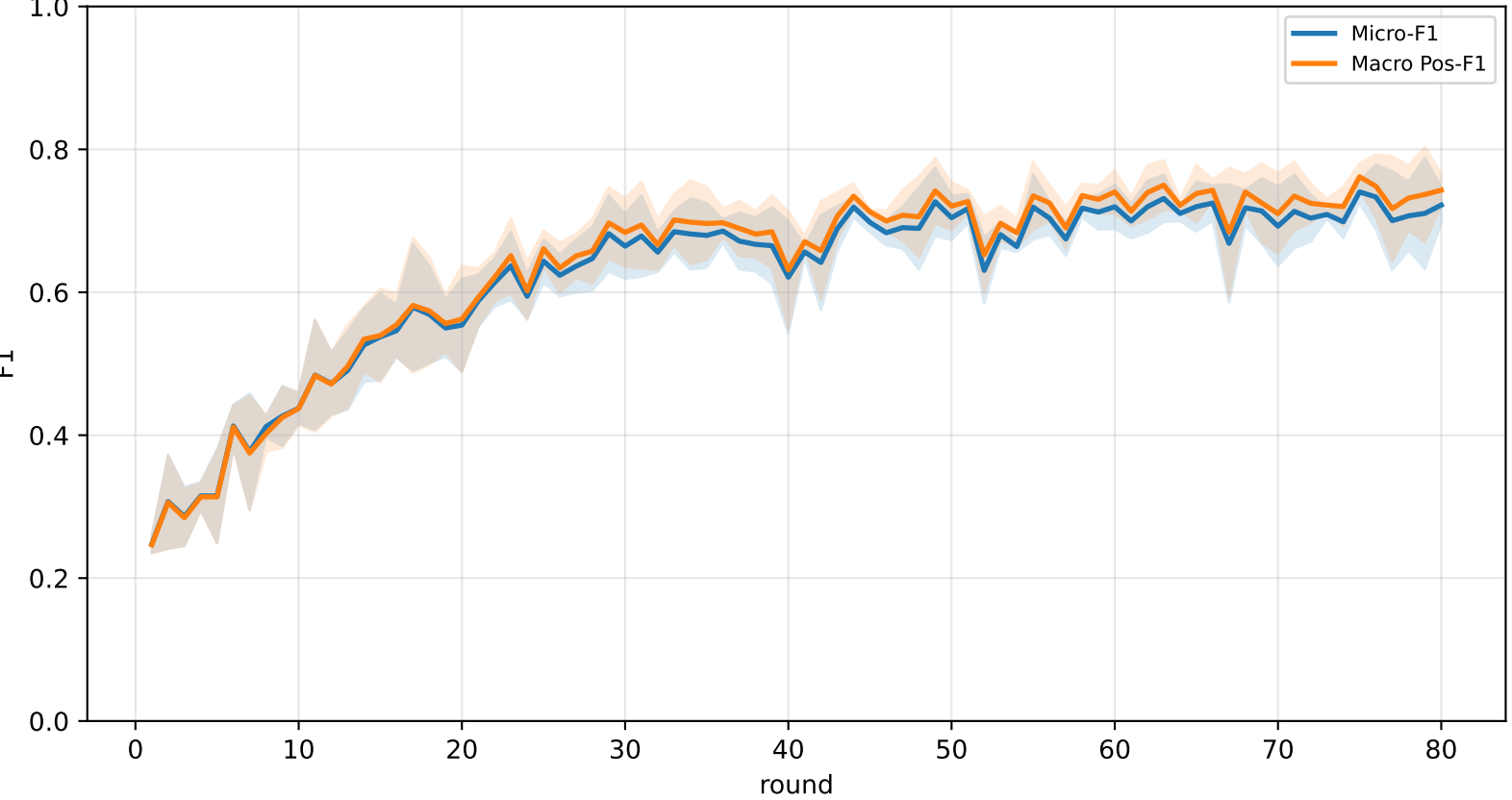
Pooled baseline micro / macro F1



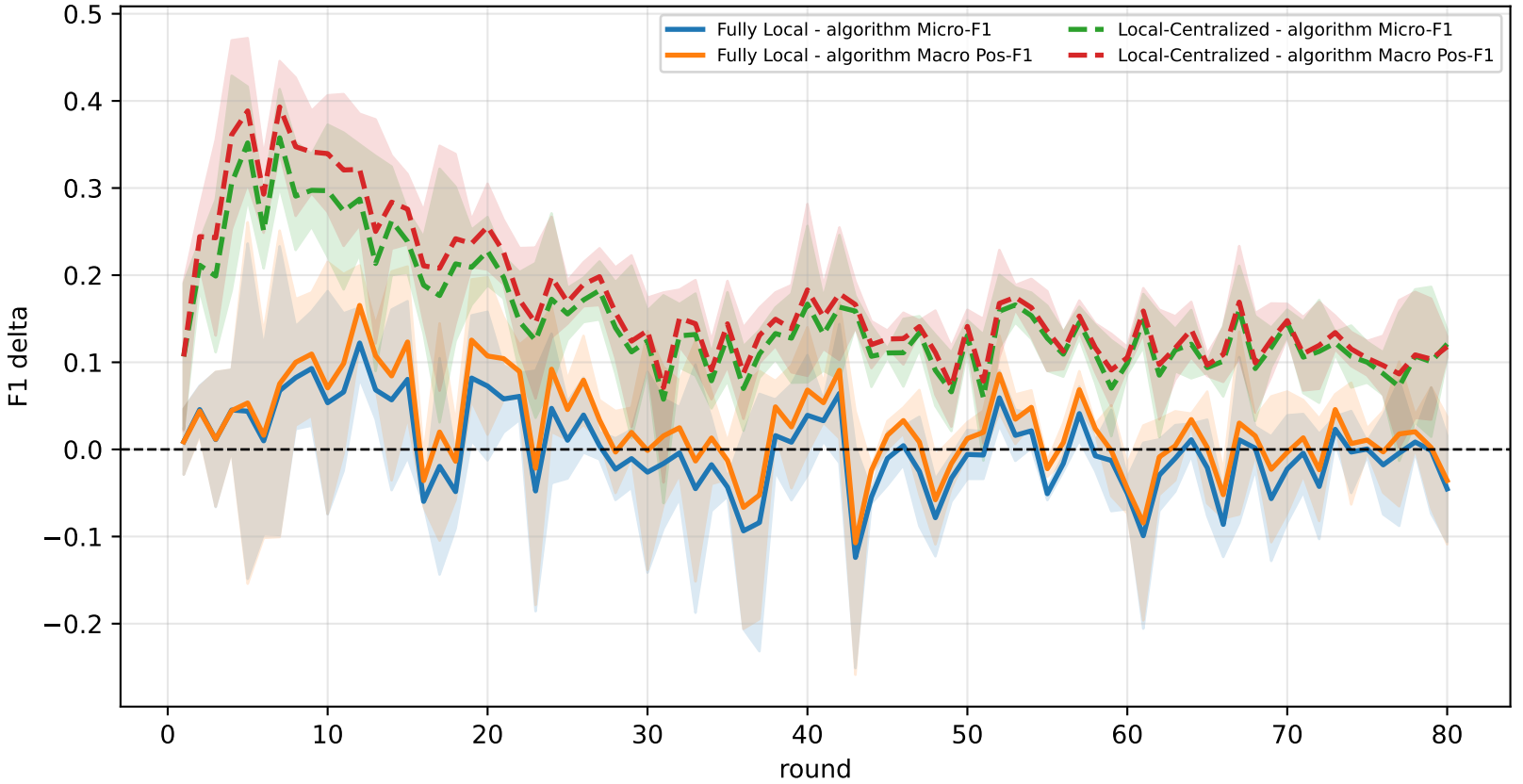
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

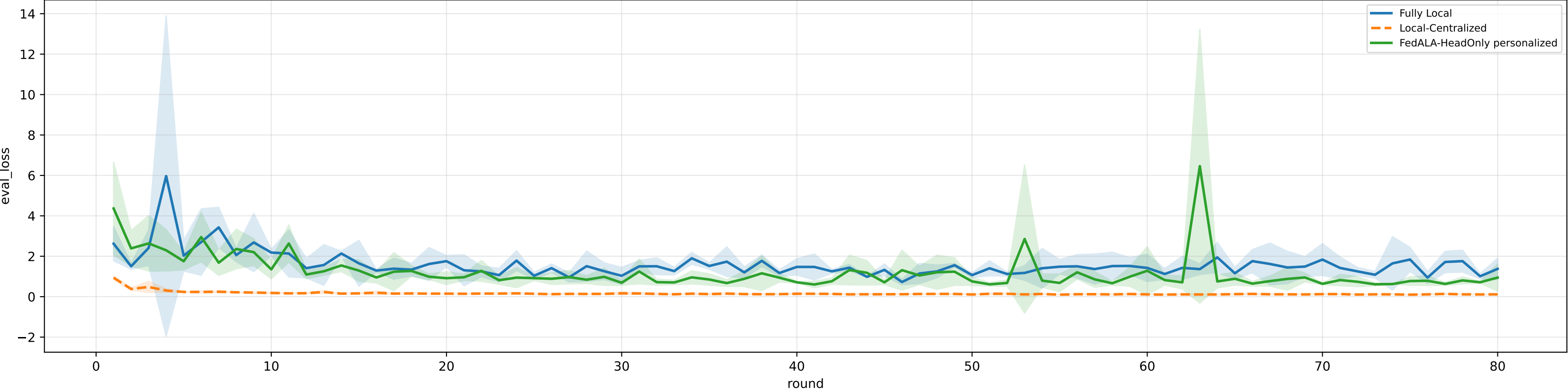


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

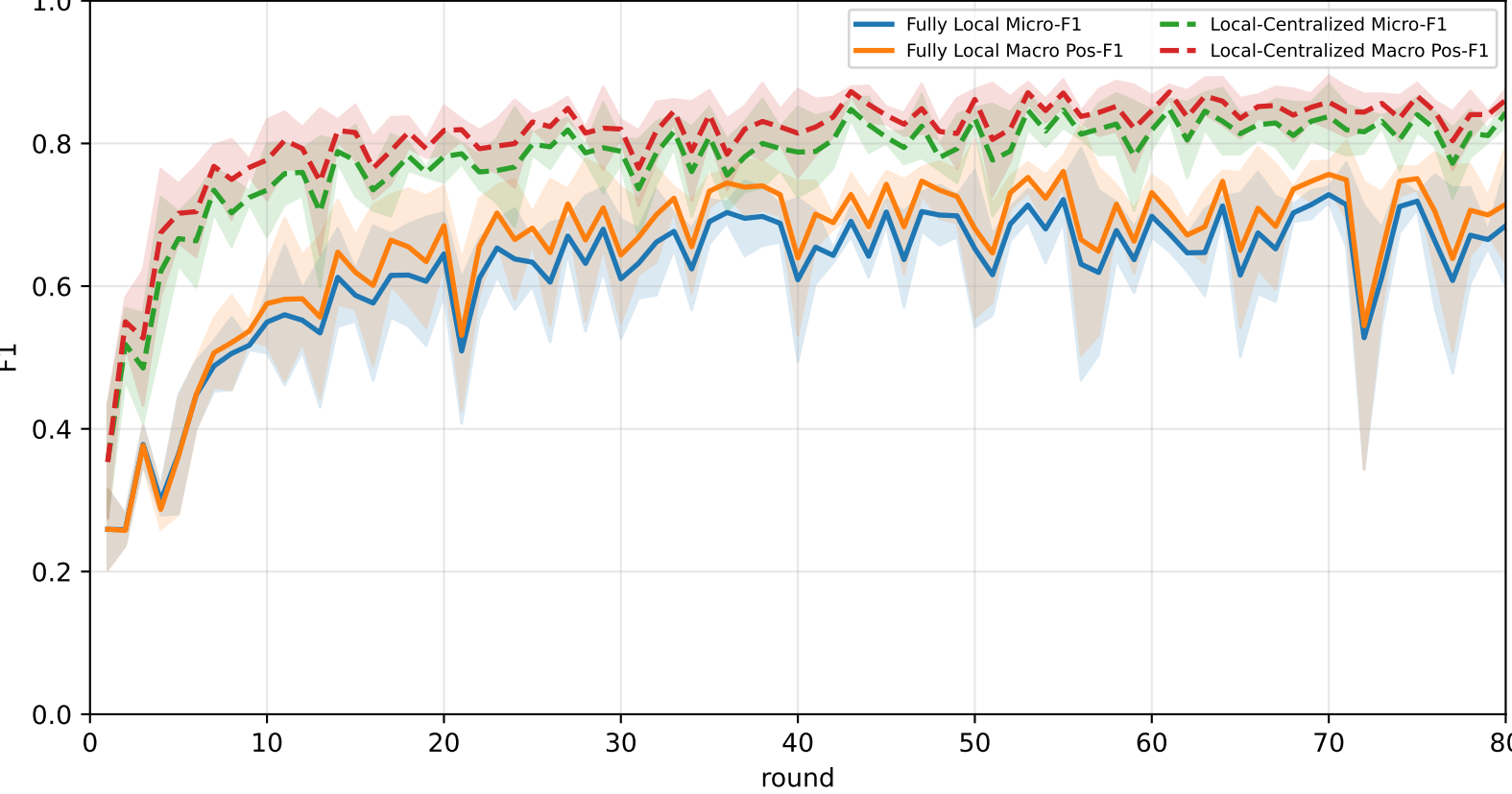
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	58.7% ± 5.5%	63.8% ± 5.1%	<u>82.7% ± 2.7%</u>	76.5% ± 5.8%	63.3% ± 5.5%	54.2% ± 7.8%	42.1% ± 3.8%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>68.0% ± 3.4%</u>	<u>69.6% ± 3.3%</u>	74.8% ± 7.1%	<u>78.7% ± 6.9%</u>	<u>68.7% ± 3.9%</u>	<u>67.7% ± 2.1%</u>	<u>58.0% ± 4.4%</u>

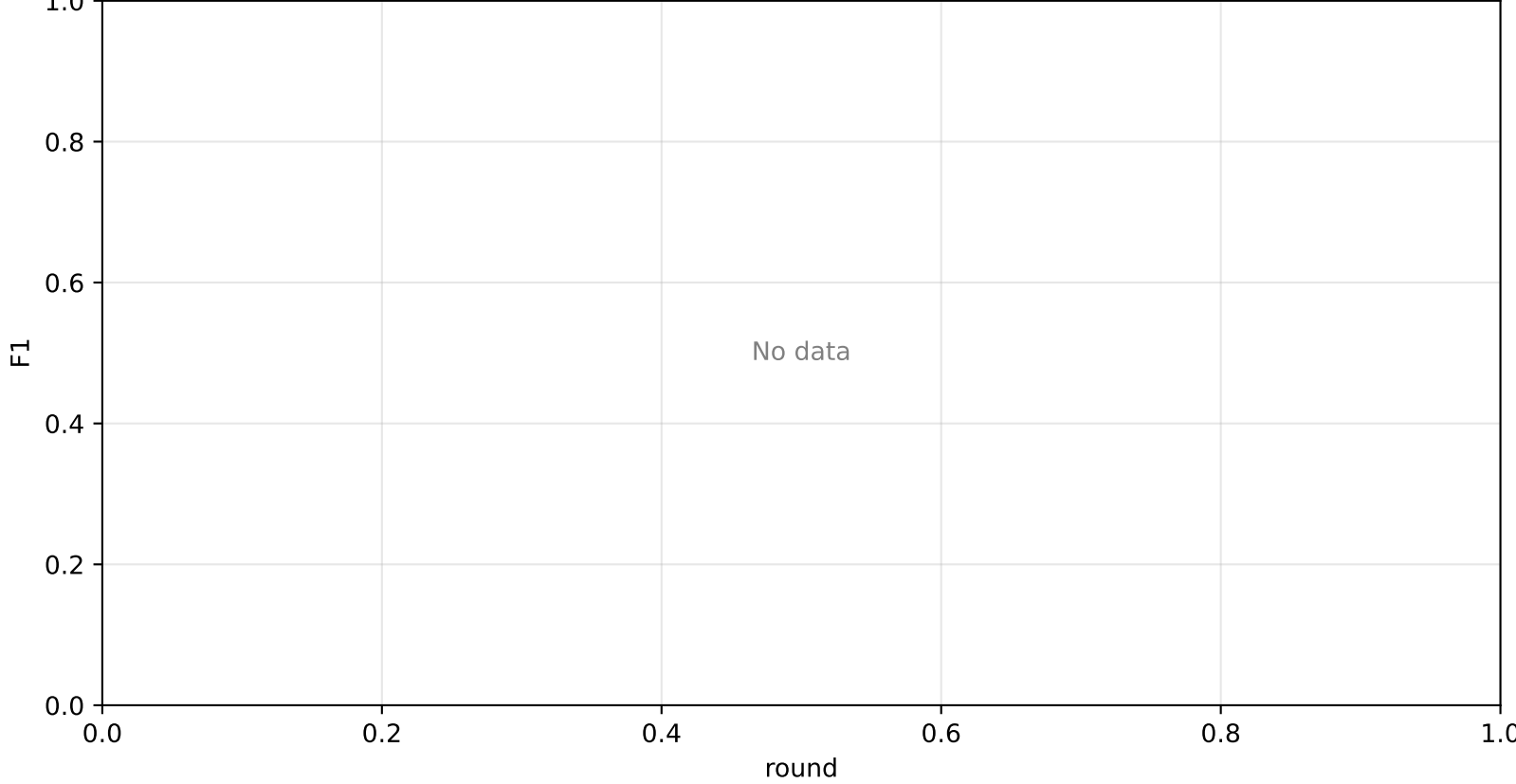
Validation loss comparison



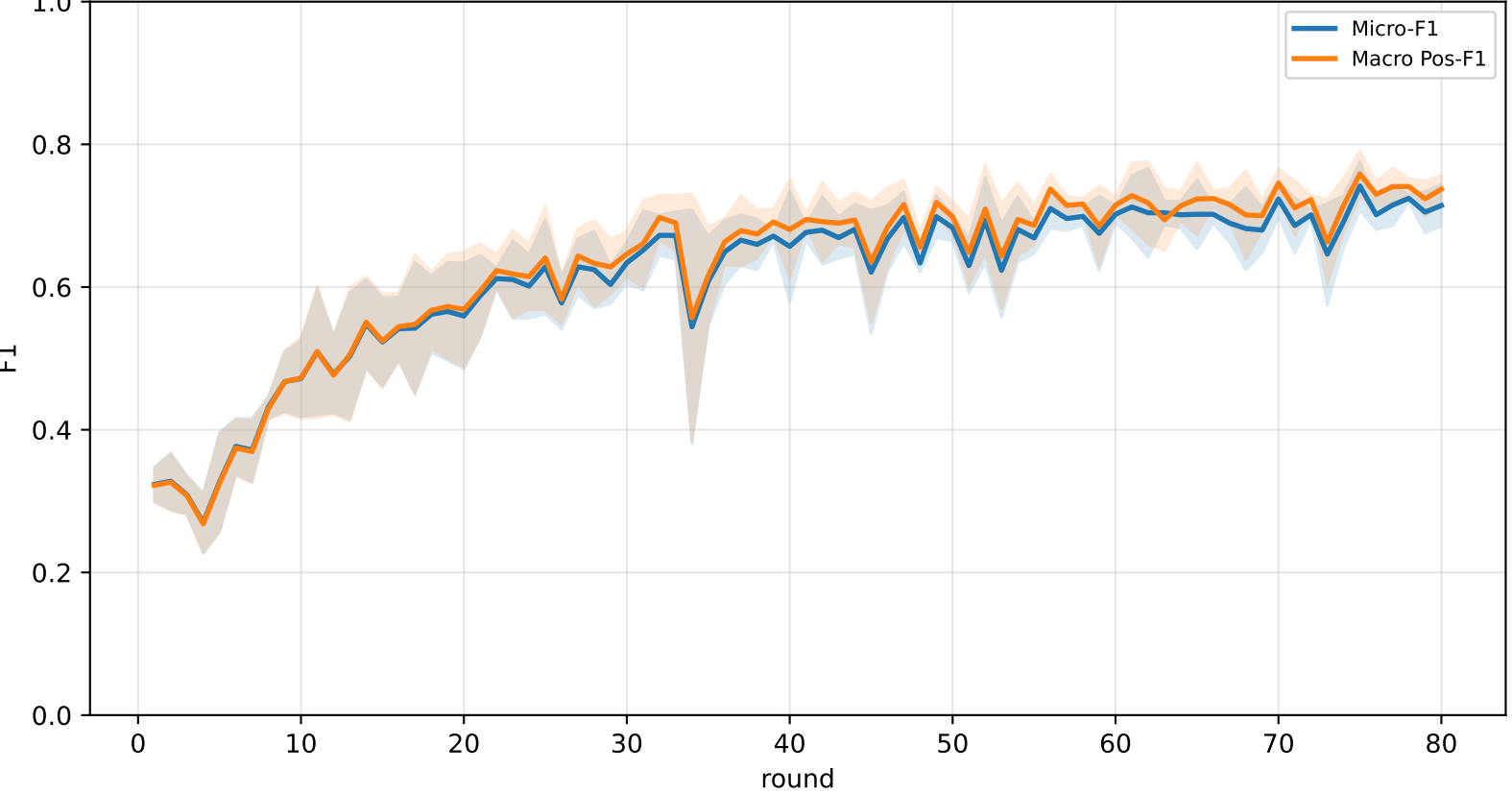
Pooled baseline micro / macro F1



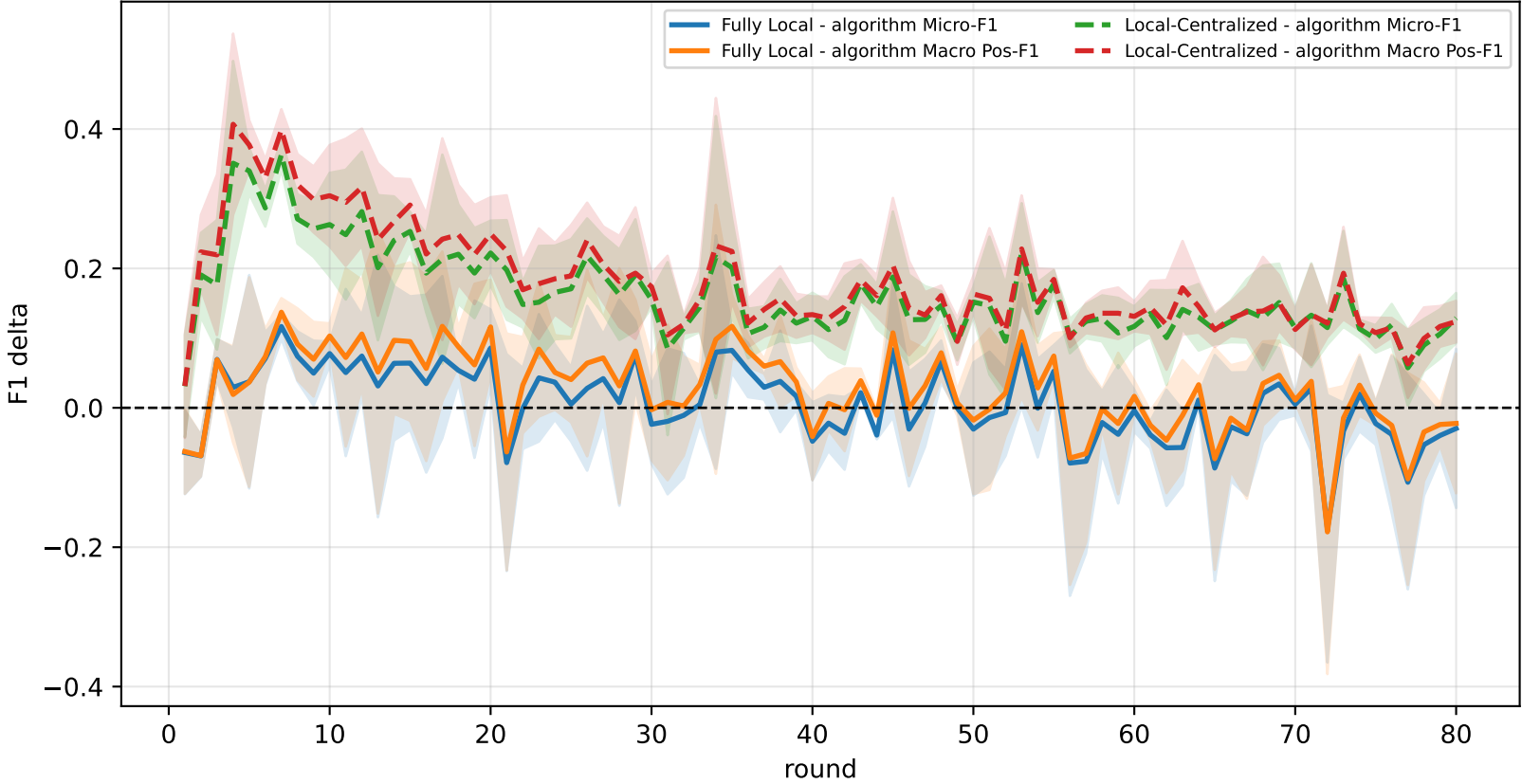
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



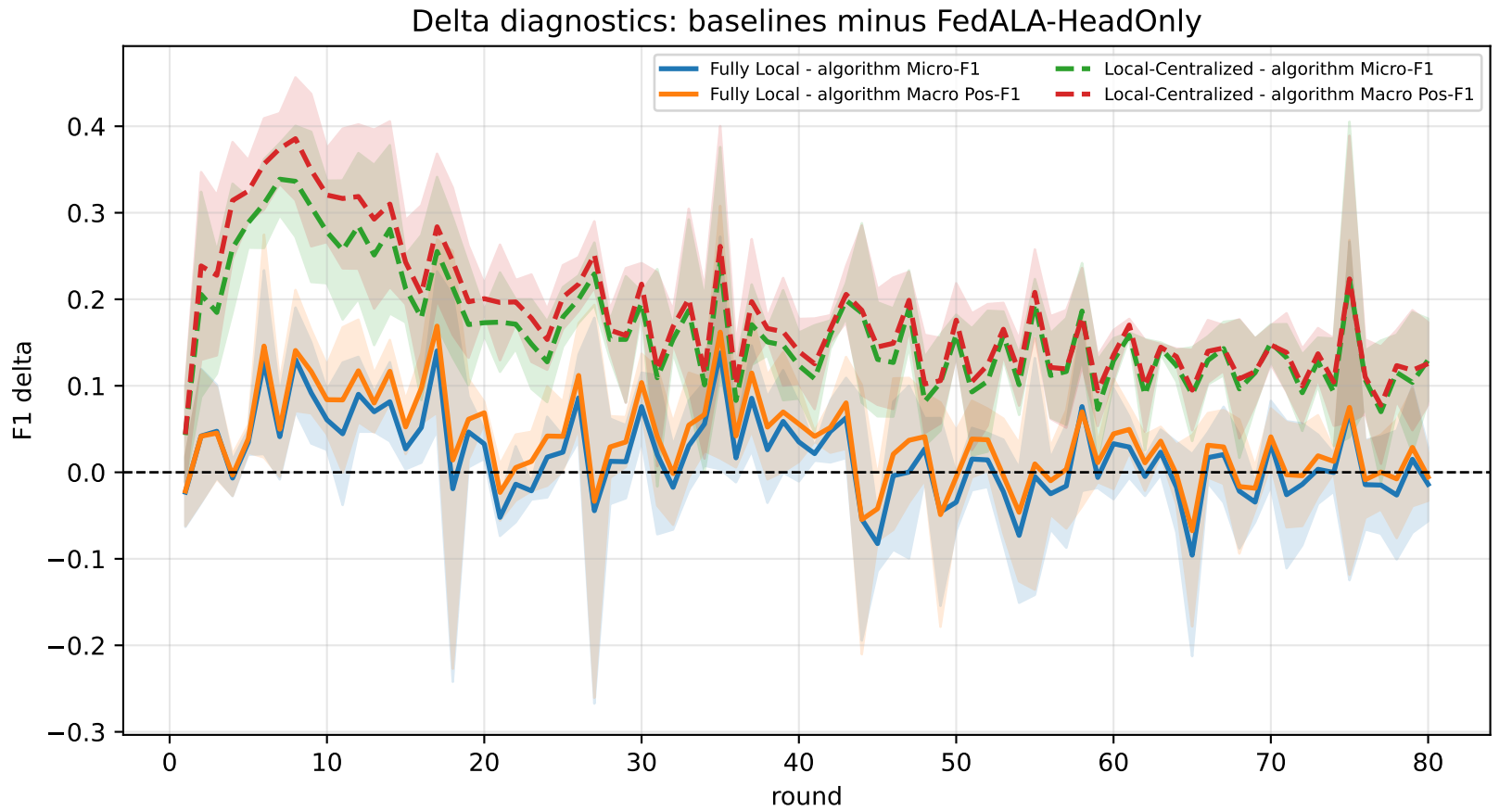
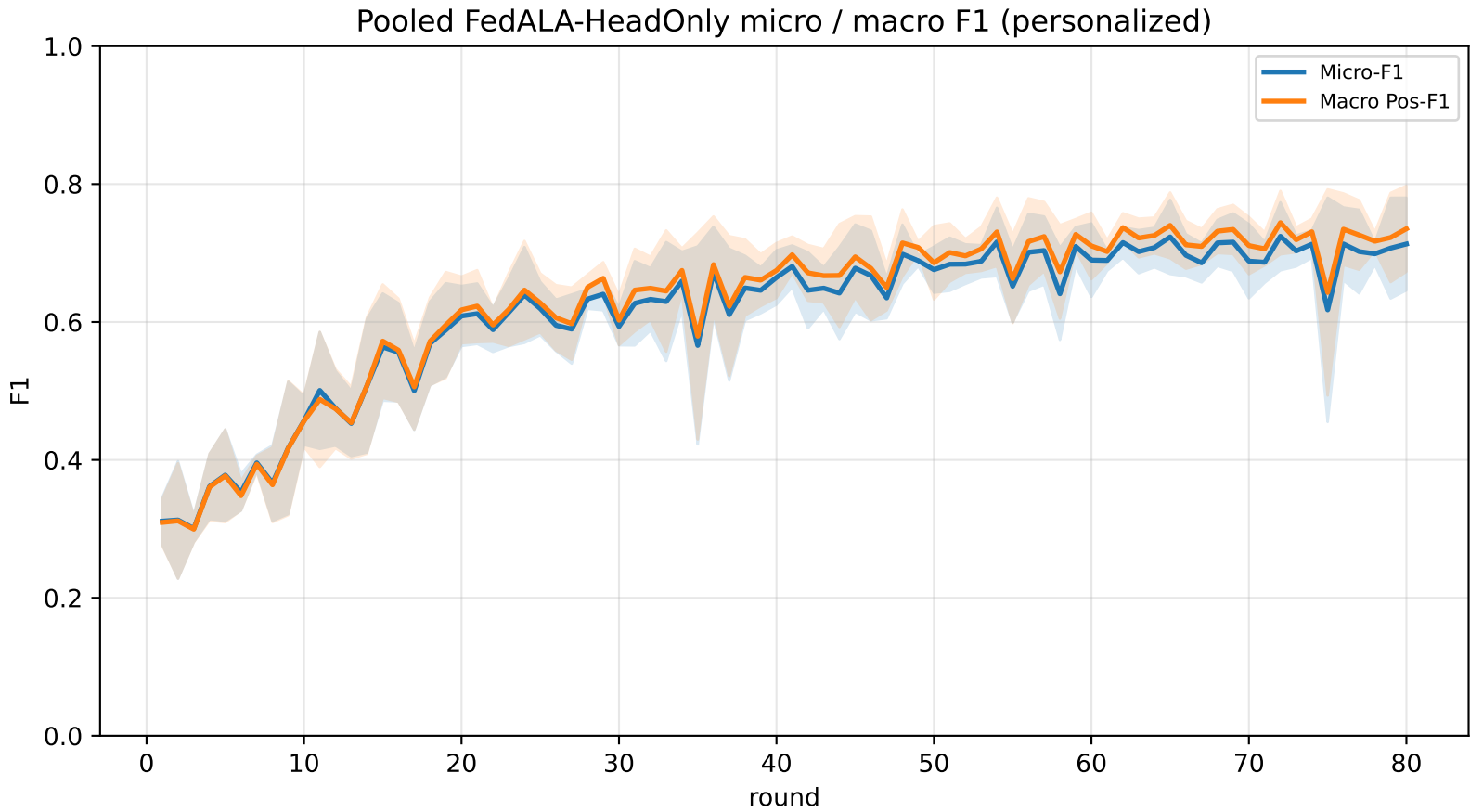
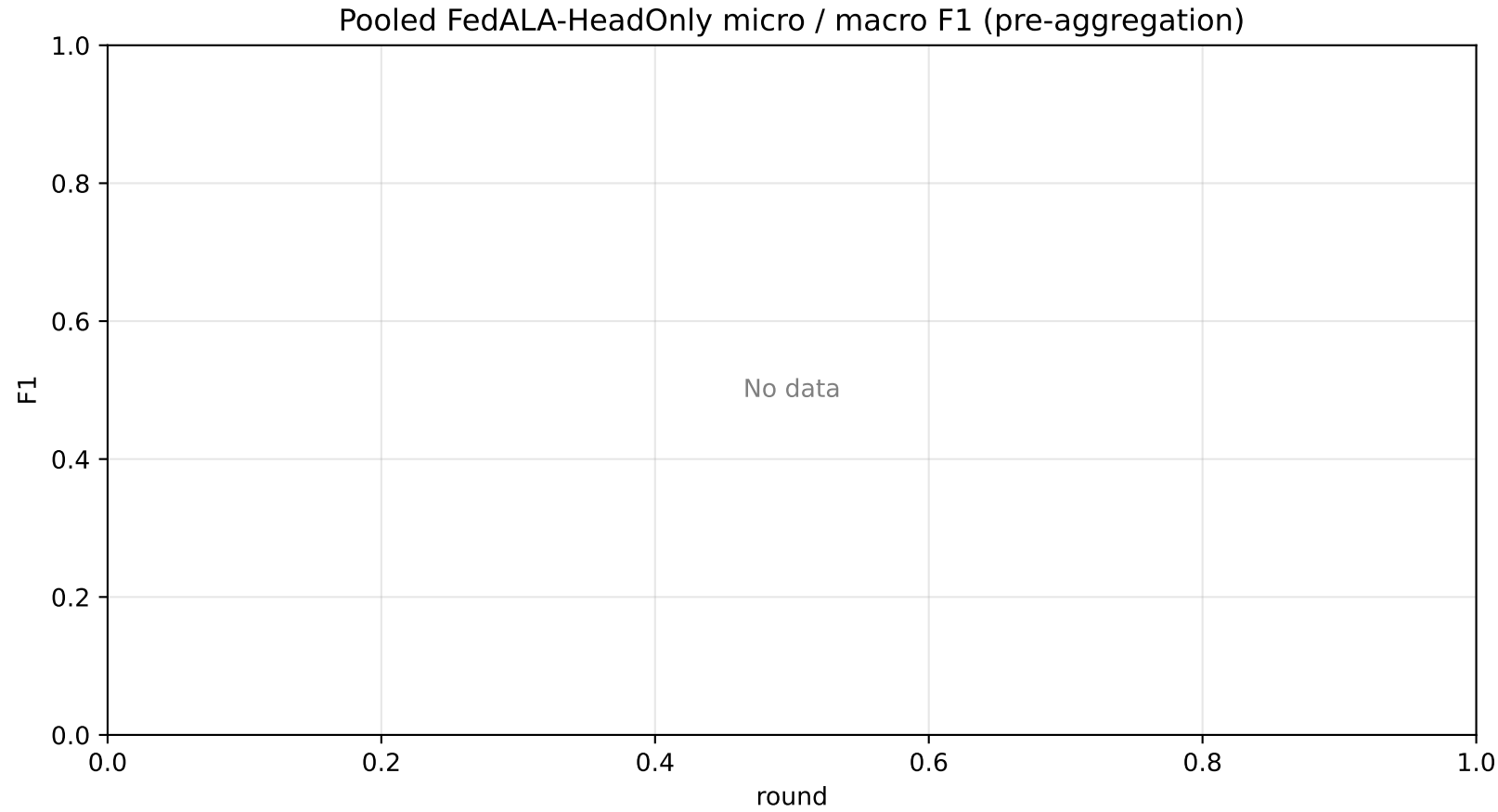
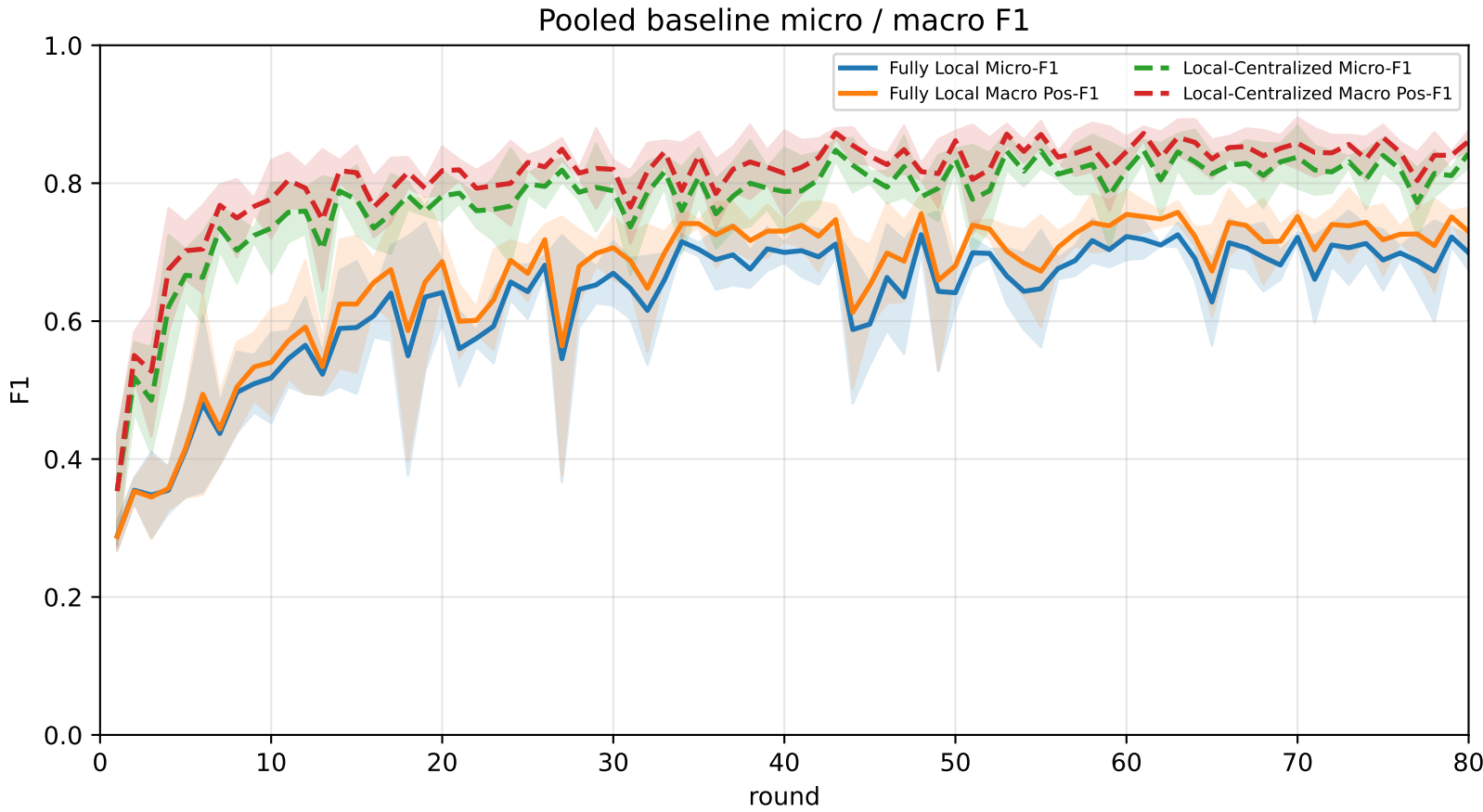
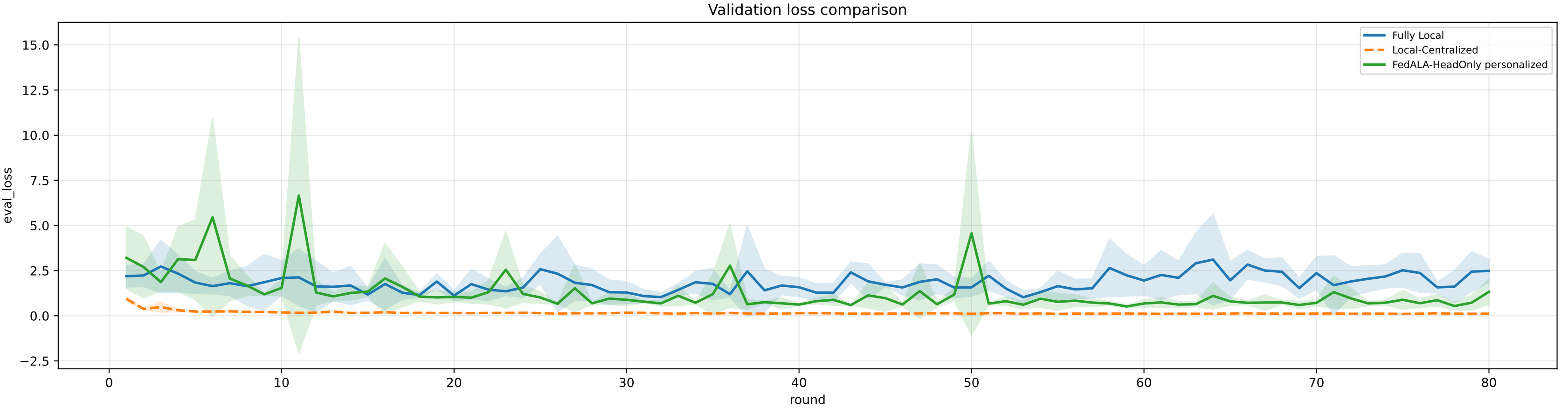
Delta diagnostics: baselines minus FedALA-HeadOnly



family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.2% ± 0.9%	66.3% ± 1.4%	<u>86.3% ± 4.7%</u>	79.8% ± 5.8%	63.6% ± 1.5%	58.1% ± 1.4%	43.8% ± 3.4%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>69.4% ± 1.4%</u>	<u>71.2% ± 1.3%</u>	74.9% ± 9.2%	<u>83.4% ± 3.1%</u>	<u>69.9% ± 3.2%</u>	<u>68.9% ± 3.9%</u>	<u>58.6% ± 1.2%</u>

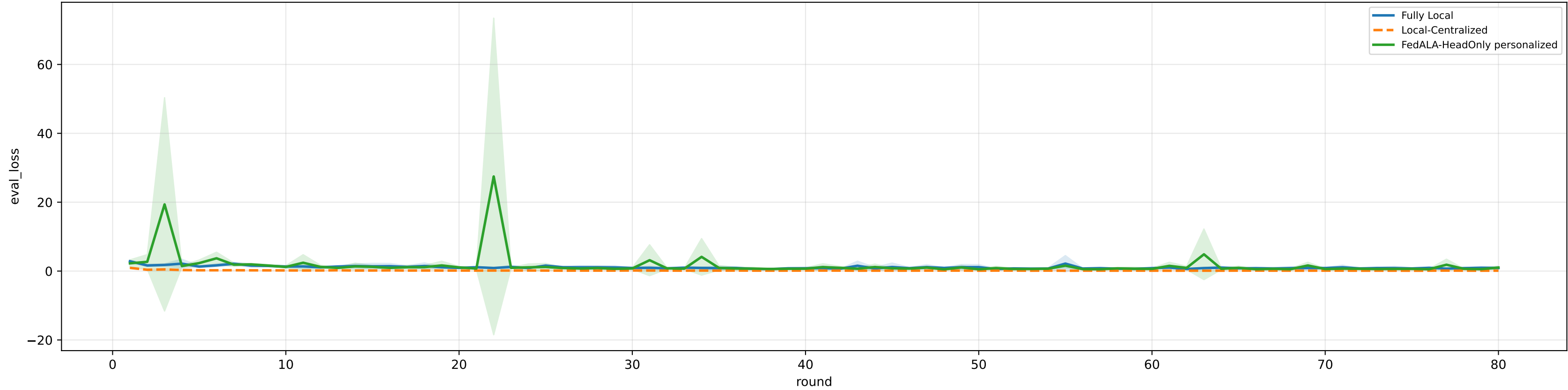


family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

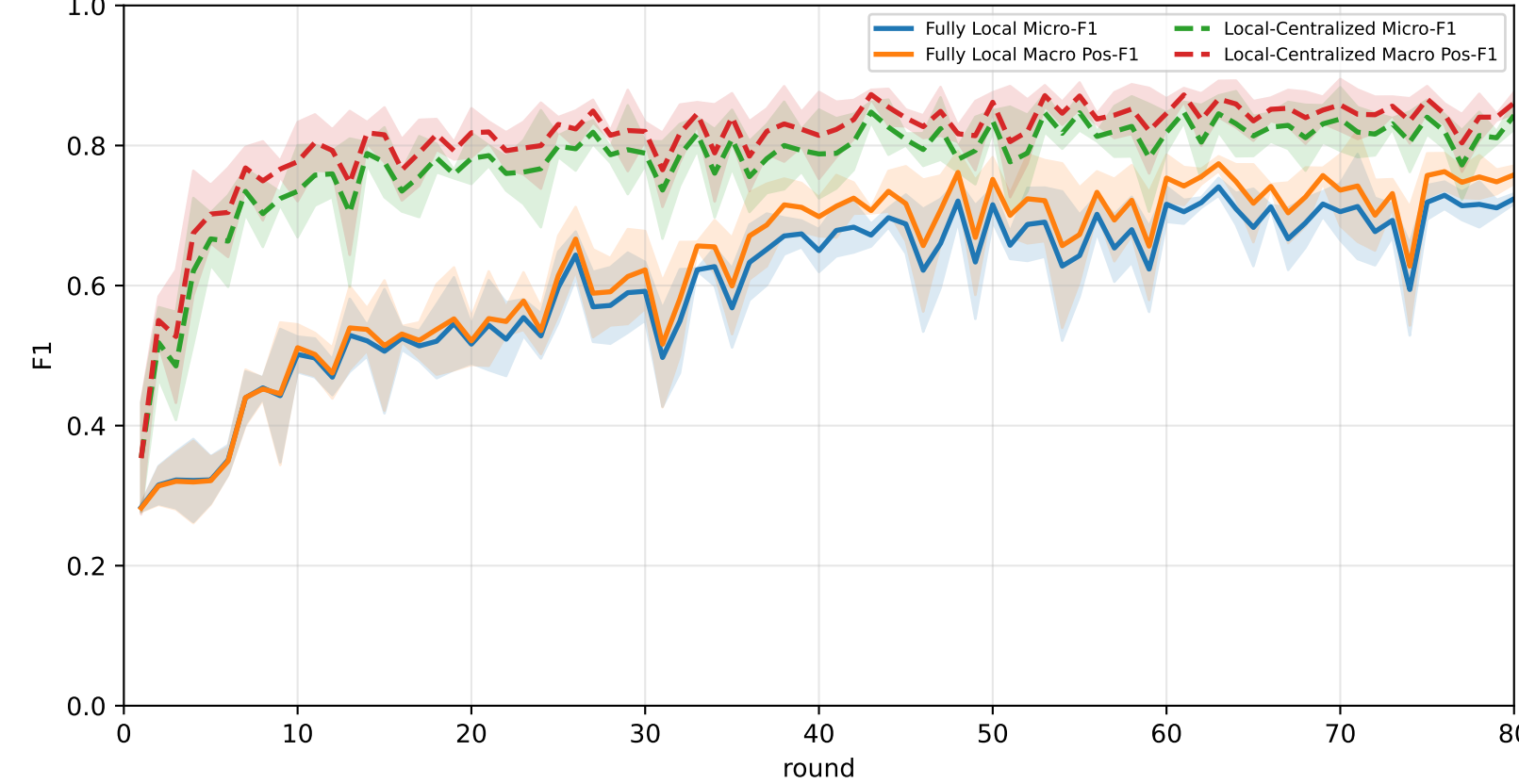
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.1% ± 1.8%	67.8% ± 2.4%	<u>81.8% ± 8.2%</u>	78.3% ± 6.8%	64.9% ± 1.4%	64.9% ± 4.4%	49.0% ± 3.2%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedALA-HeadOnly	<u>69.0% ± 2.5%</u>	<u>70.6% ± 2.5%</u>	79.0% ± 2.4%	<u>79.9% ± 3.3%</u>	<u>66.2% ± 2.2%</u>	<u>69.0% ± 3.6%</u>	<u>59.0% ± 1.9%</u>

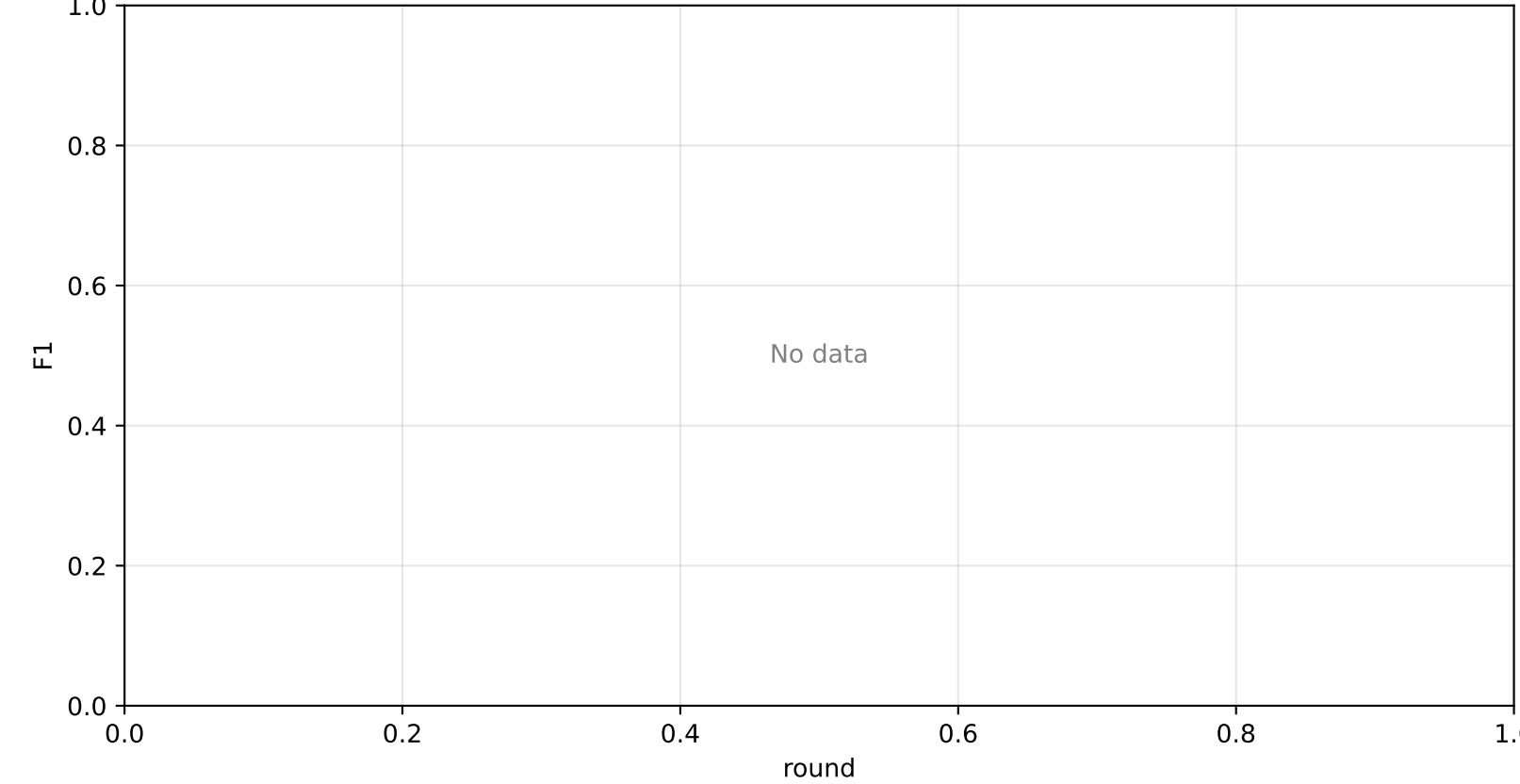
Validation loss comparison



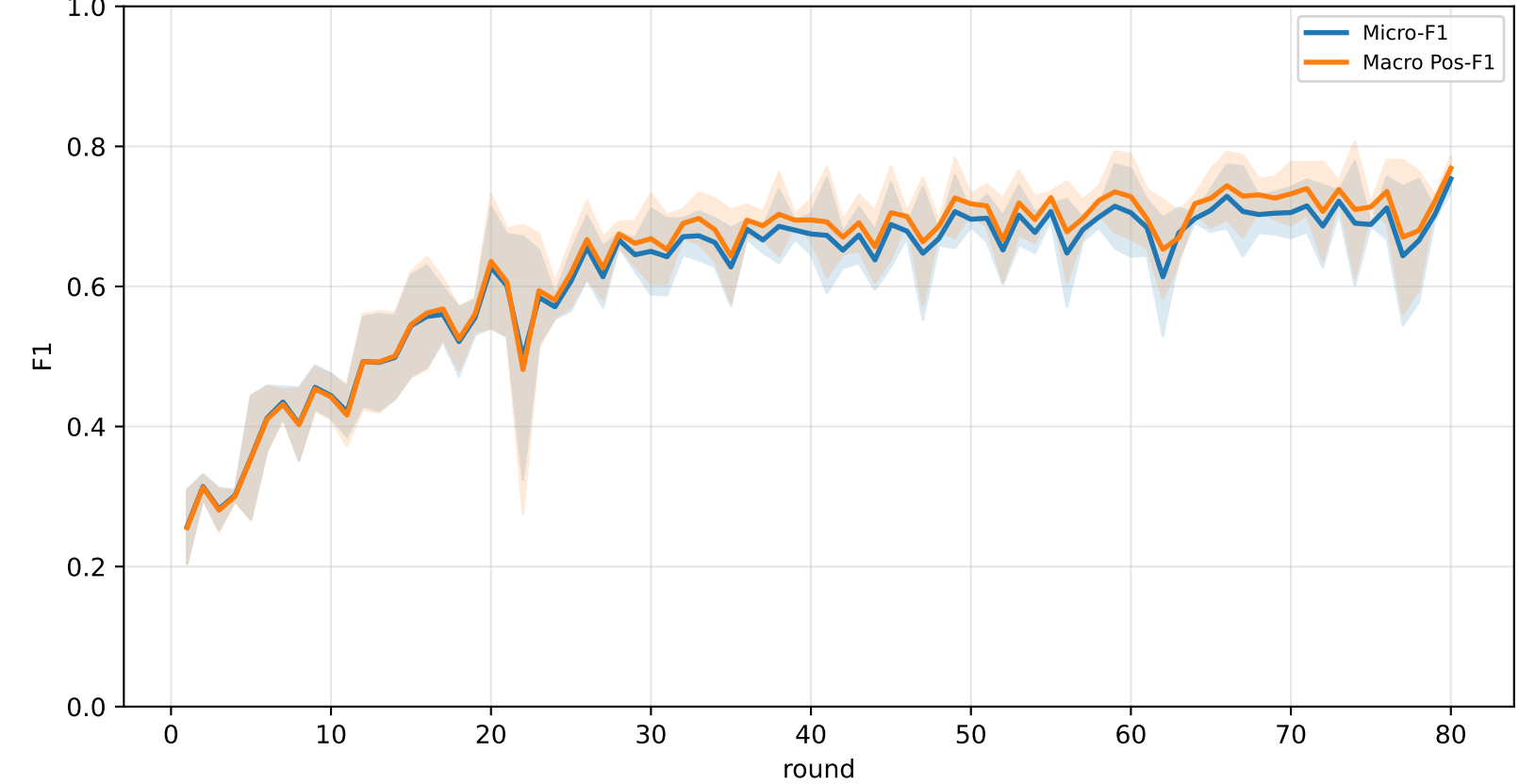
Pooled baseline micro / macro F1



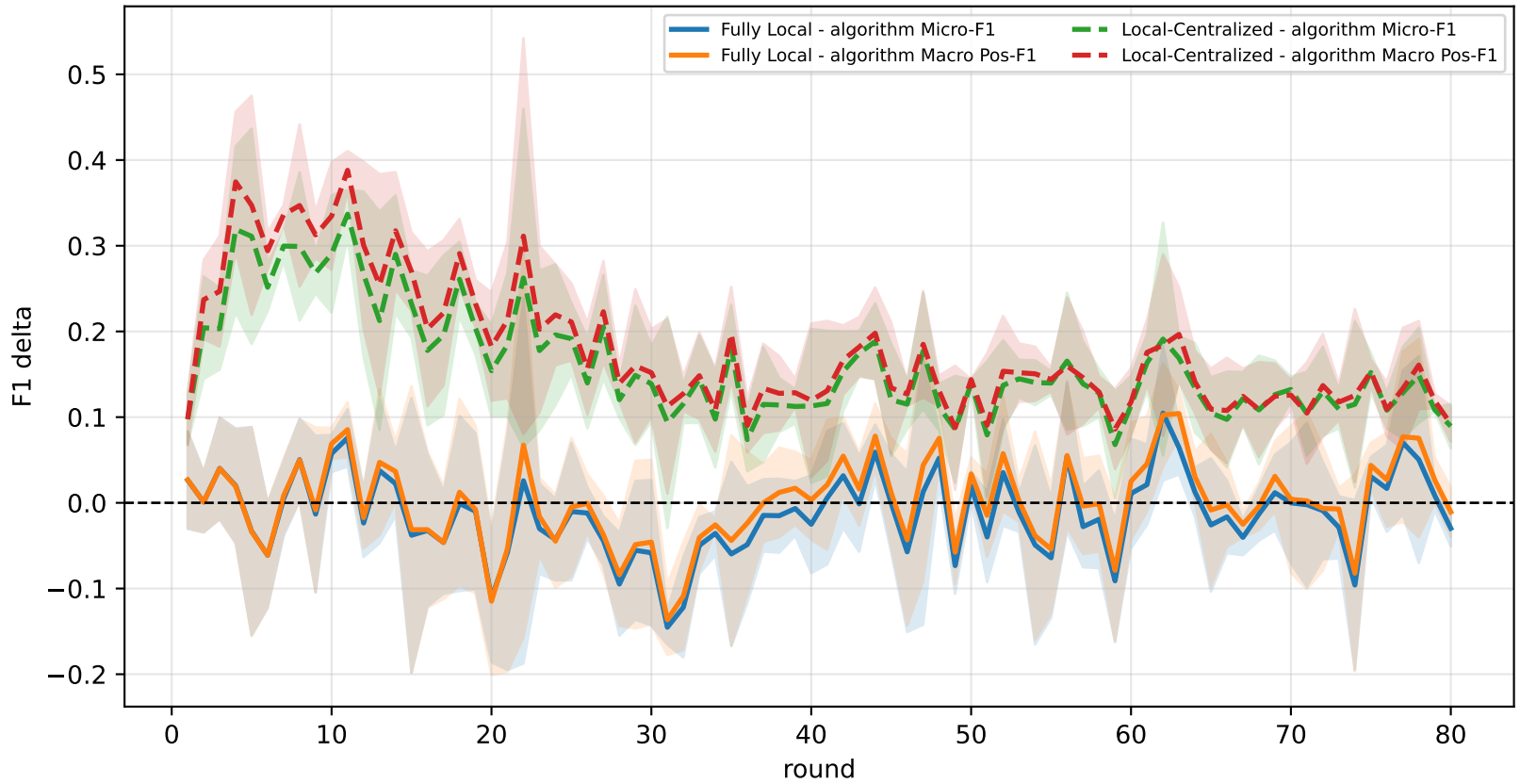
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



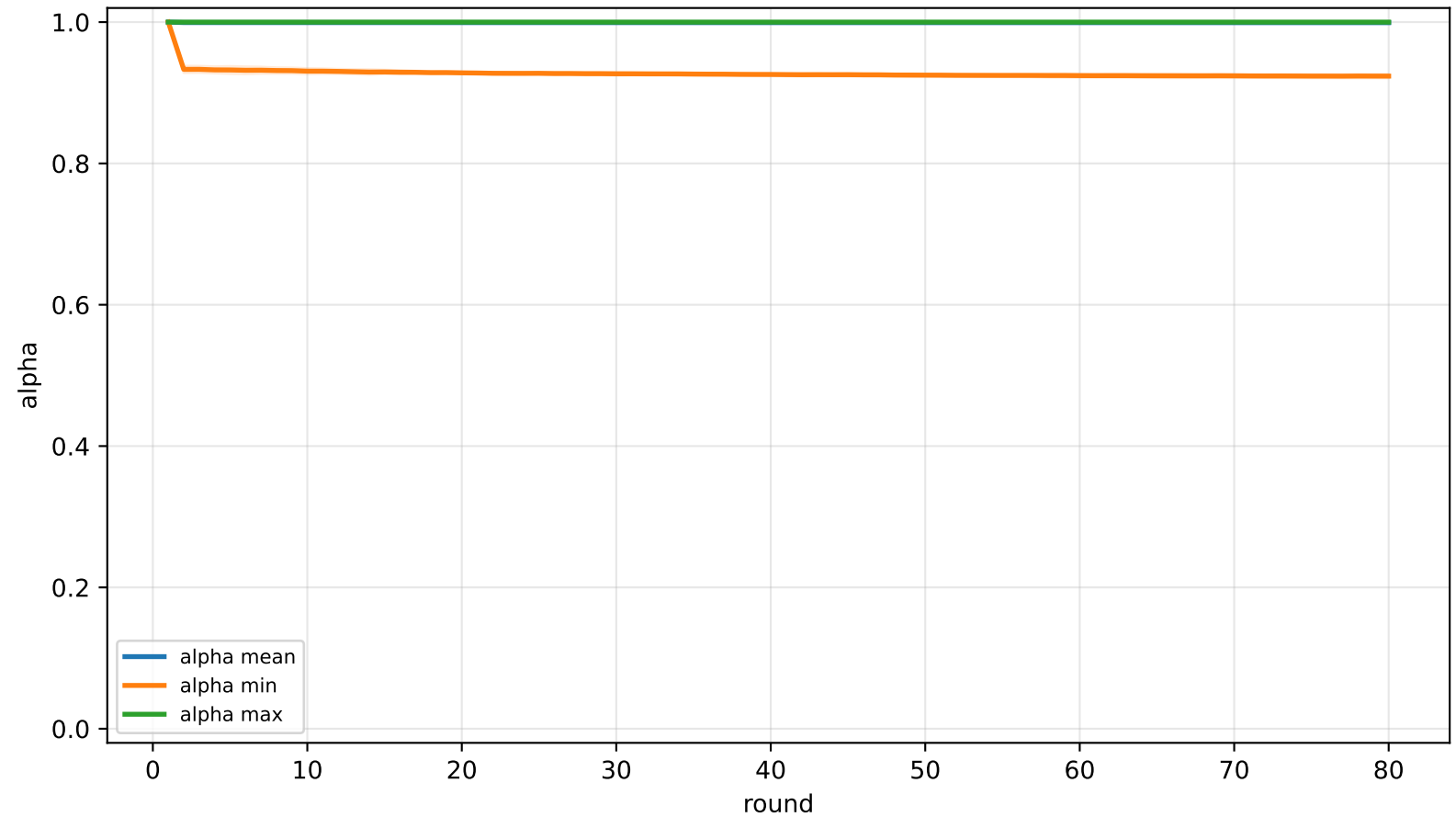
Delta diagnostics: baselines minus FedALA-HeadOnly



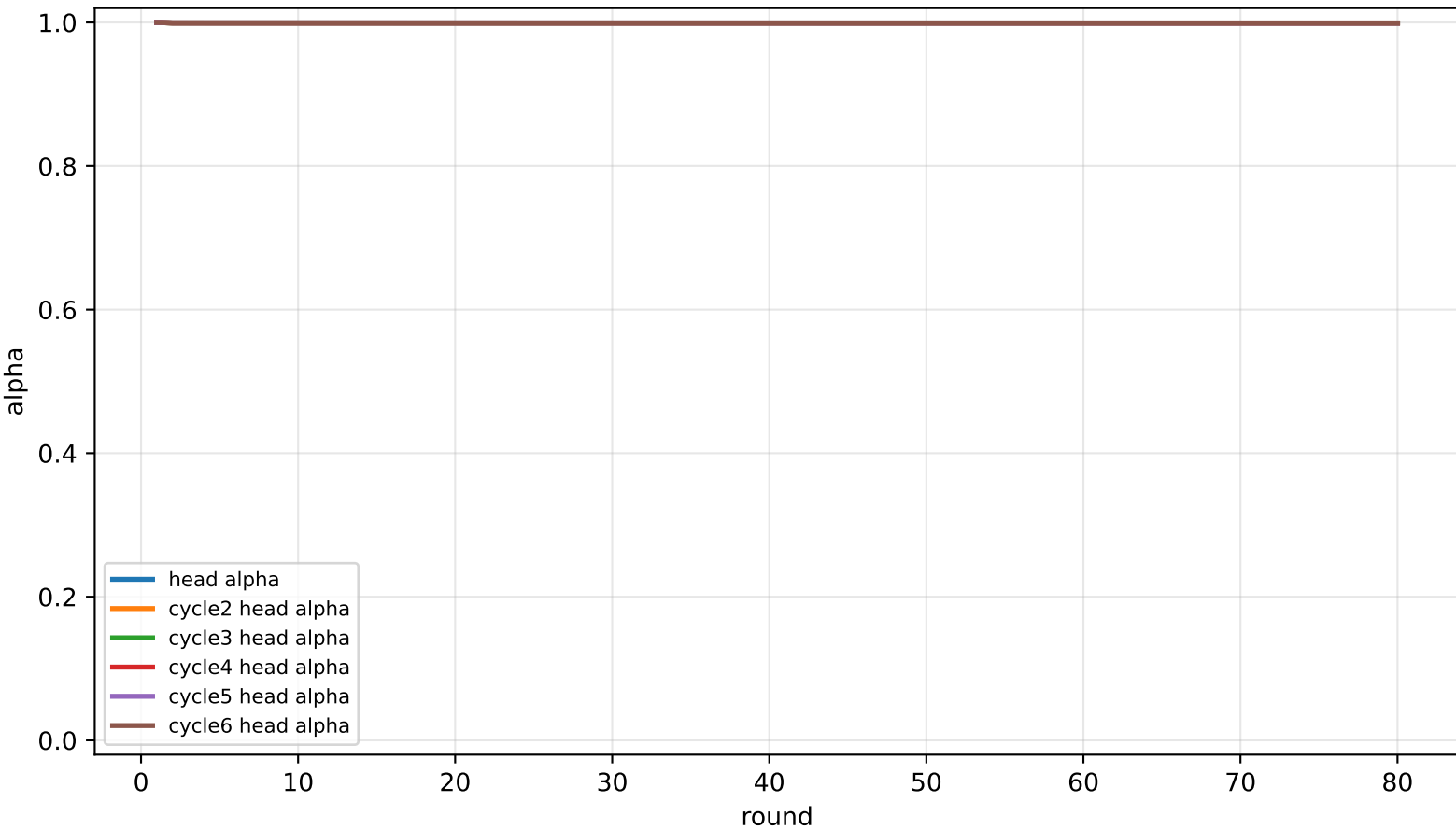
Controlled q-label heterogeneity: Mid | FedALA-HeadOnly ALA diagnostics | aggregated / pooled | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

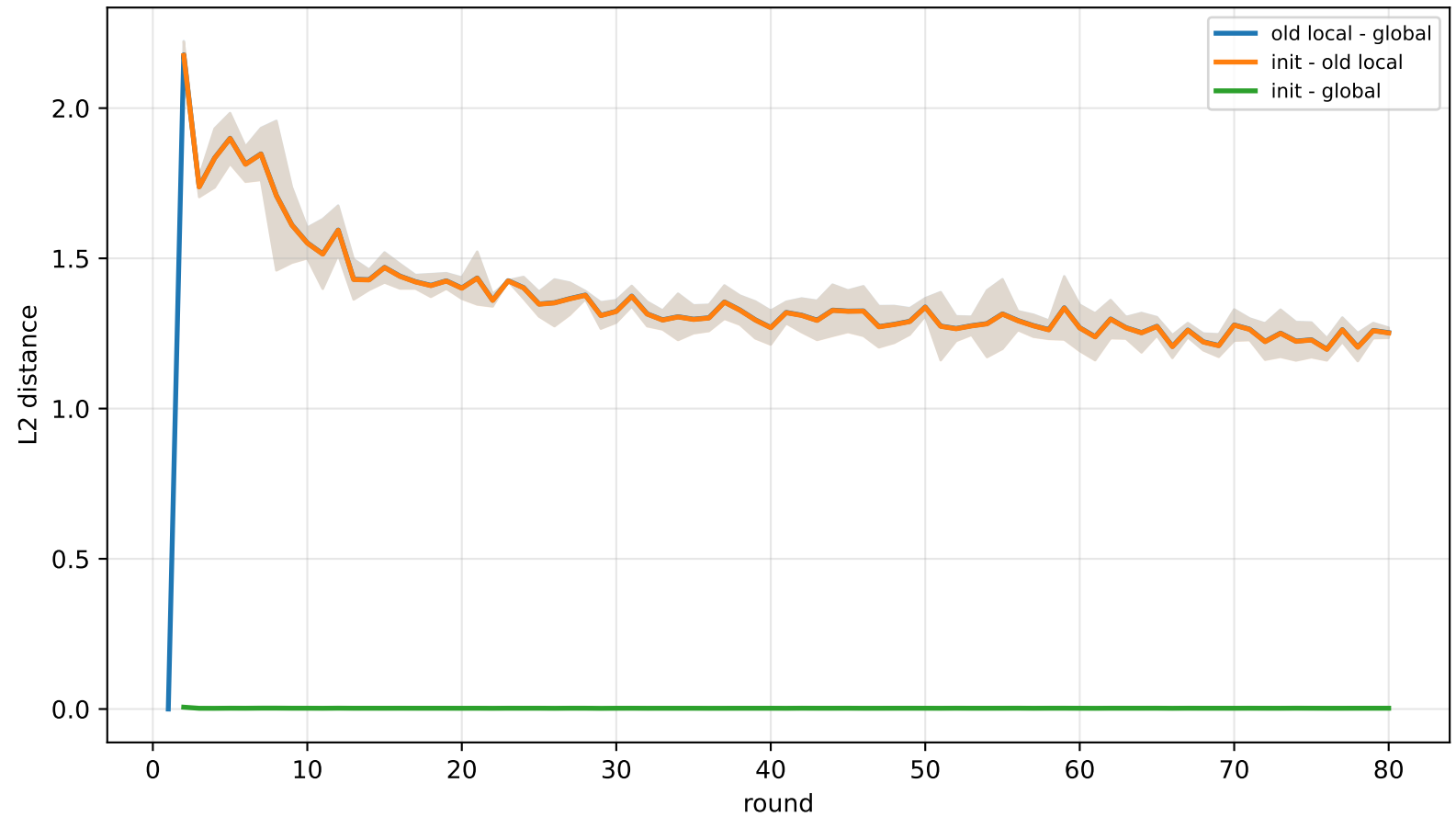
ALA interpolation weights



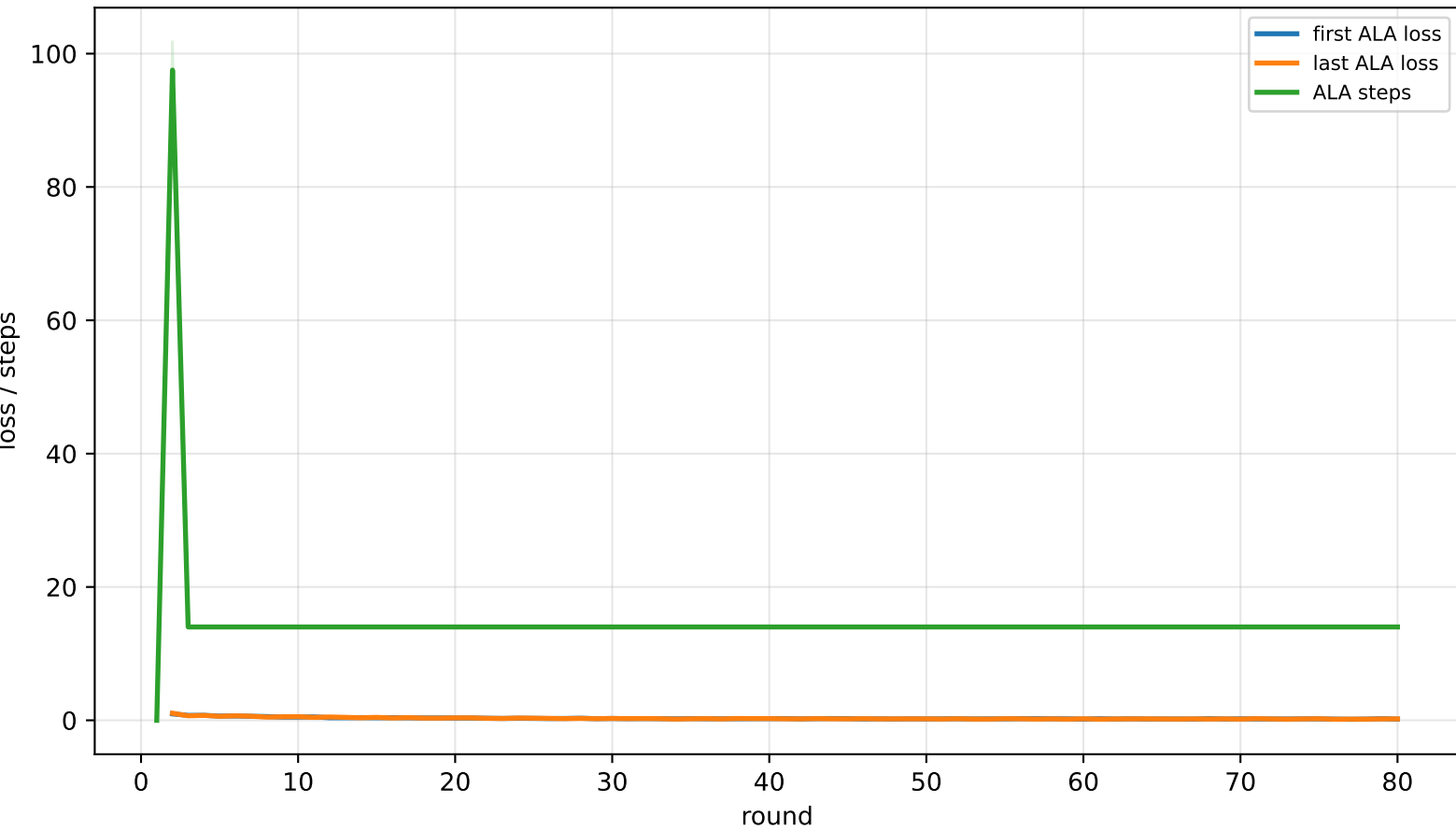
Backbone/head/task alpha diagnostics



Model distance diagnostics



ALA optimization diagnostics



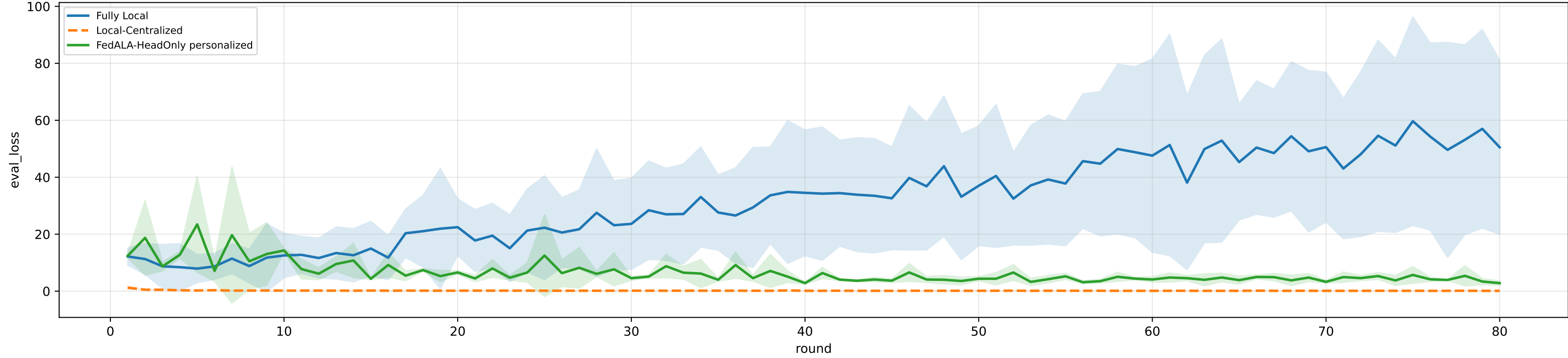
Controlled q-label heterogeneity: High | FedALA-HeadOnly diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

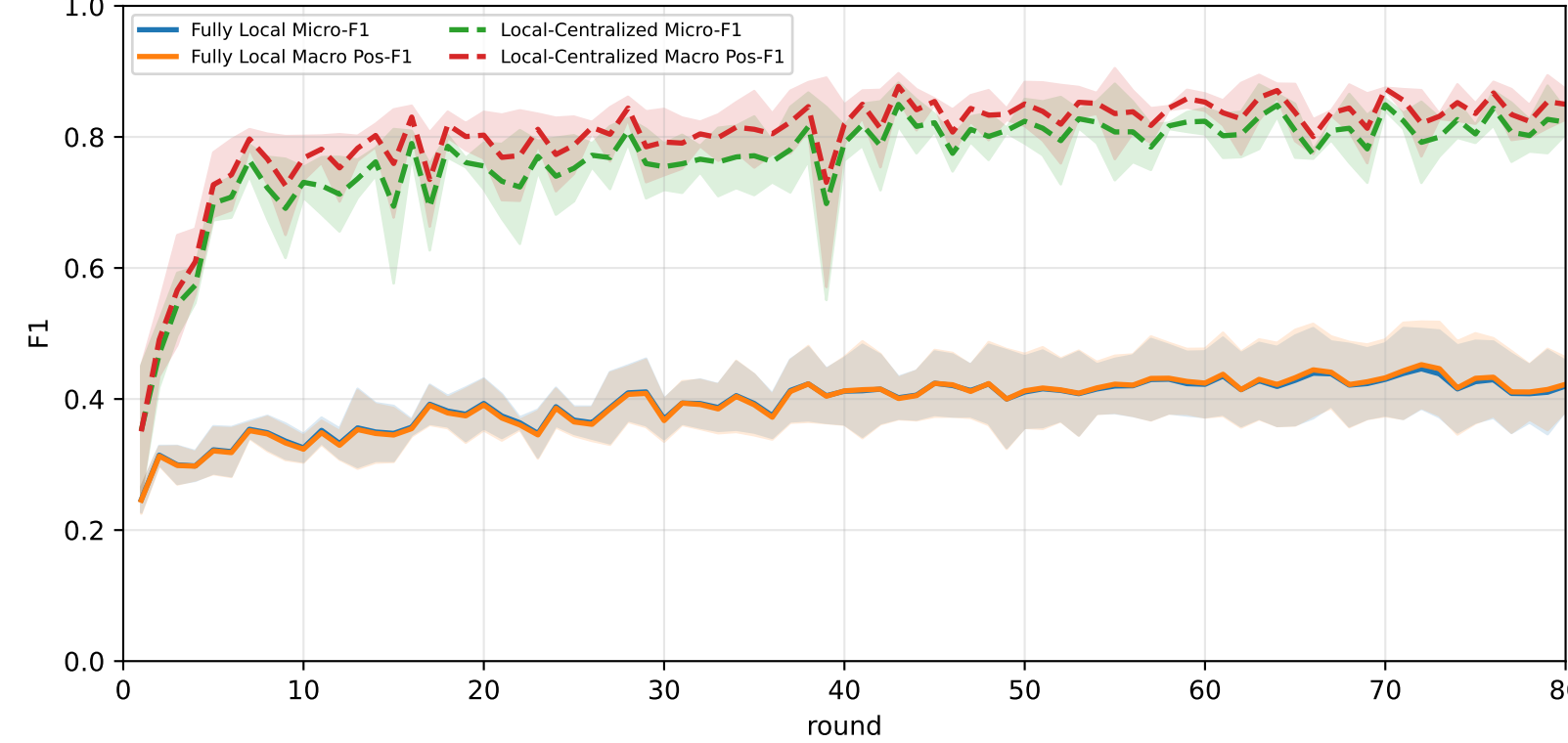
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.6% ± 1.8%	27.6% ± 1.8%	22.4% ± 1.5%	27.5% ± 2.1%	27.3% ± 1.2%	26.7% ± 1.2%	30.2% ± 1.5%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	<u>56.2% ± 2.8%</u>	<u>56.7% ± 2.8%</u>	<u>55.6% ± 4.6%</u>	<u>60.7% ± 1.7%</u>	<u>57.2% ± 2.9%</u>	<u>56.0% ± 5.0%</u>	<u>51.8% ± 1.1%</u>

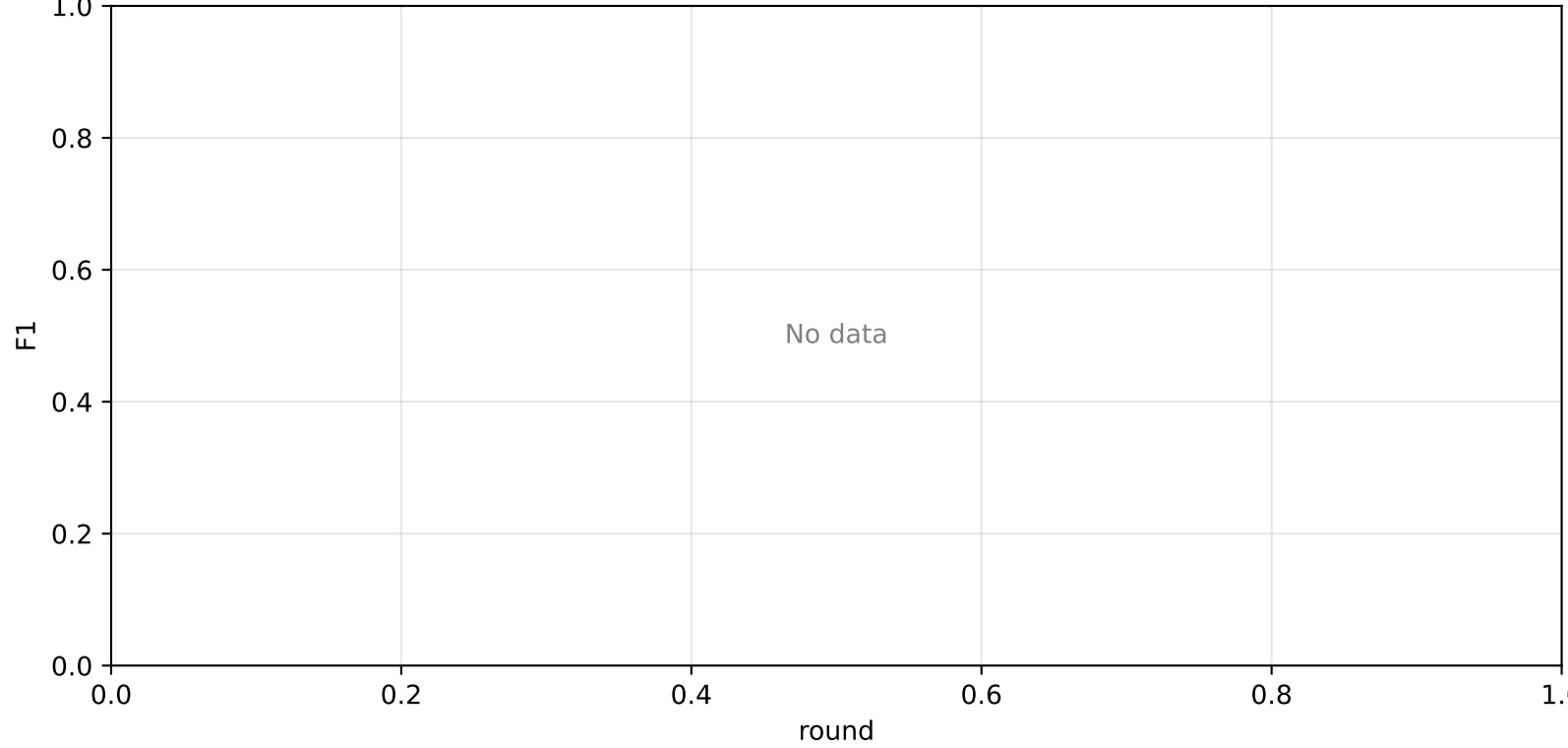
Validation loss comparison



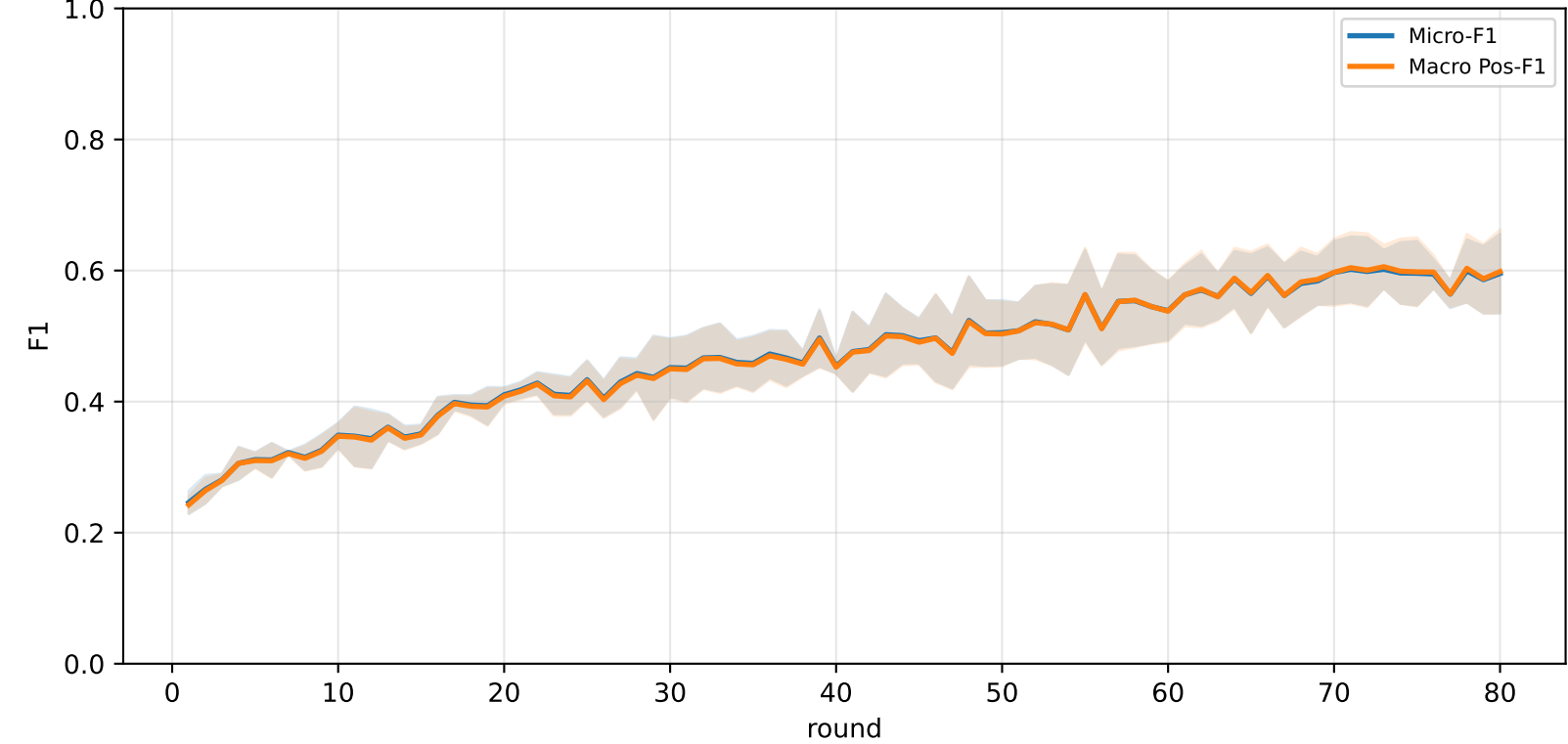
Pooled baseline micro / macro F1



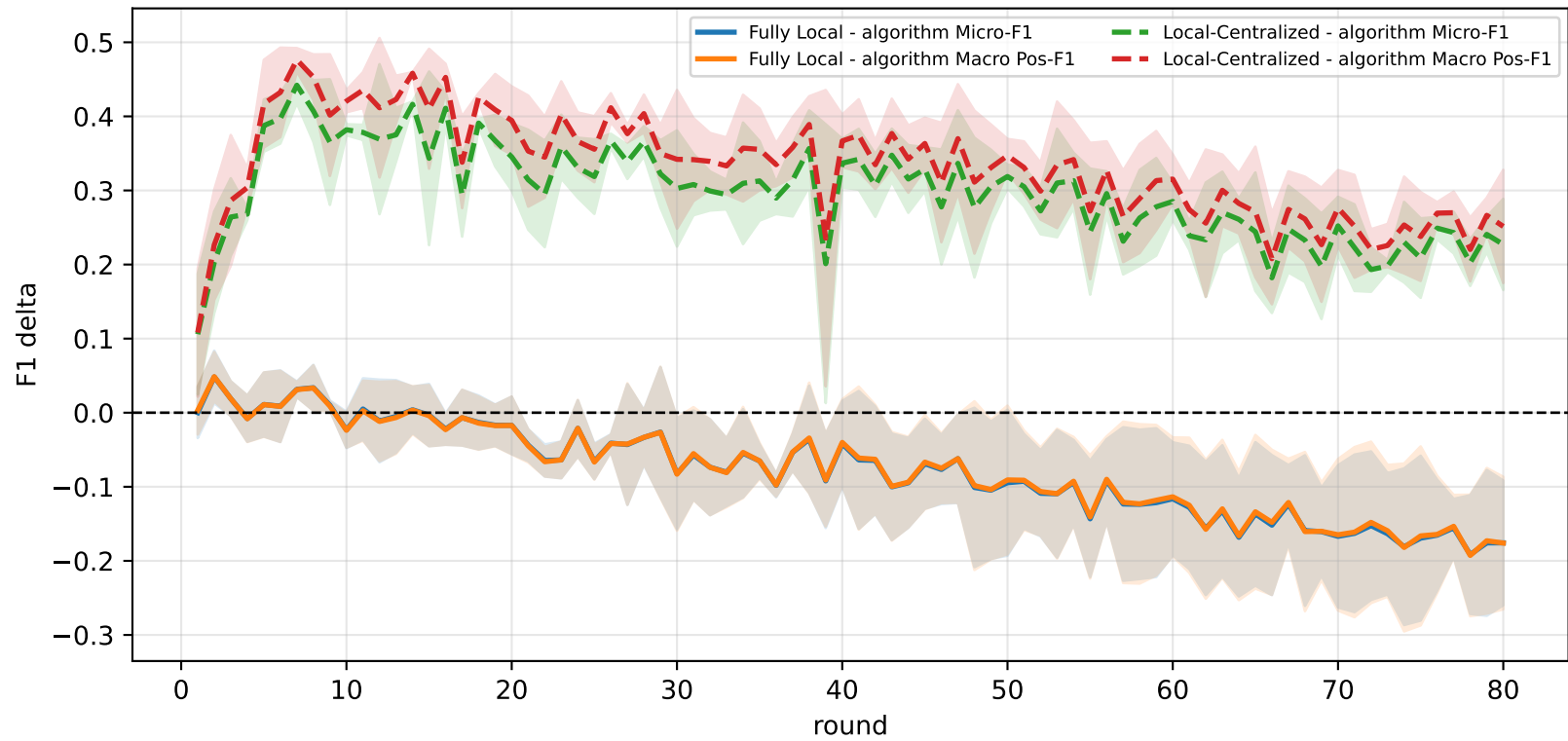
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

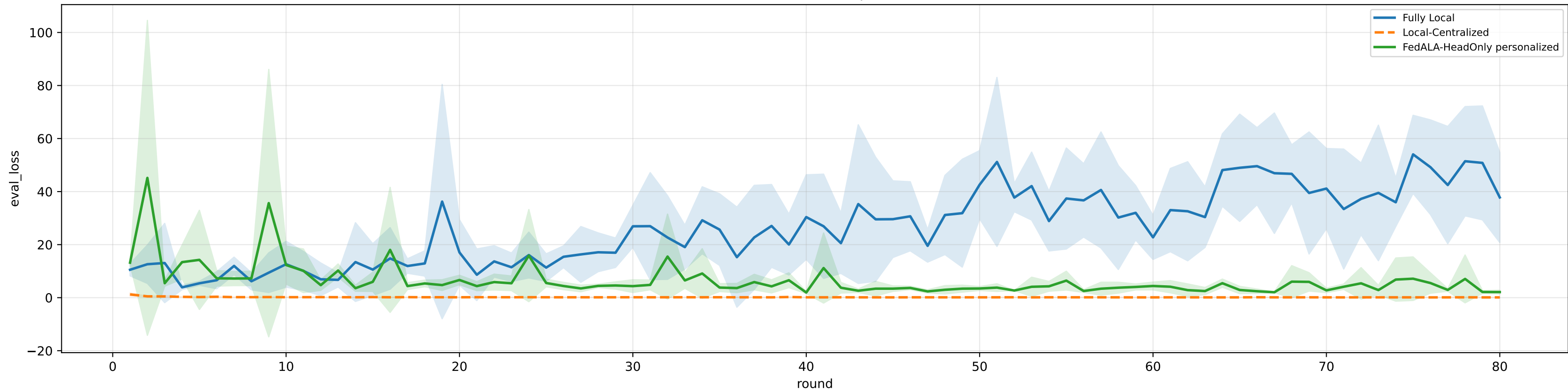


family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

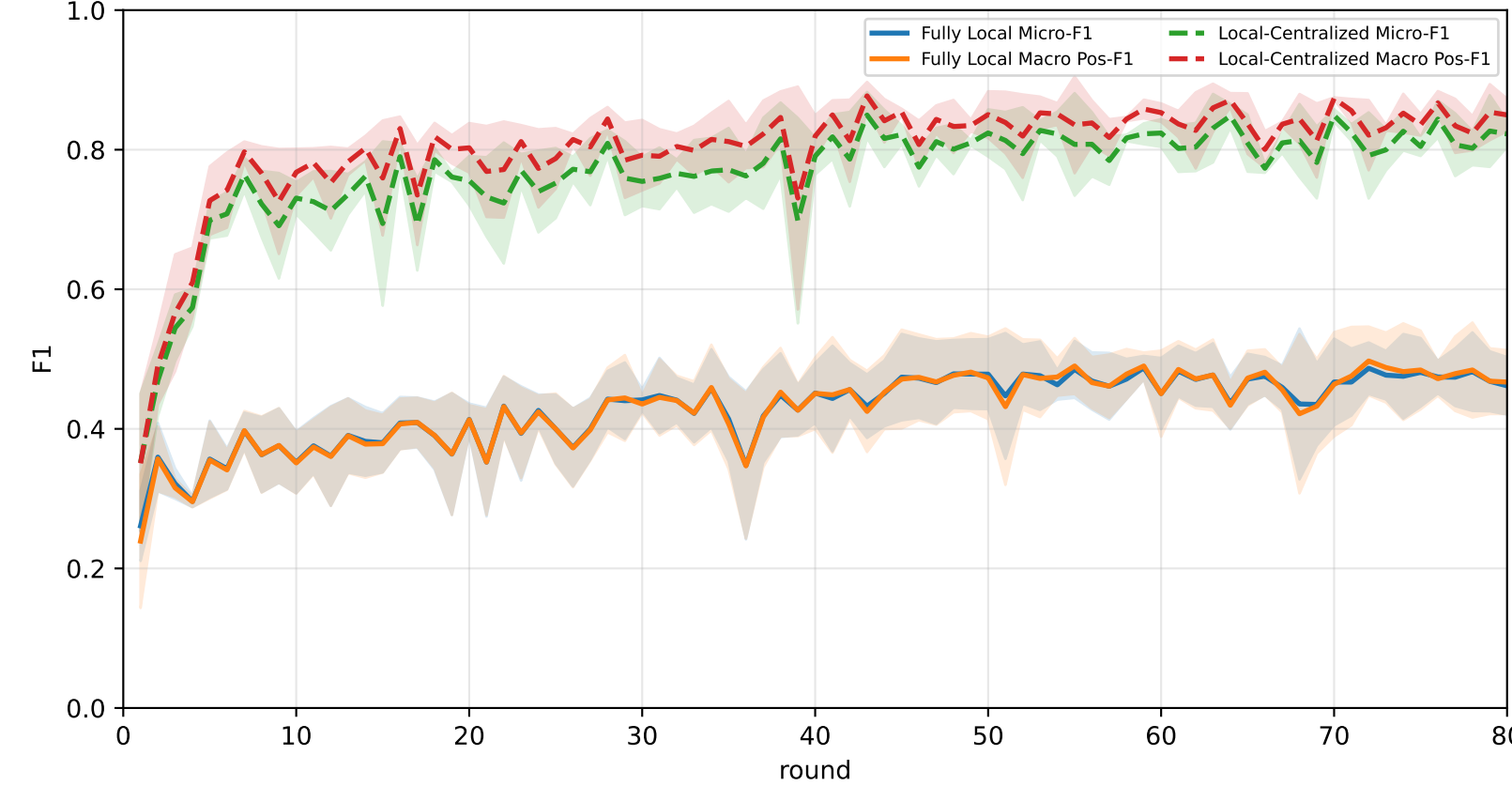
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	28.6% ± 0.7%	28.6% ± 0.7%	23.4% ± 1.7%	29.1% ± 1.2%	30.2% ± 0.8%	28.5% ± 1.4%	32.0% ± 1.3%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	55.1% ± 2.1%	55.3% ± 2.1%	55.4% ± 0.6%	58.1% ± 2.5%	56.2% ± 2.3%	55.0% ± 3.8%	51.8% ± 1.9%

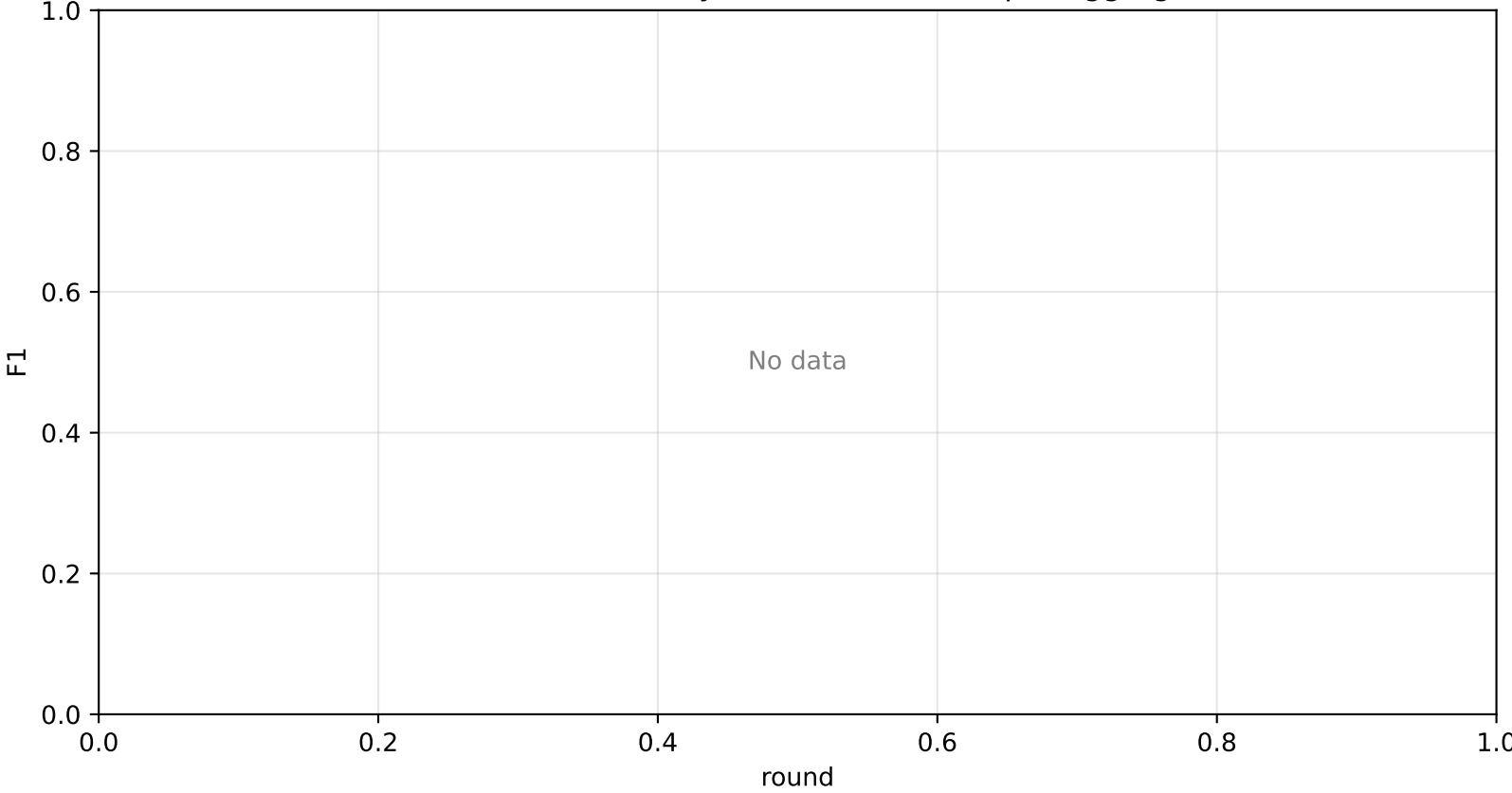
Validation loss comparison



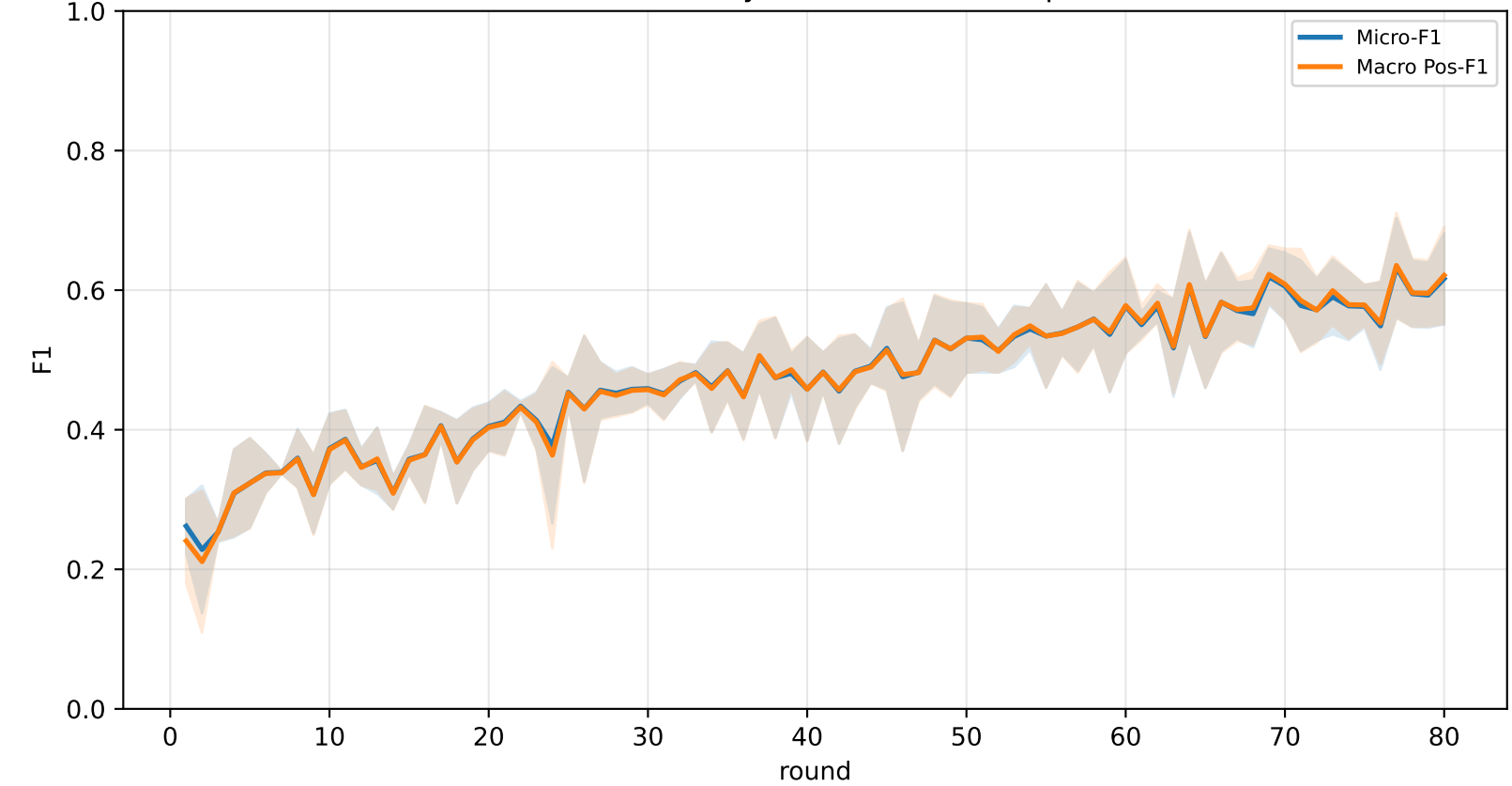
Pooled baseline micro / macro F1



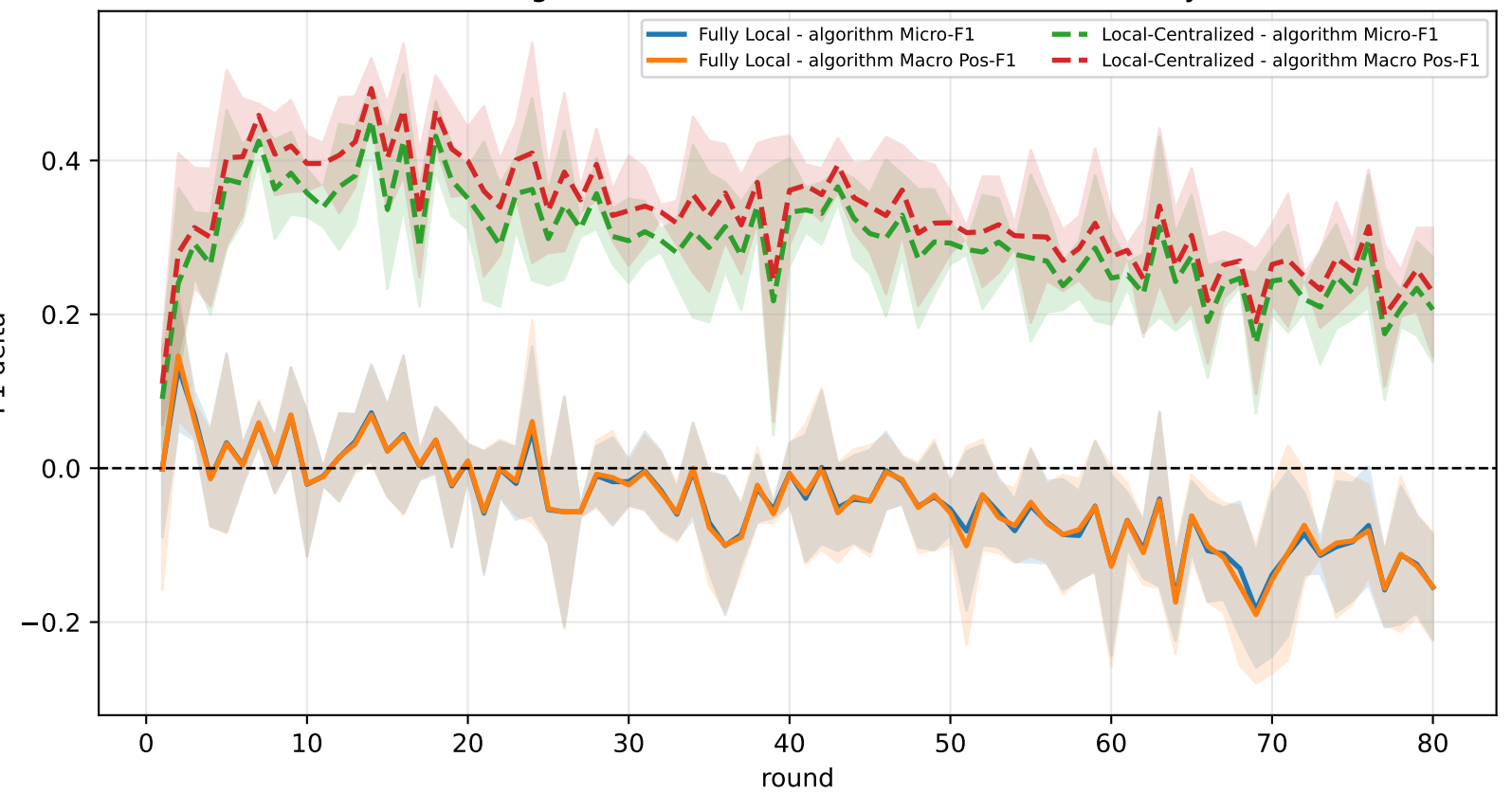
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

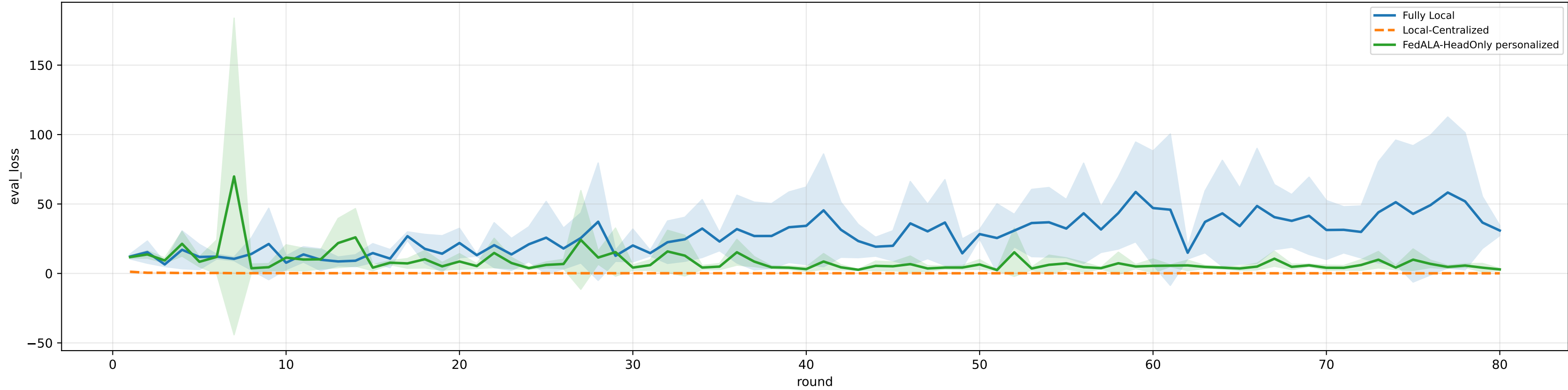


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

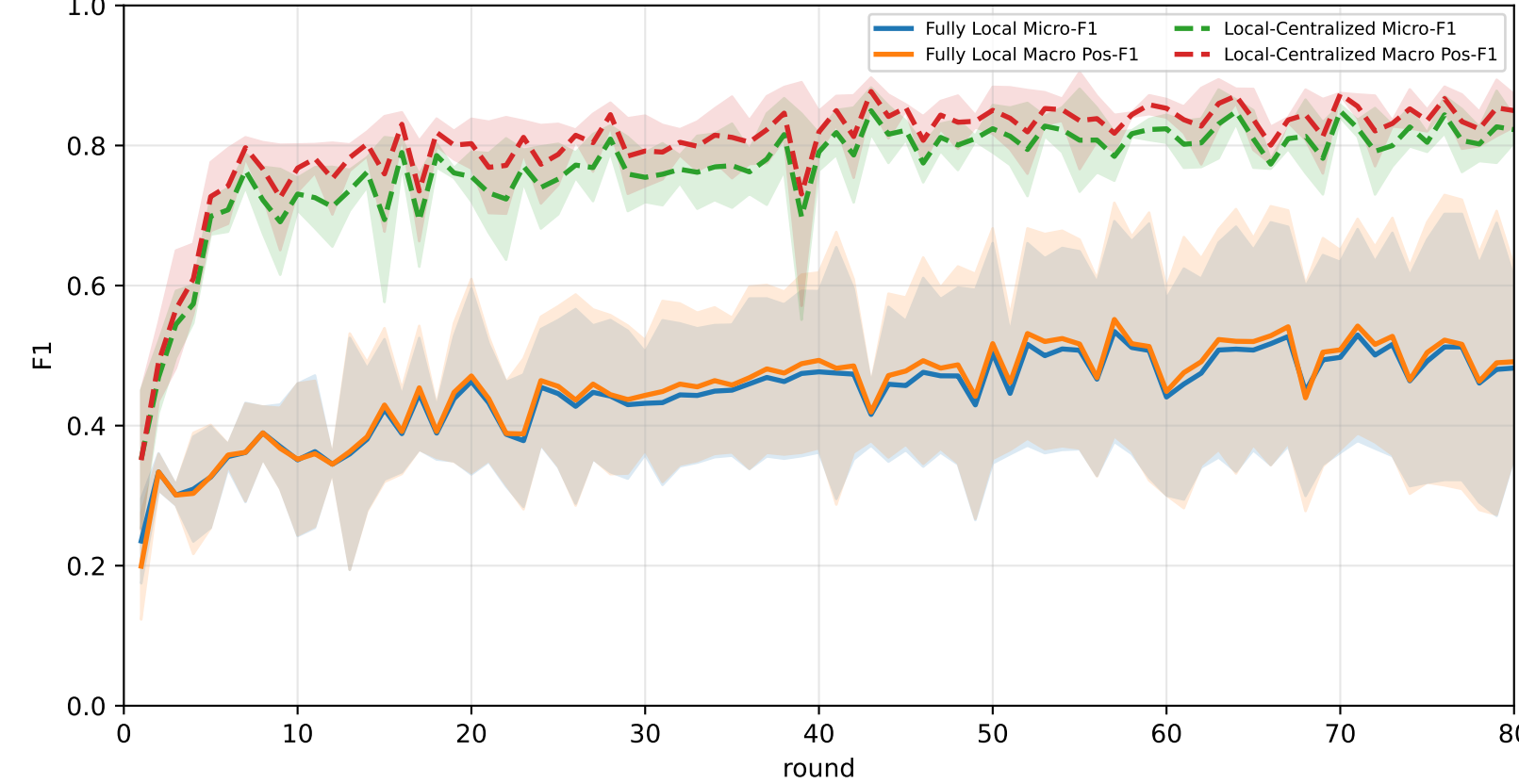
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	26.5% ± 4.7%	26.6% ± 4.7%	22.3% ± 4.2%	27.0% ± 7.0%	26.9% ± 5.3%	27.0% ± 4.1%	29.5% ± 3.2%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	54.5% ± 5.7%	55.2% ± 5.9%	53.0% ± 12.6%	60.5% ± 6.4%	54.8% ± 6.5%	55.8% ± 7.7%	51.8% ± 2.0%

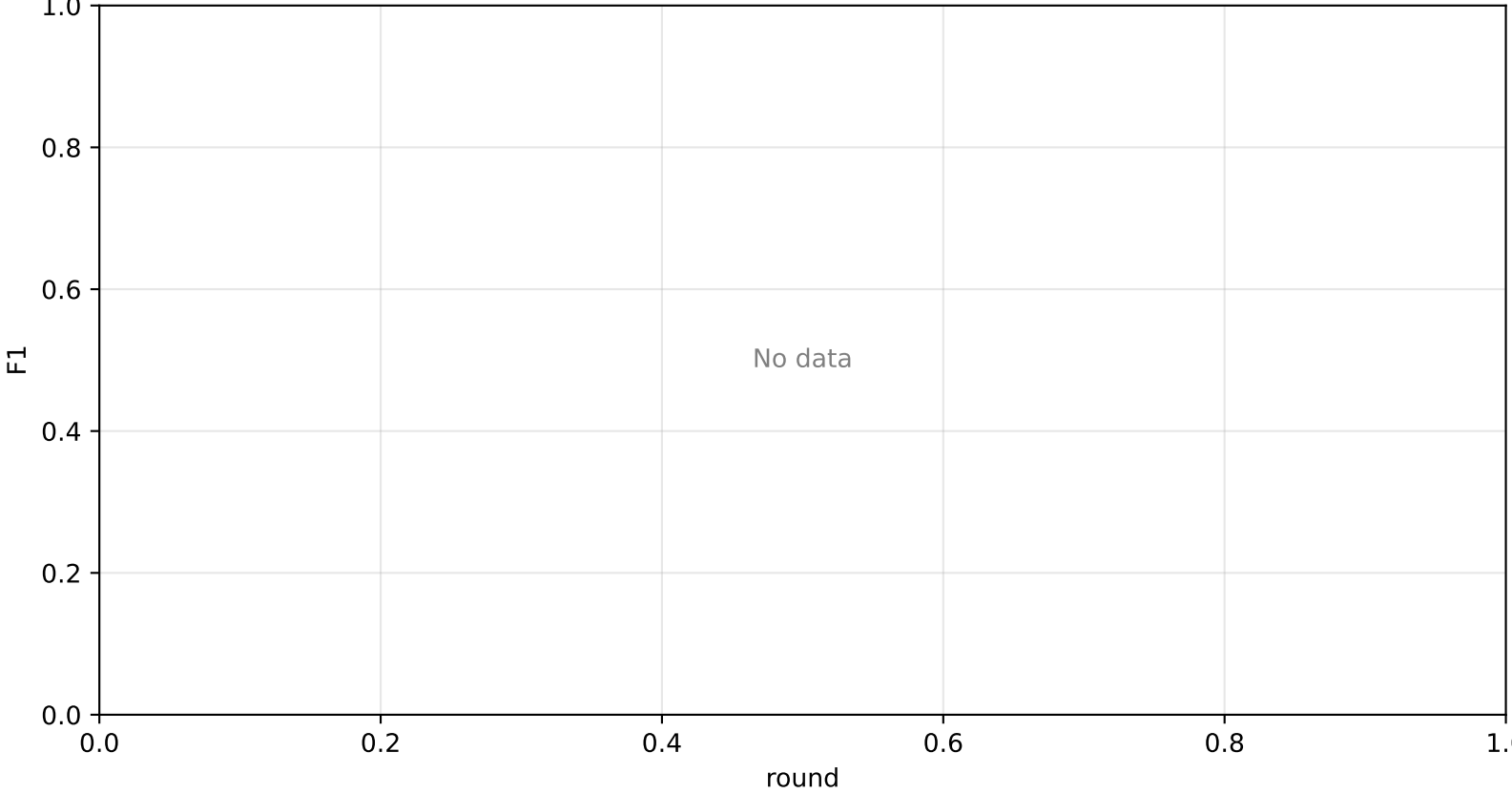
Validation loss comparison



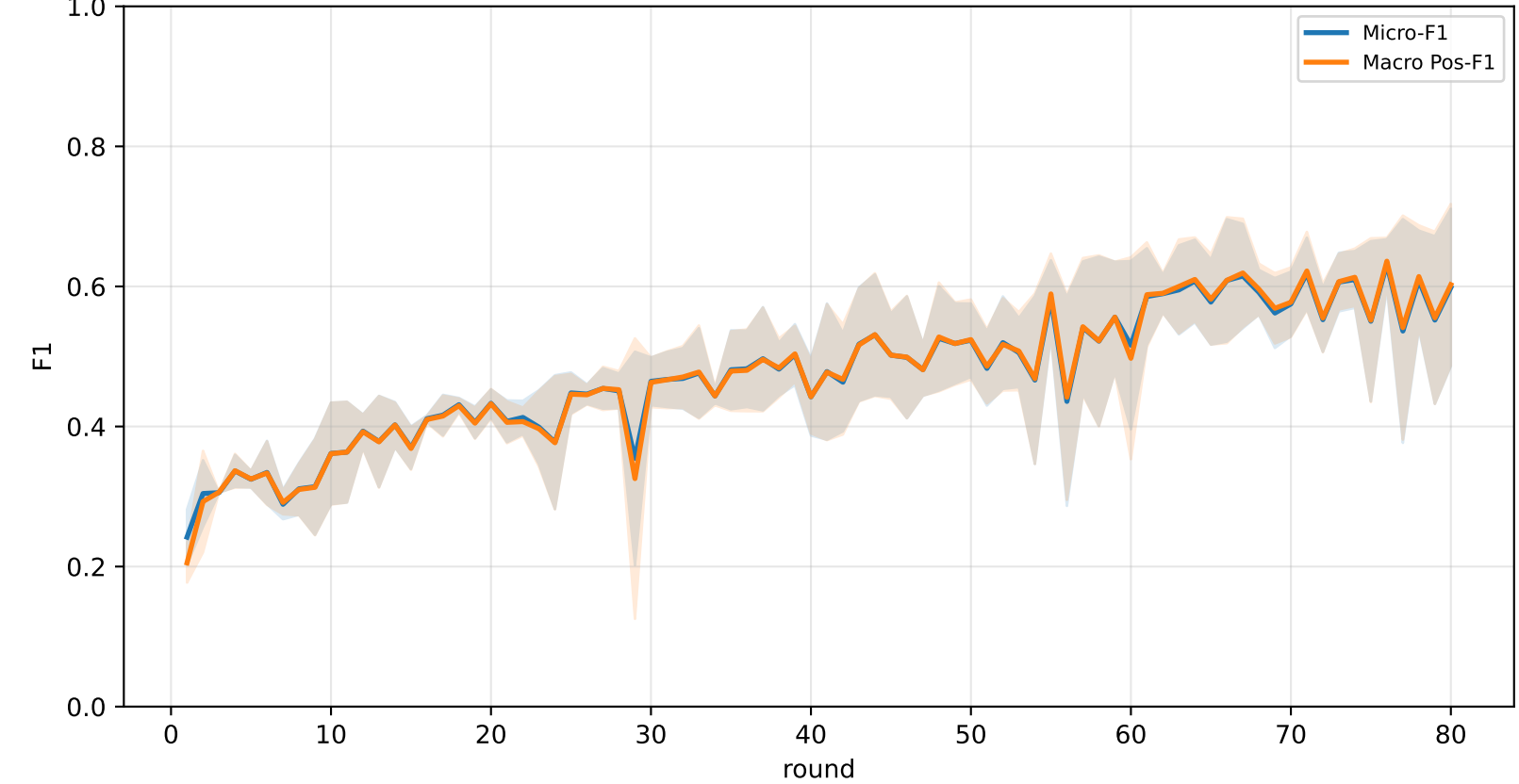
Pooled baseline micro / macro F1



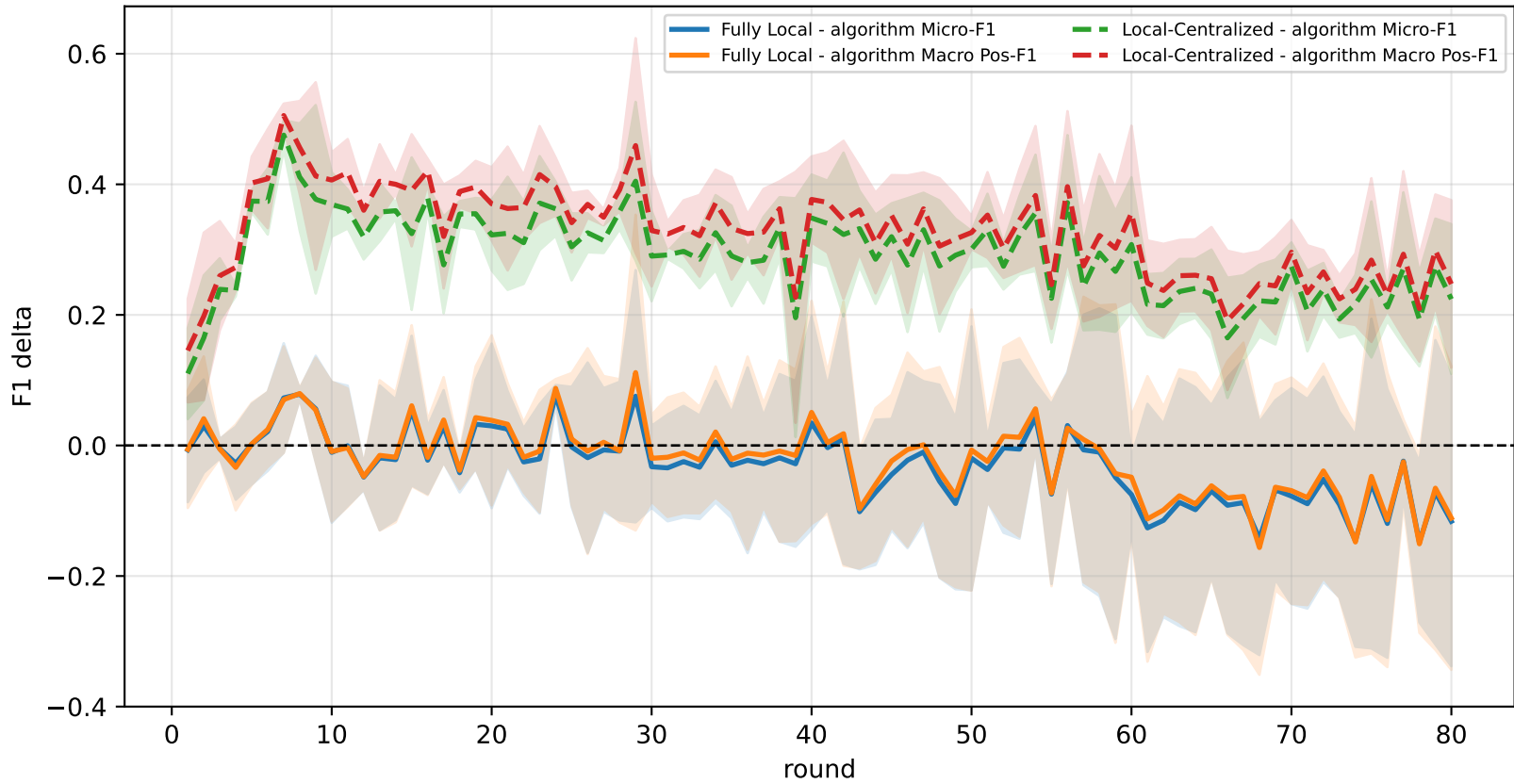
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

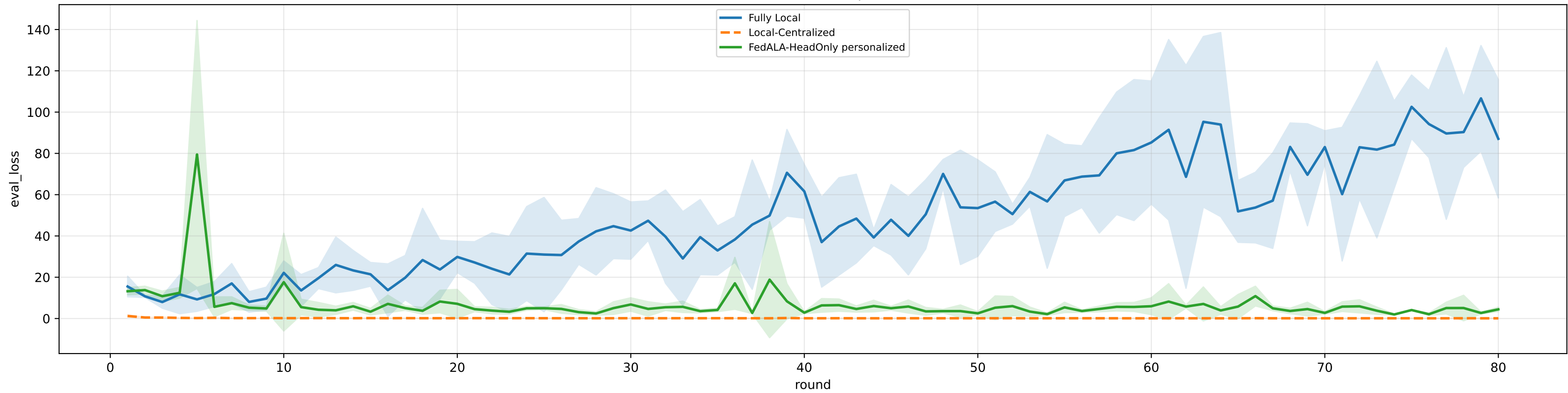


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

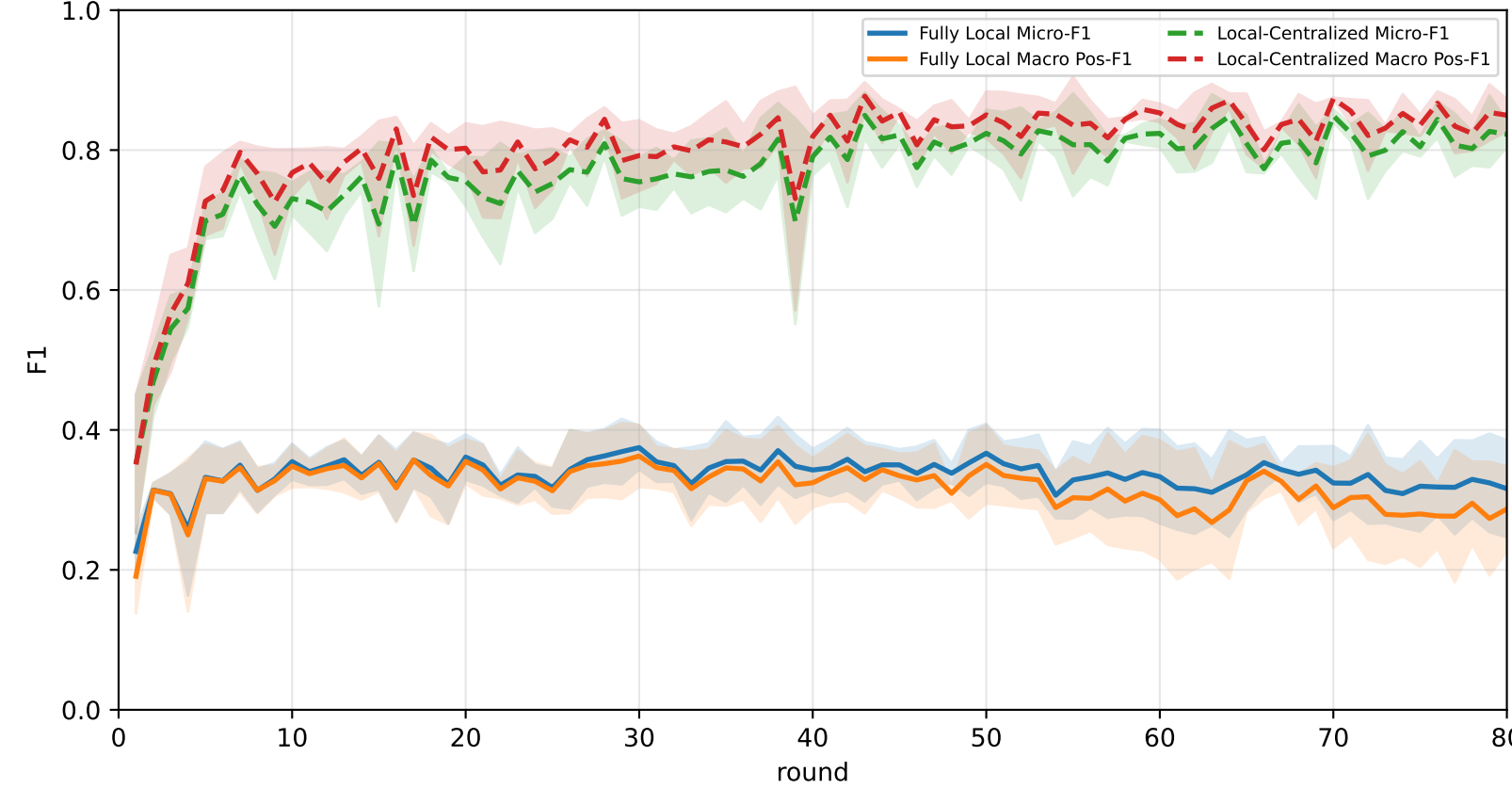
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.5% ± 1.5%	27.5% ± 1.4%	23.1% ± 1.8%	28.2% ± 0.7%	26.4% ± 1.8%	28.3% ± 1.7%	31.5% ± 2.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	55.9% ± 2.8%	56.3% ± 3.0%	56.9% ± 4.8%	57.6% ± 4.9%	59.5% ± 3.4%	56.6% ± 3.2%	51.1% ± 1.9%

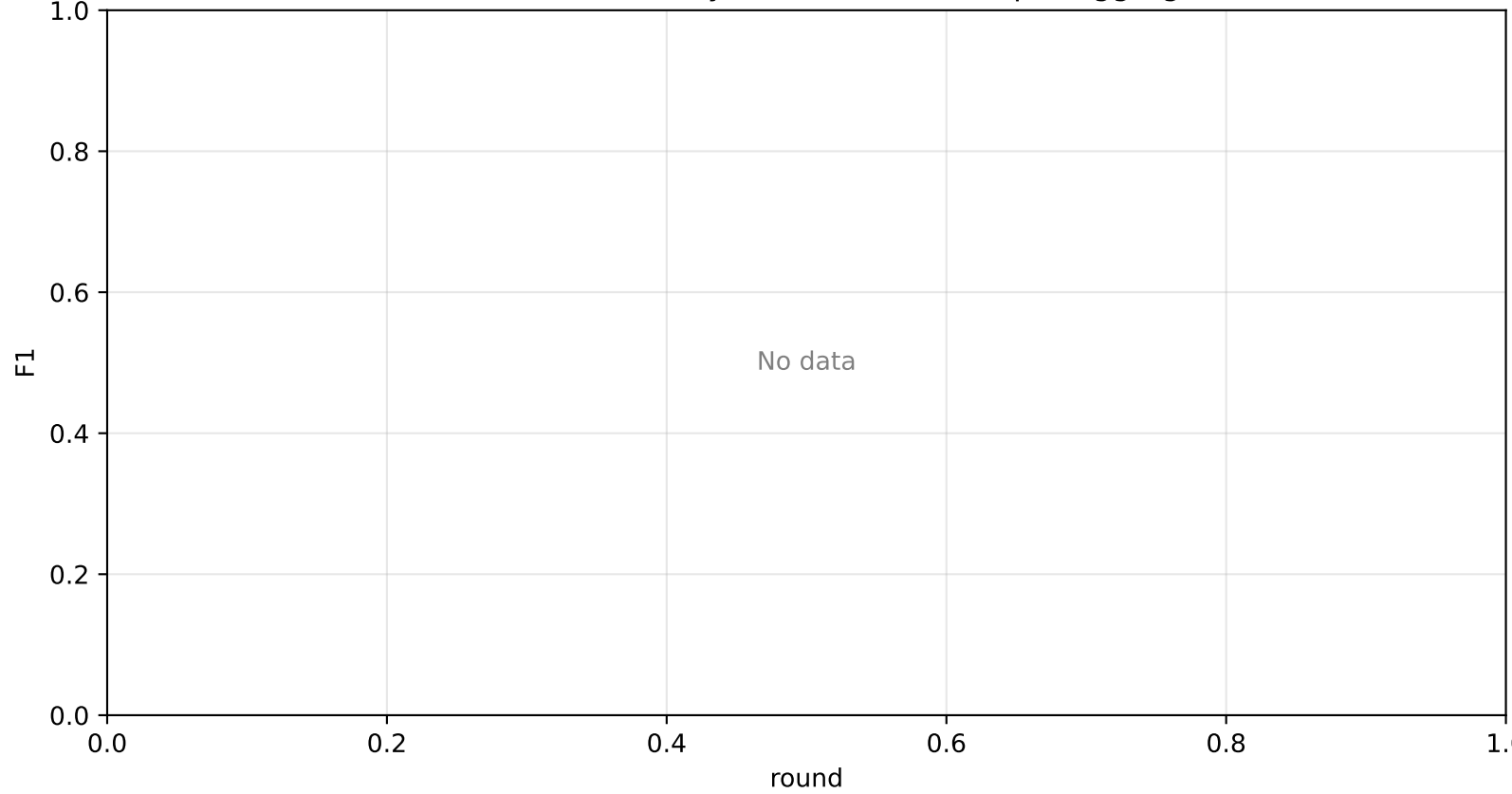
Validation loss comparison



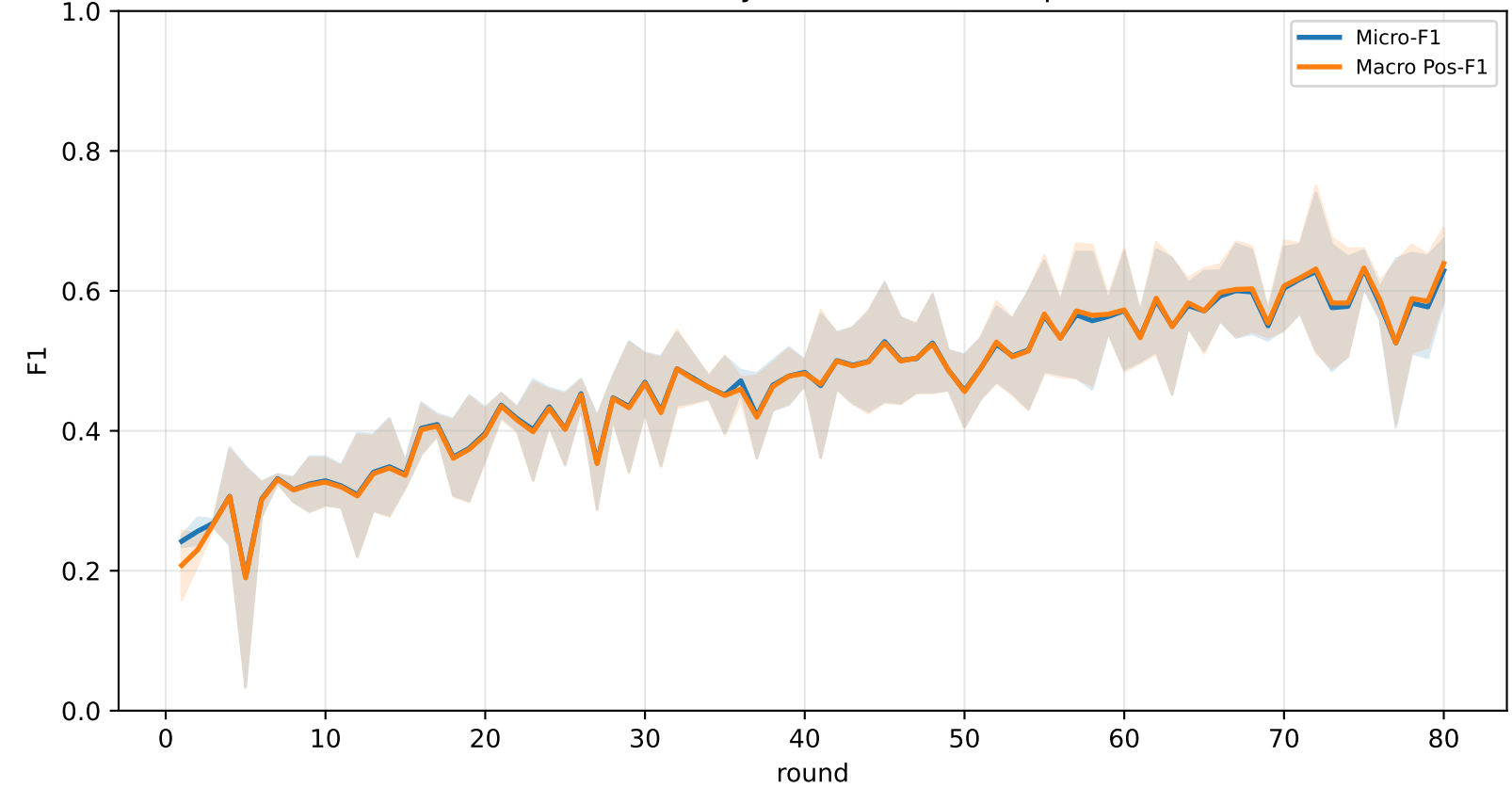
Pooled baseline micro / macro F1



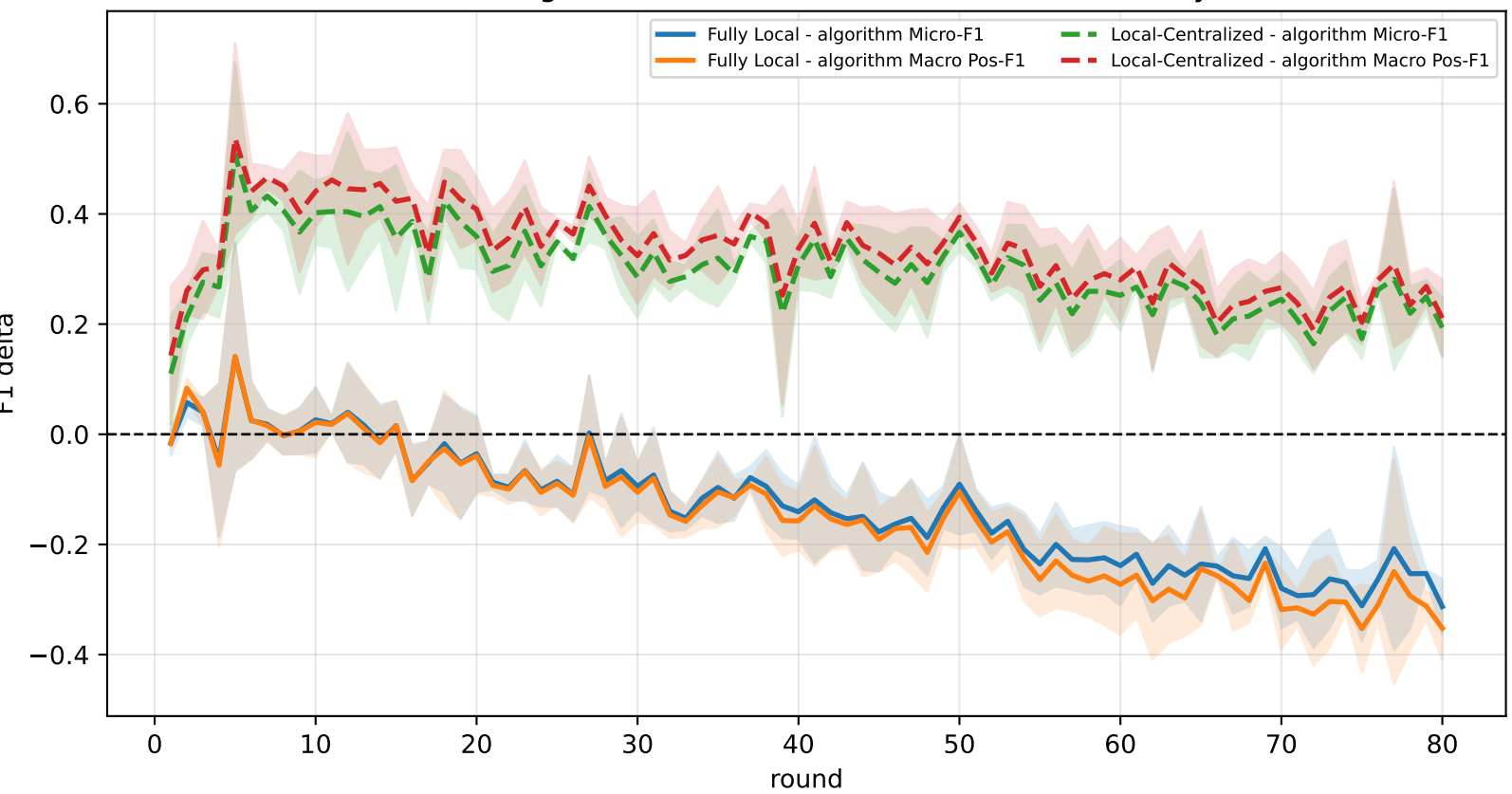
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

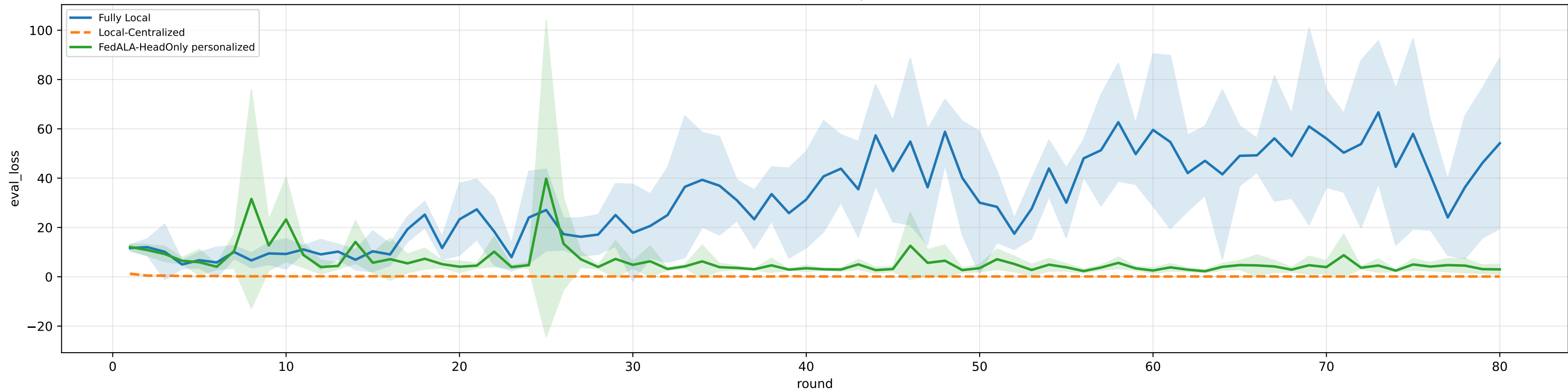


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

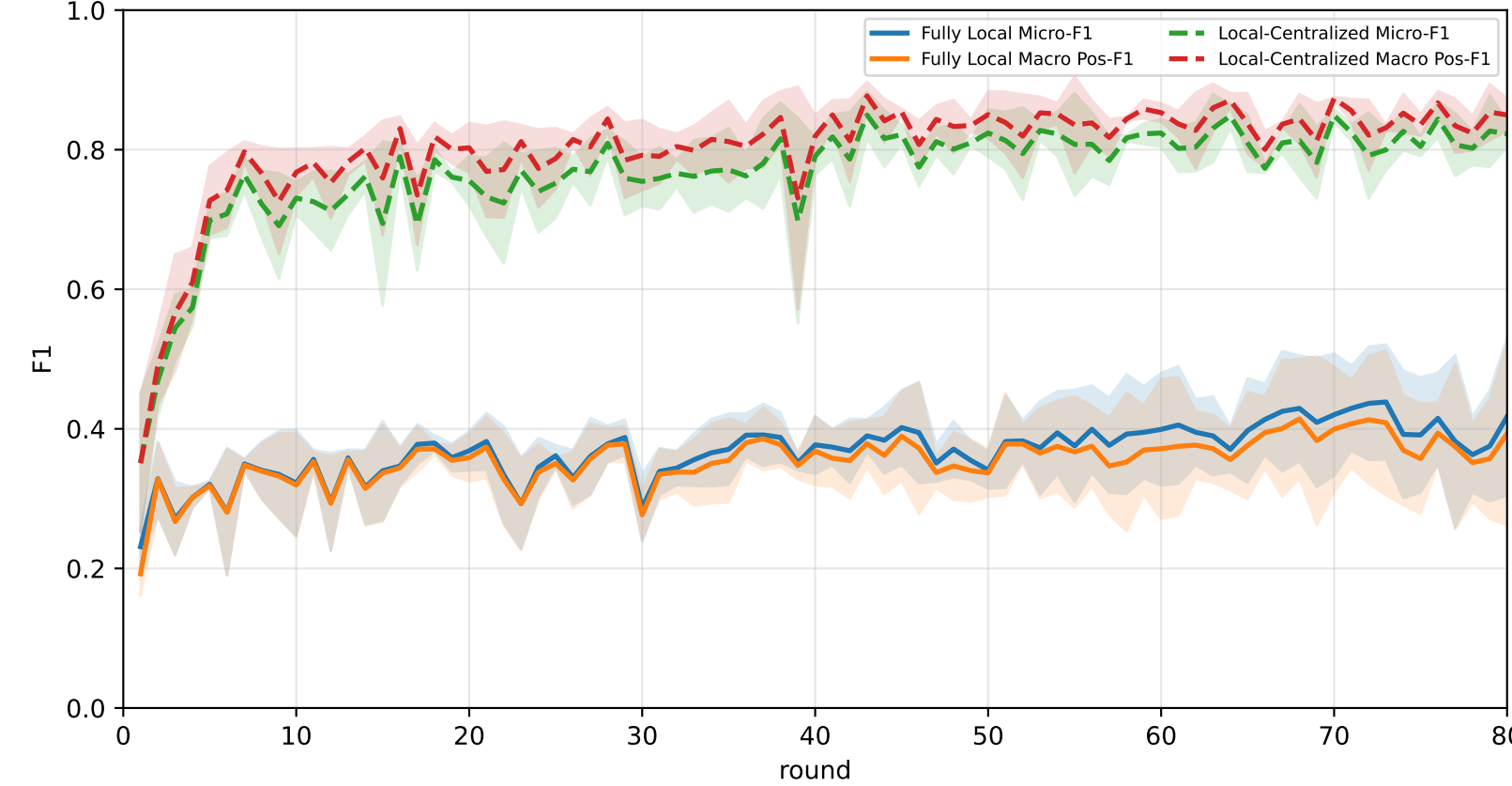
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	24.2% ± 1.1%	24.2% ± 1.1%	20.1% ± 0.7%	25.3% ± 1.1%	24.5% ± 1.6%	22.9% ± 1.3%	28.3% ± 1.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	56.4% ± 3.3%	57.1% ± 3.0%	56.9% ± 1.7%	62.8% ± 4.2%	58.0% ± 2.1%	55.7% ± 8.9%	51.9% ± 2.8%

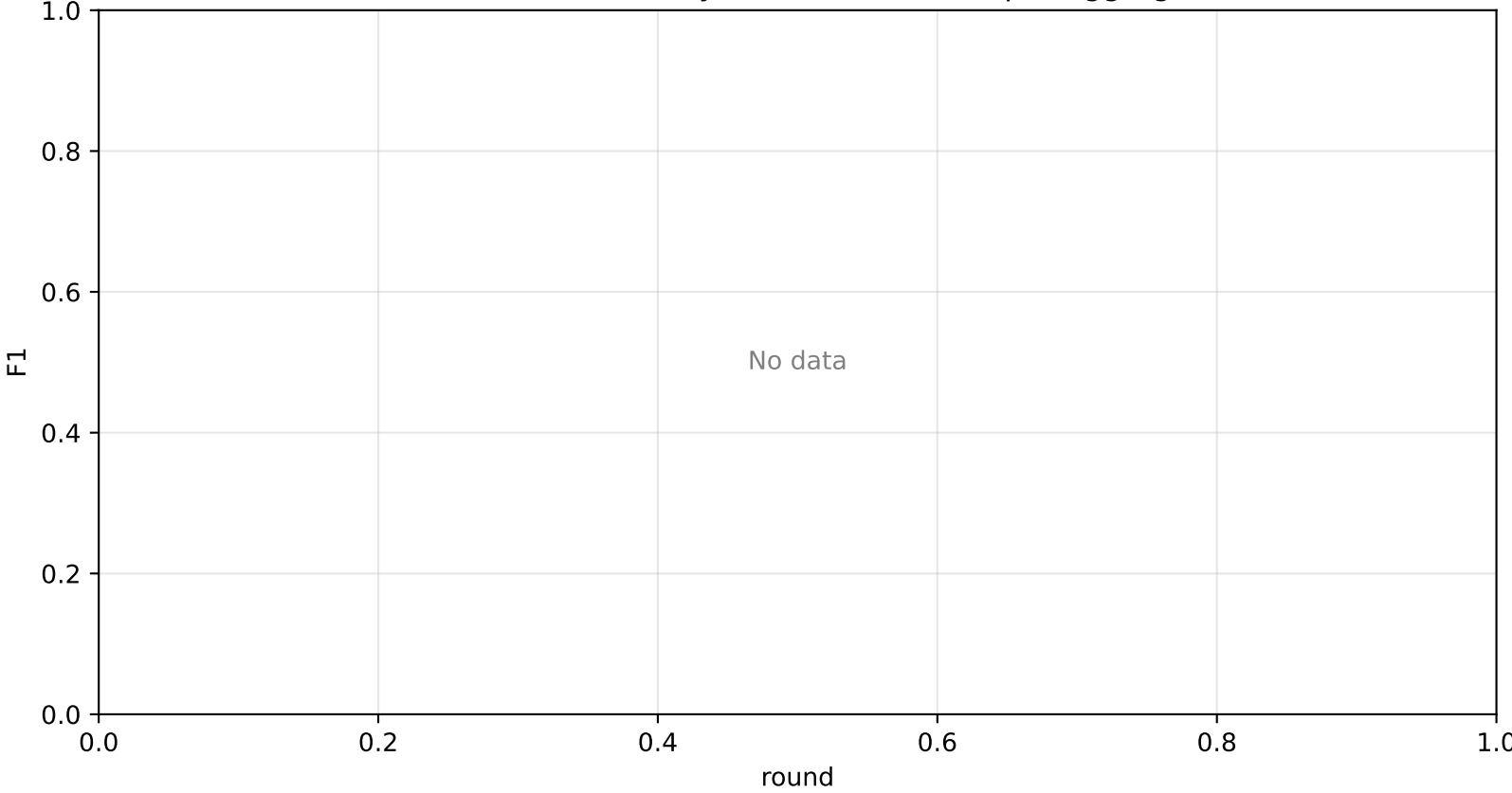
Validation loss comparison



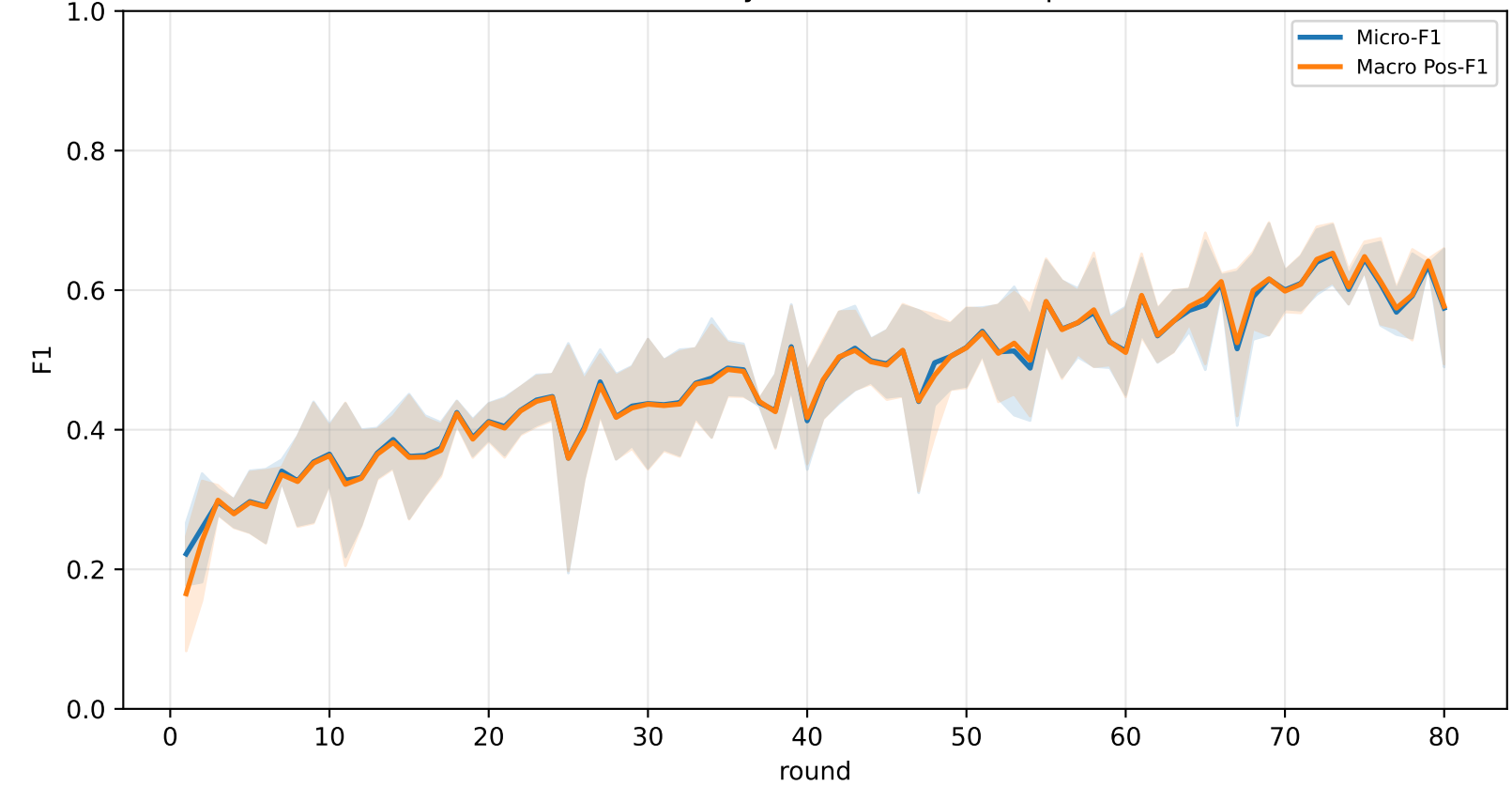
Pooled baseline micro / macro F1



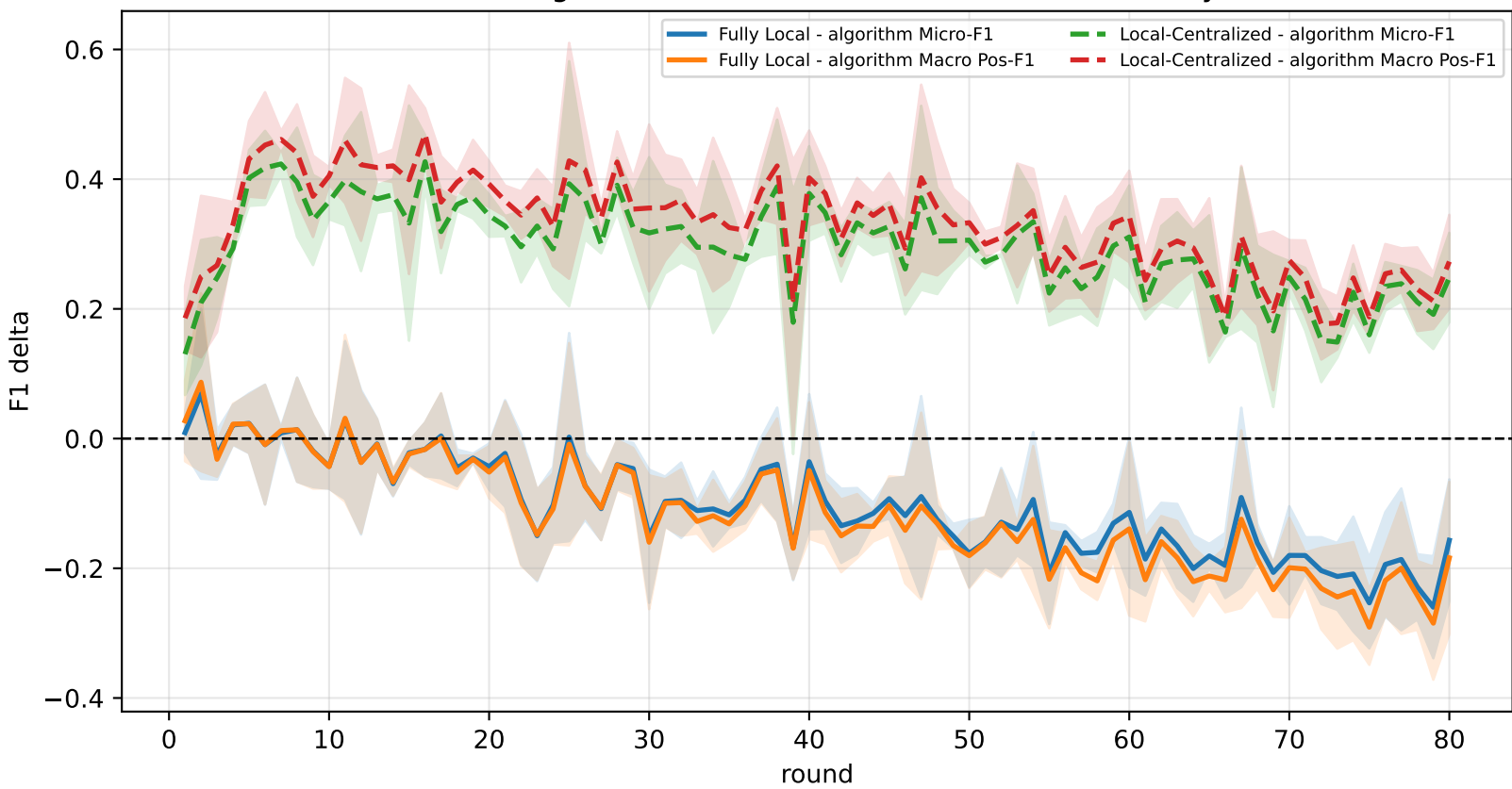
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly

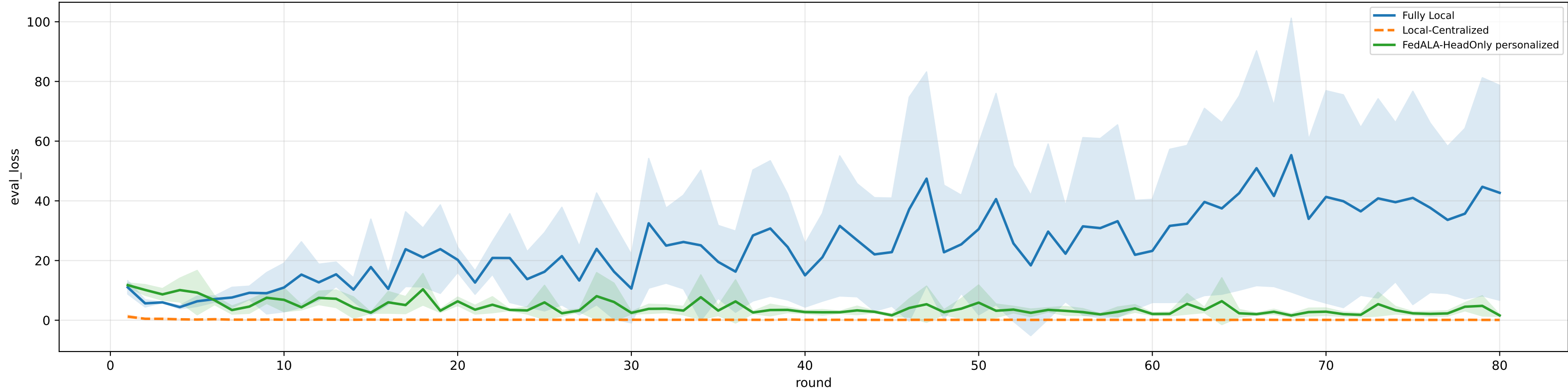


family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

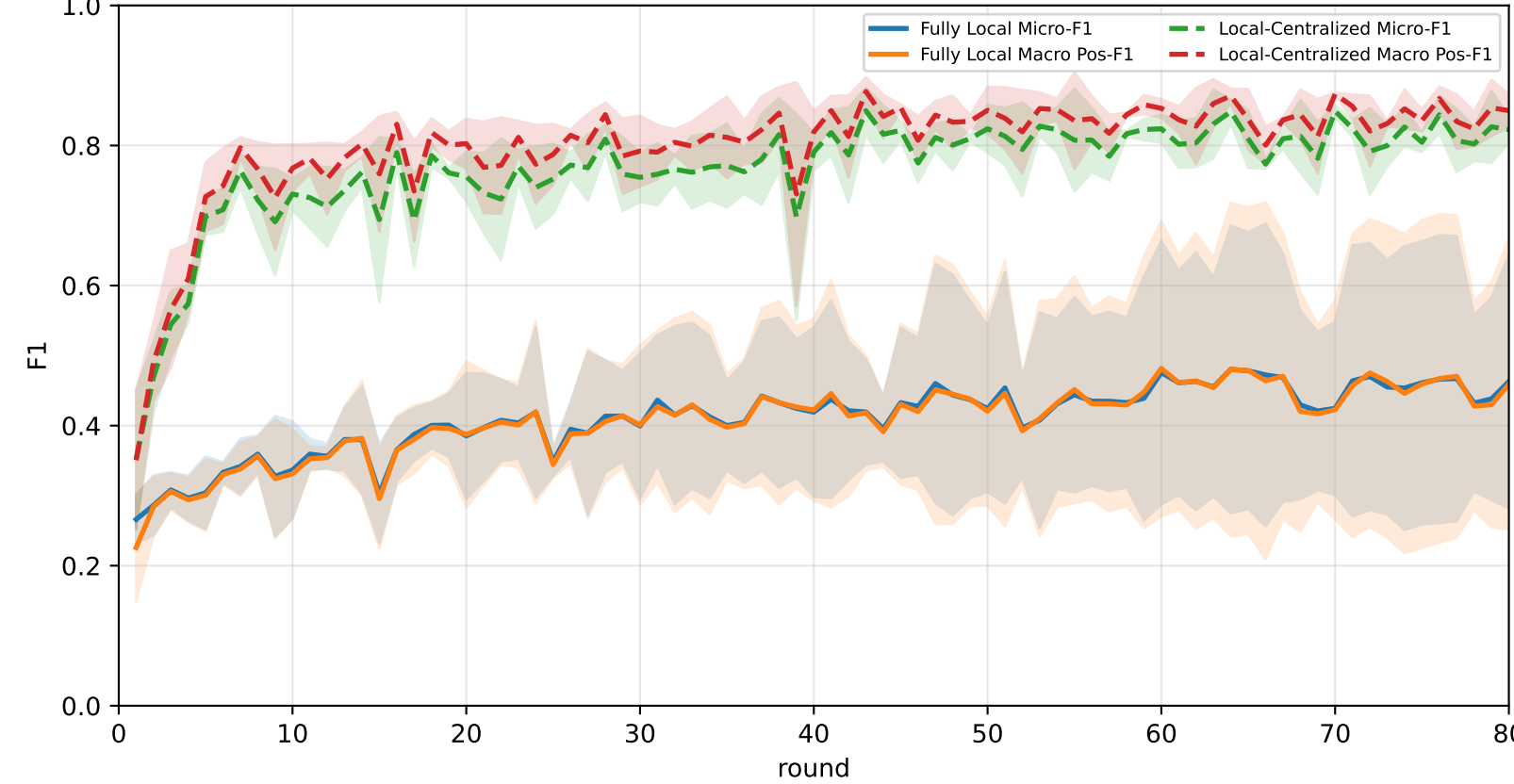
Best logged test performance (Fully Local vs Local-Centralized vs FedALA-HeadOnly)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	31.2% ± 8.8%	31.1% ± 8.9%	26.7% ± 8.1%	32.1% ± 9.5%	33.6% ± 11.2%	31.3% ± 9.2%	32.1% ± 6.8%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedALA-HeadOnly	<u>59.0% ± 5.3%</u>	<u>59.9% ± 5.5%</u>	<u>61.7% ± 12.2%</u>	<u>67.1% ± 2.3%</u>	<u>58.8% ± 5.7%</u>	<u>58.7% ± 5.9%</u>	<u>53.1% ± 2.9%</u>

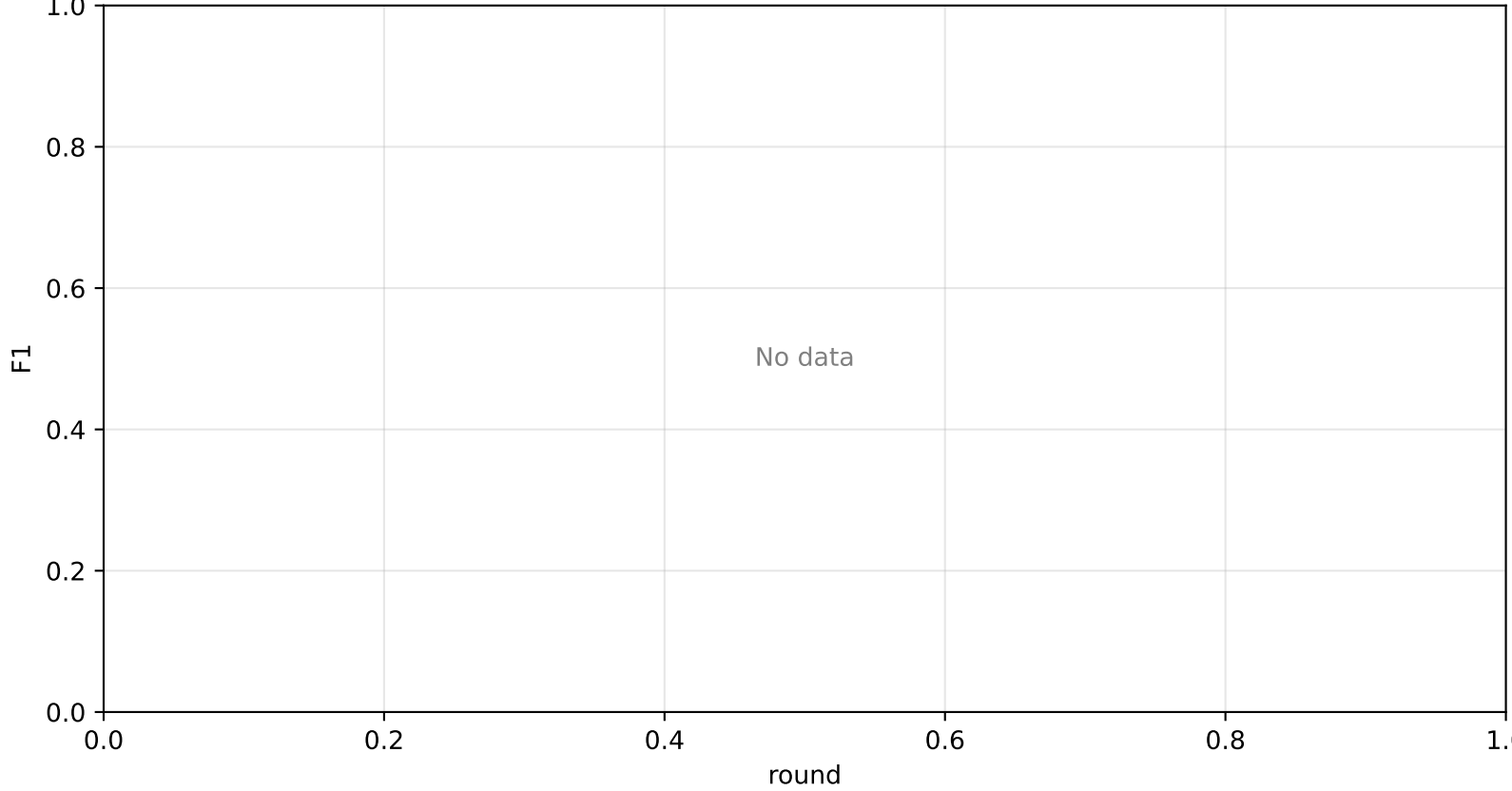
Validation loss comparison



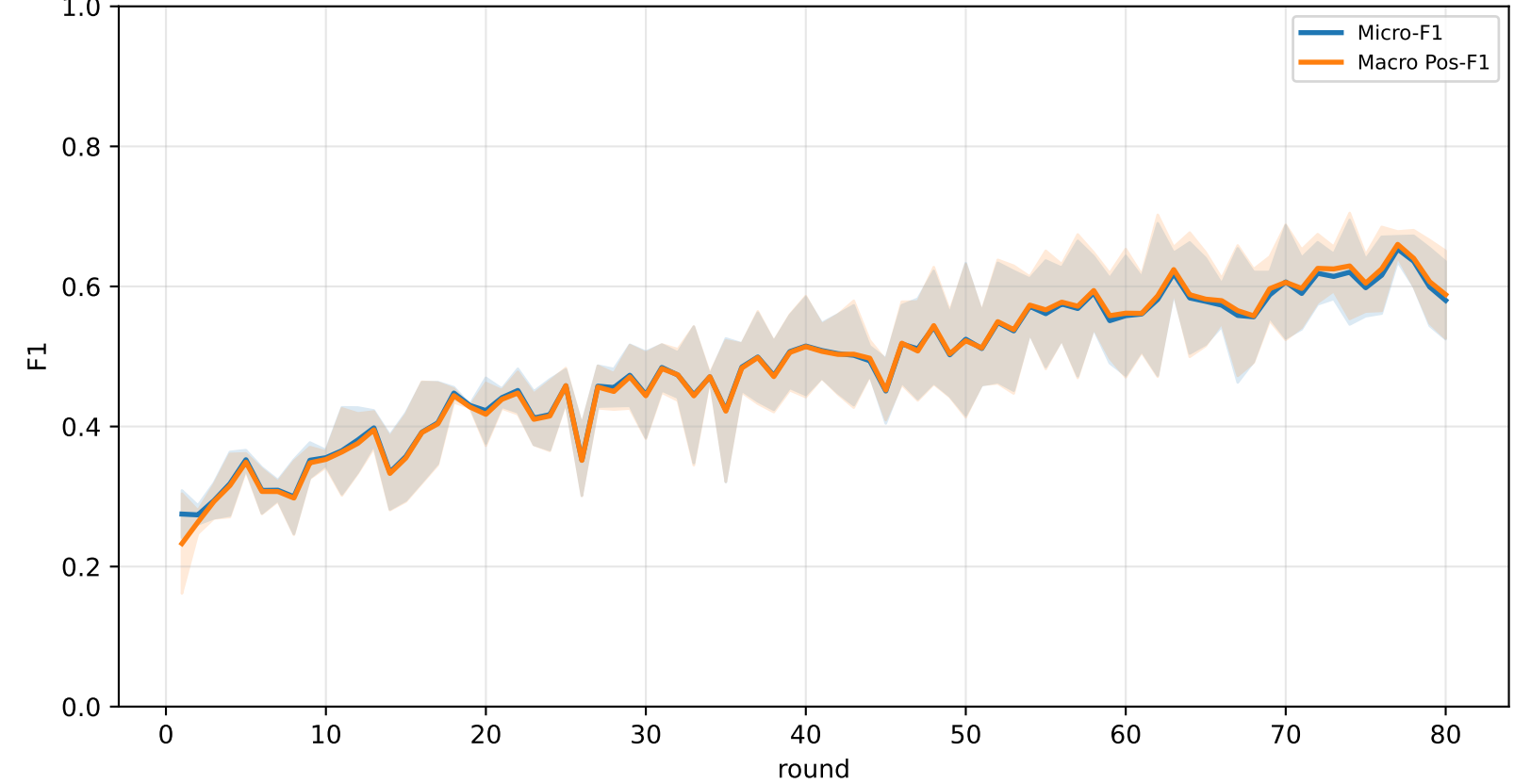
Pooled baseline micro / macro F1



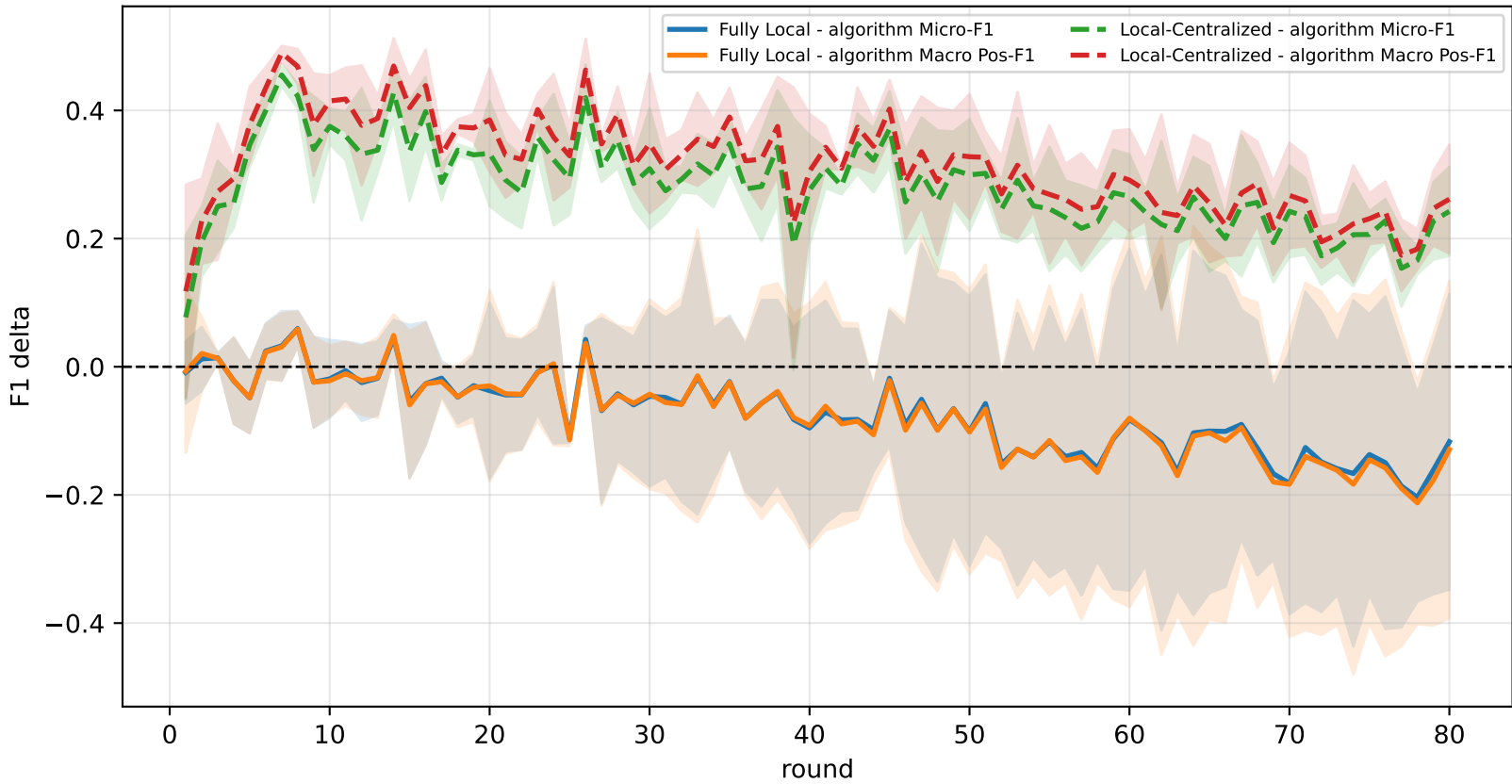
Pooled FedALA-HeadOnly micro / macro F1 (pre-aggregation)



Pooled FedALA-HeadOnly micro / macro F1 (personalized)



Delta diagnostics: baselines minus FedALA-HeadOnly



Controlled q-label heterogeneity: High | FedALA-HeadOnly ALA diagnostics | aggregated / pooled | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

